



Fundamentals of neurobiology

ch I : fundamental principles of neuronal excitability

R.C. Lambert

Neuronal networks and physiopathological rhythms

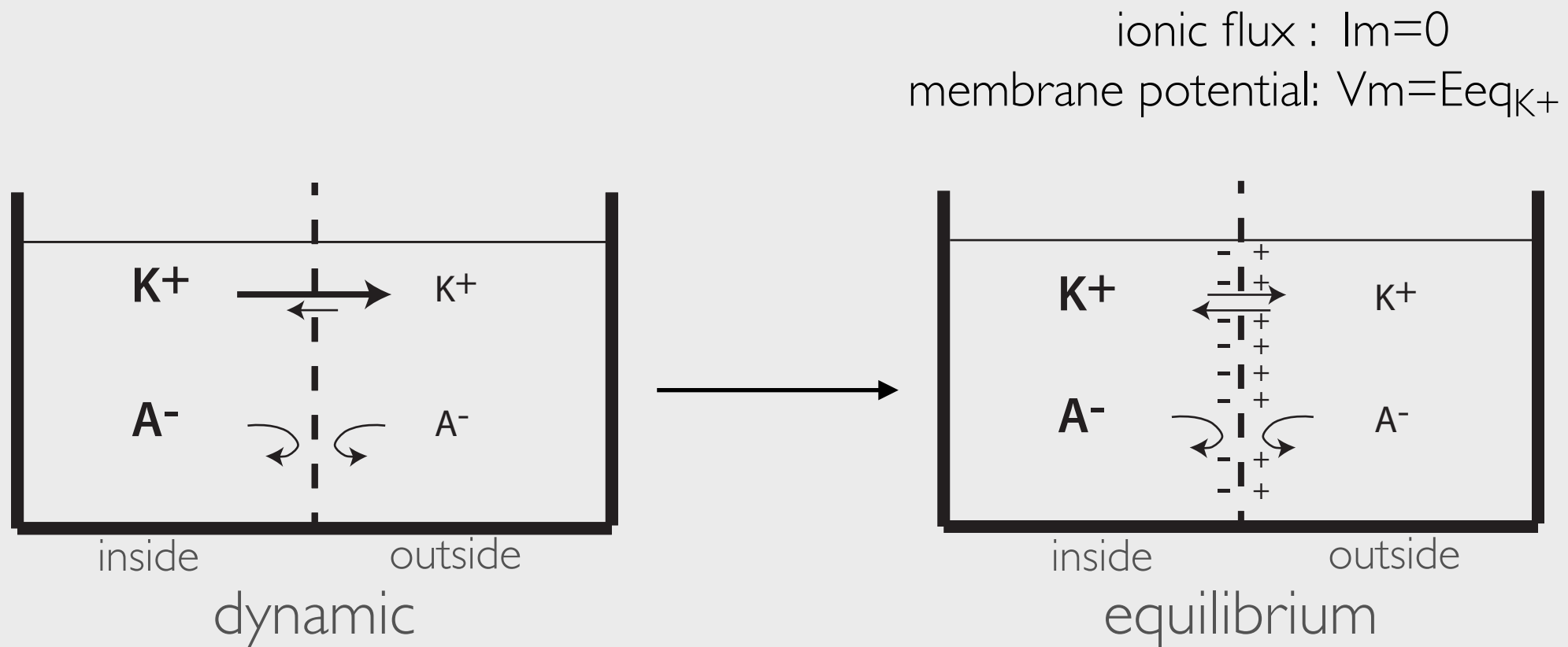
<http://www.ibps.upmc.fr/fr/Recherche/umr-8246/reseaux-de-neurones-et-rythmes-physiopathologiques>

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return to basics ...



2 opposite forces :

- concentration gradient
- ionic repulsion

— — —> equilibrium is reached when the ionic repulsion is sufficient to cancel the concentration gradient

— — —> excess of + charges in the outside compartment (default in the inside), $V_m \neq 0 \text{ mV} = E_{eq_{K^+}}$

Nernst equation : $E_{eq} =$
$$E_{eq} = -\frac{RT}{ZF} \ln \frac{[ion]_i}{[ion]_o}$$

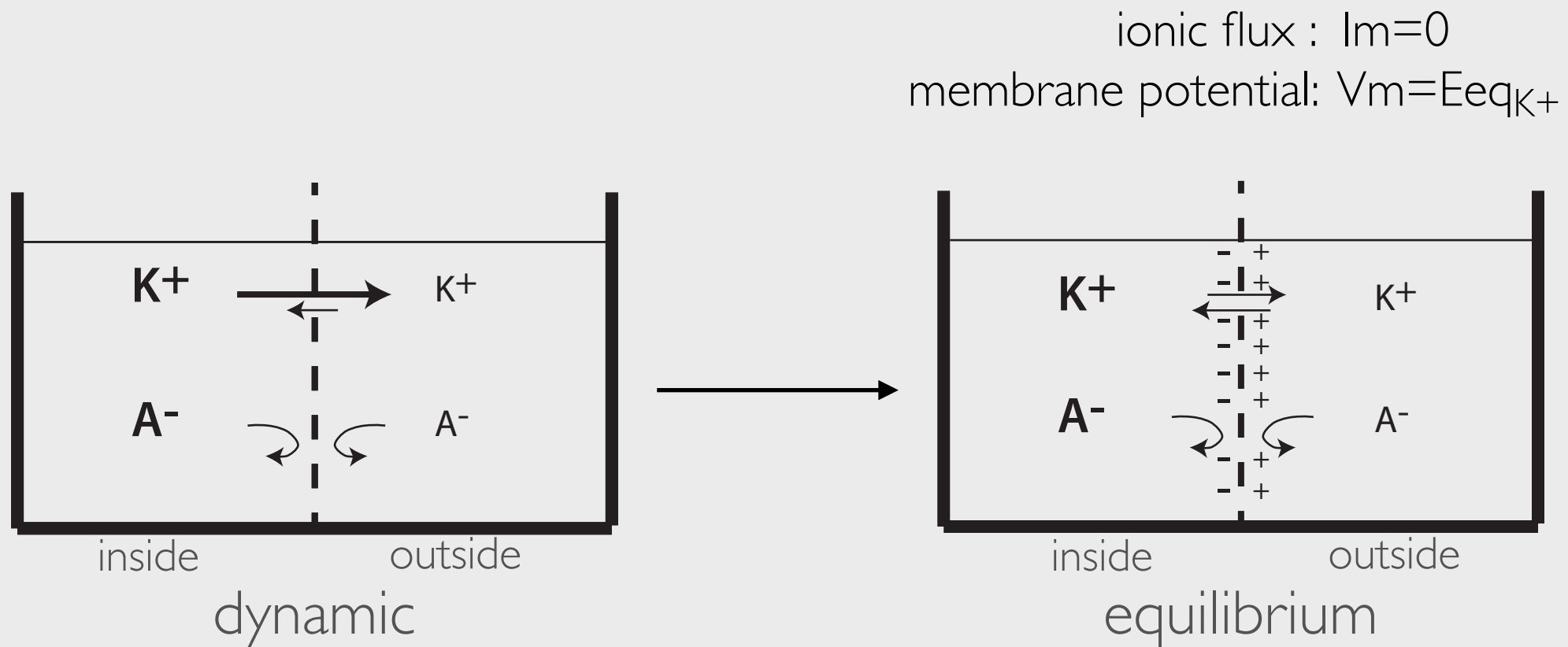
R is the universal gas constant: $R = 8.314472 \text{ JK}^{-1} \text{ mol}^{-1}$

T is temperature in Kelvins

Z is ion valence

F is the Faraday constant: $F = 9.64853399 \times 10^4 \text{ C mol}^{-1}$

return to basics ...



2 opposite forces :

- concentration gradient
- ionic repulsion

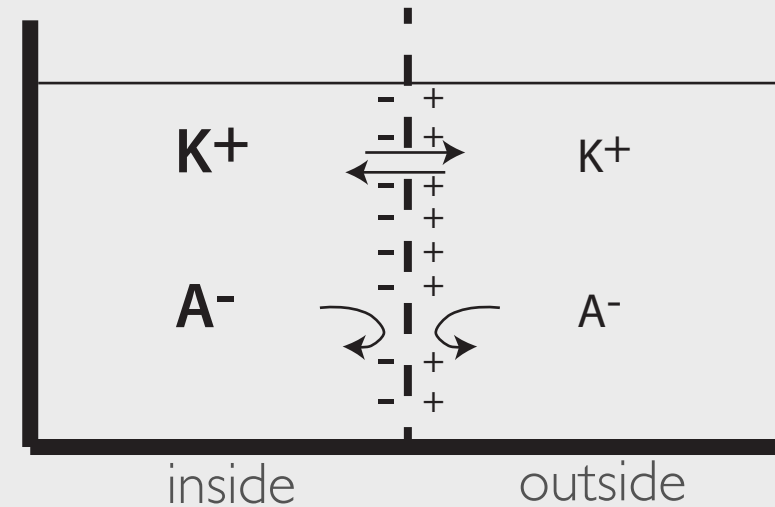
— — —> equilibrium is reached when the ionic repulsion is sufficient to cancel the concentration gradient
 — — —> excess of + charges in the outside compartment (default in the inside), $V_m \neq 0\text{mV} = E_{eq_{K^+}}$

golden rule of electrophysiology: when a channel opens, the ionic flux will only stop when the membrane potential is equal to the ion equilibrium potential -> the membrane potential always evolves towards this value.

however ...

ionic flux : $I_m = 0$

membrane potential: $V_m = E_{eq_{K^+}}$



excess of + charges in the outside compartment (default in the inside)

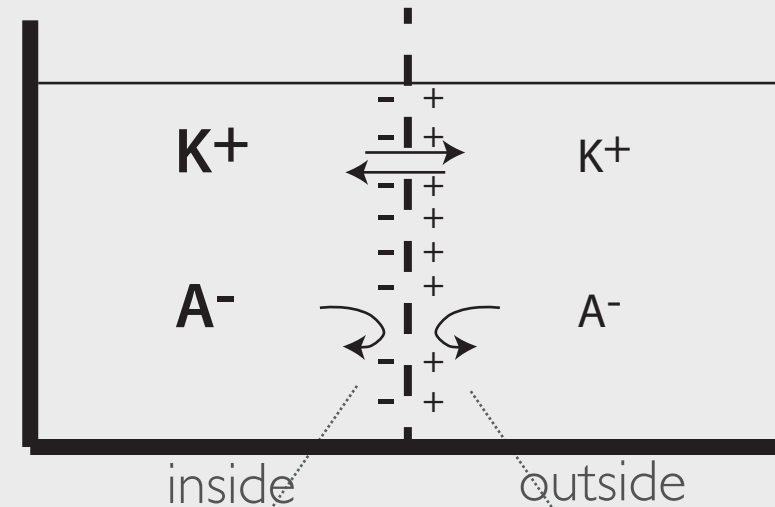
$\neq$

In an electrolyte solution, the concentrations of all the ions are such that the solution is **electrically neutral**.

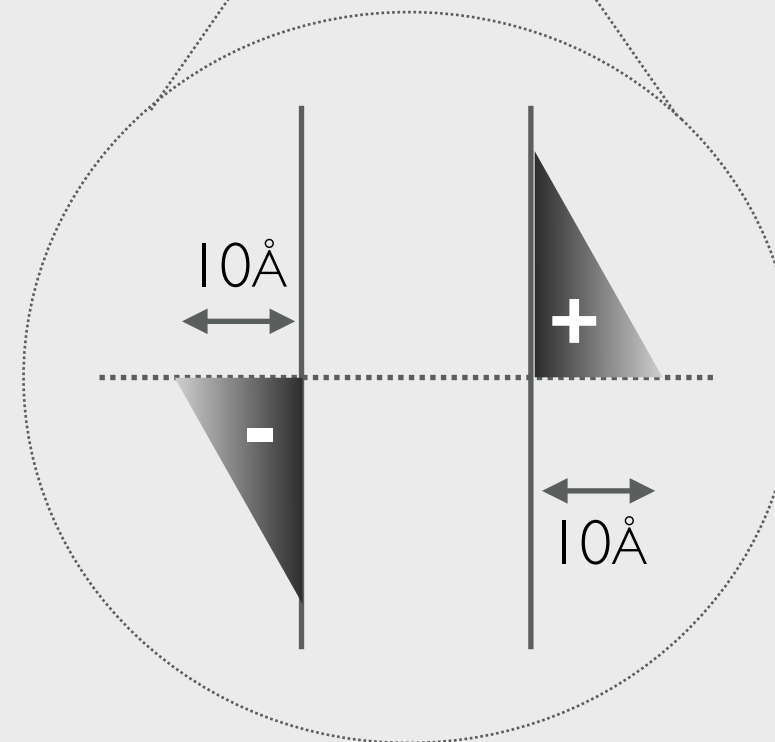
charges accumulate at the membrane

ionic flux : $I_m = 0$

membrane potential: $V_m = E_{eq_{K^+}}$



Each ion flux through the membrane will change the concentration of charges on a short distance from the membrane, creating the membrane potential



charges accumulate at the membrane

ionic flux : $I_m = 0$

membrane potential: $V_m = E_{eq_{K^+}}$

« A capacitor consists of two conductors separated by a non-conductive region » (Wikipedia :-))

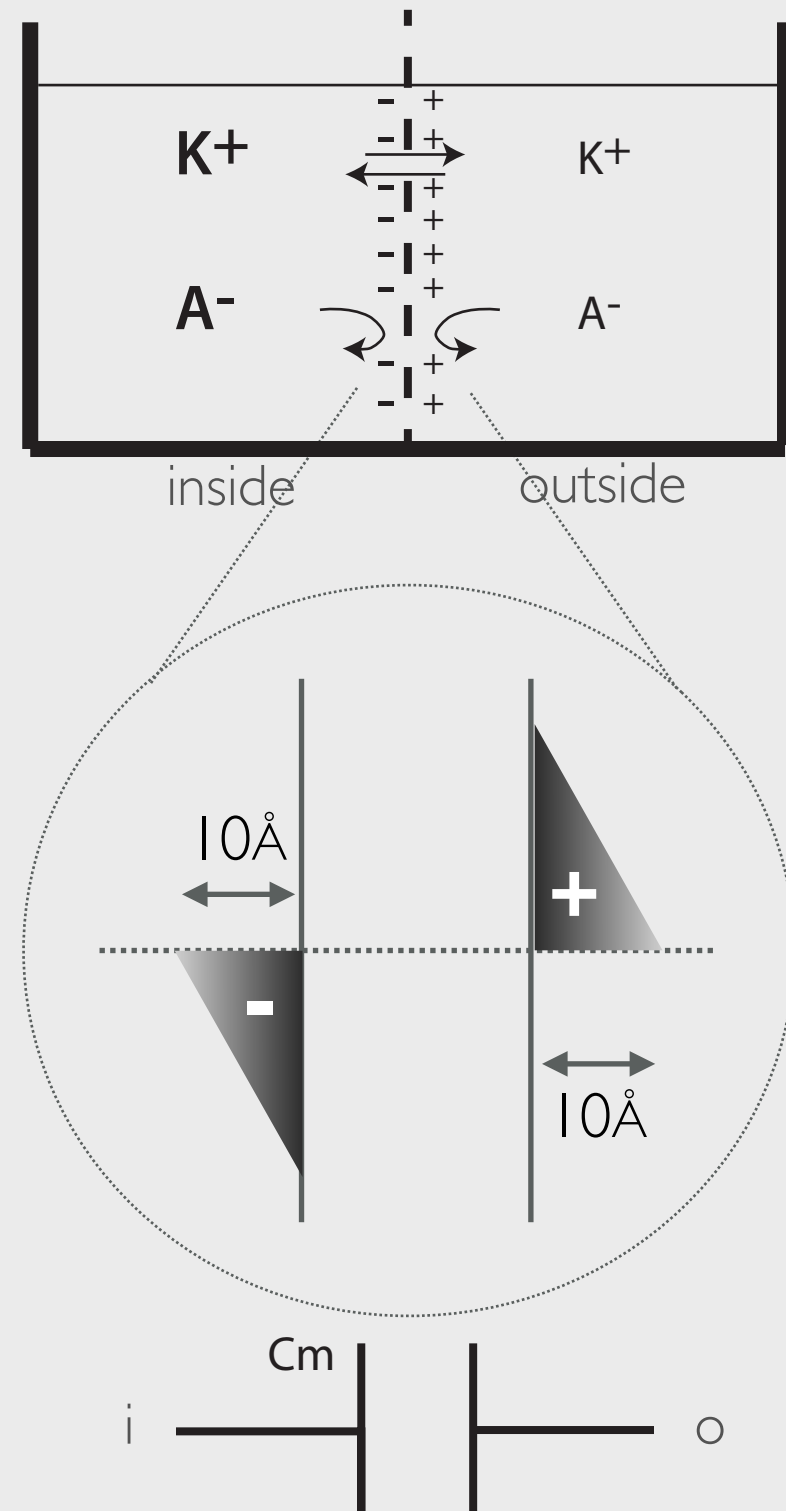
- The capacitor behaves as a charge storage device
- The **charge accumulation (Q) in the capacitor plates is not instantaneous. Potential across the capacitor exponentially changes**

$$Q = C \cdot E_{eq} \rightarrow I_c = \frac{dQ}{dt} = C \cdot \frac{dV}{dt}$$

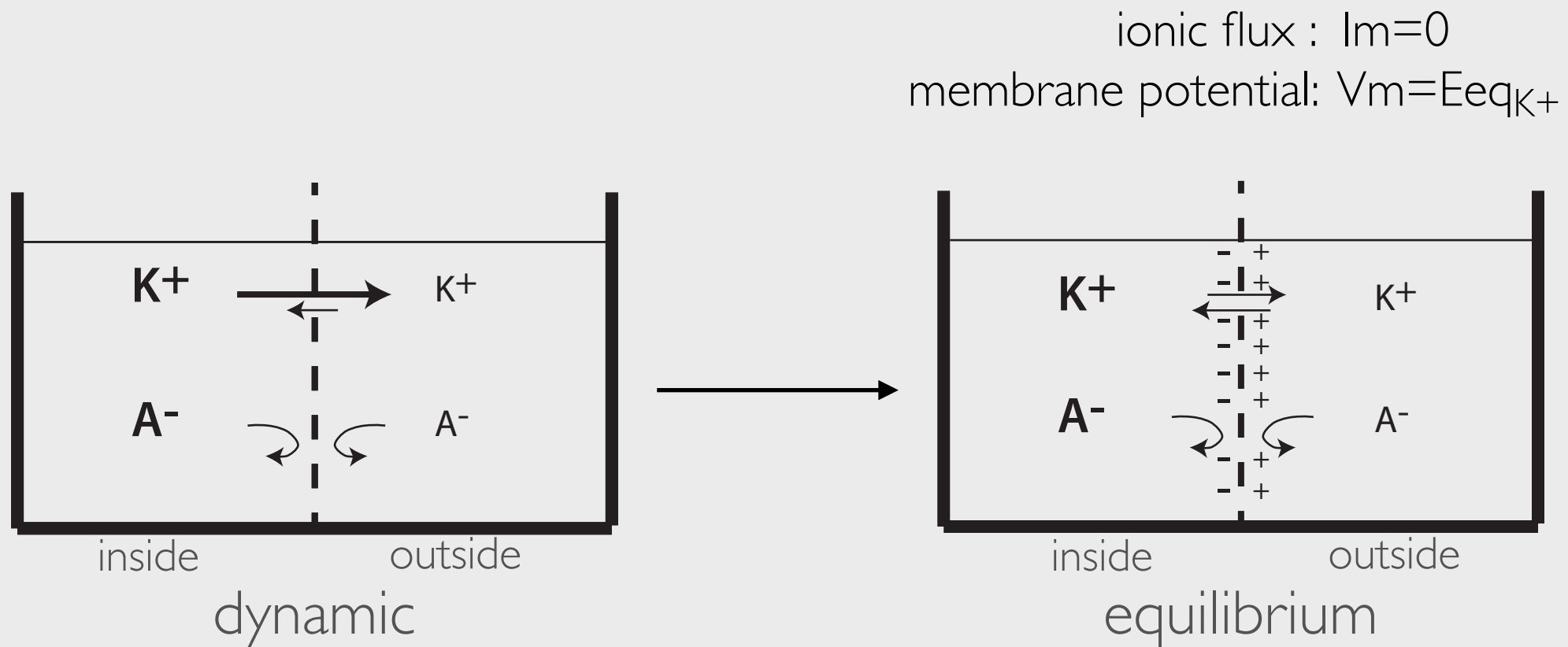
The value of the capacitance C (in F) is proportional to the nature of the dielectric (cste), its thickness (cste) and the area of the « plates » (size of the cell).

$$C_m = cmA$$

with cm (specific capacitance) = 10nF/mm² and A in mm²
(0.01 to 0.1 mm²)



ionic currents



2 opposite forces :

- concentration gradient
- ionic repulsion

$$I_m = G_m \cdot (V_m - E_{eq_{K^+}})$$

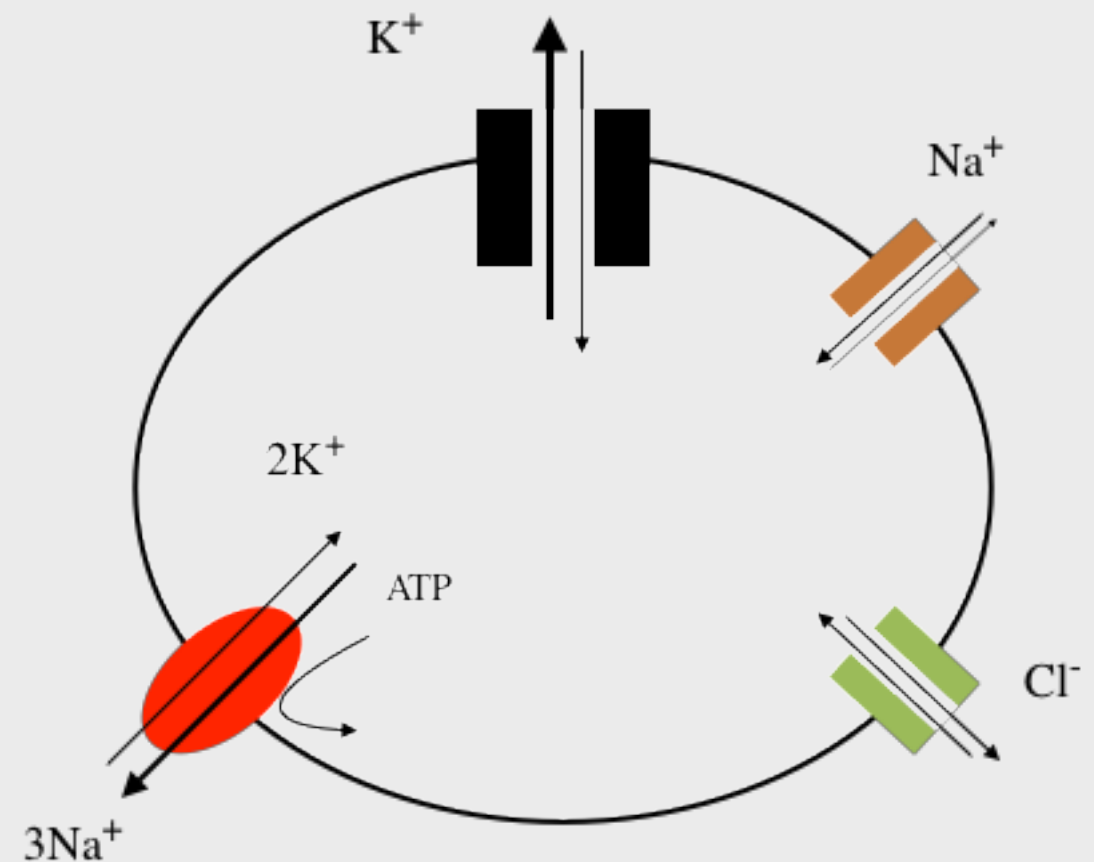
conductance (nS) electrical gradient « concentration » gradient

$V_m > E_{eq} \Rightarrow I_m > 0$ outgoing flux of +
 $V_m < E_{eq} \Rightarrow I_m < 0$ incoming flux of +

resting membrane potential

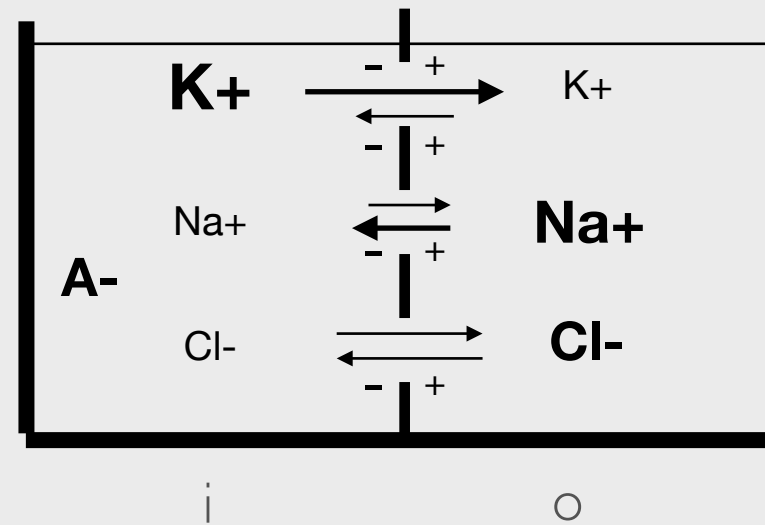
Ion	Ext. conc. (mM)	Int. conc. (mM)	Nernst potential (mV)
Na^+	150	5	60
K^+	3	150	-90
Ca^{2+}	2	50 nM	120
Cl^-	125	5	-75

- other intracellular anions (non-permeant except bicarbonate) : proteins, metabolites, phosphates, bicarbonate
- gradients maintained by pumps (Na/K-ATPase: electrogenic 3Na/2K)



resting membrane potential

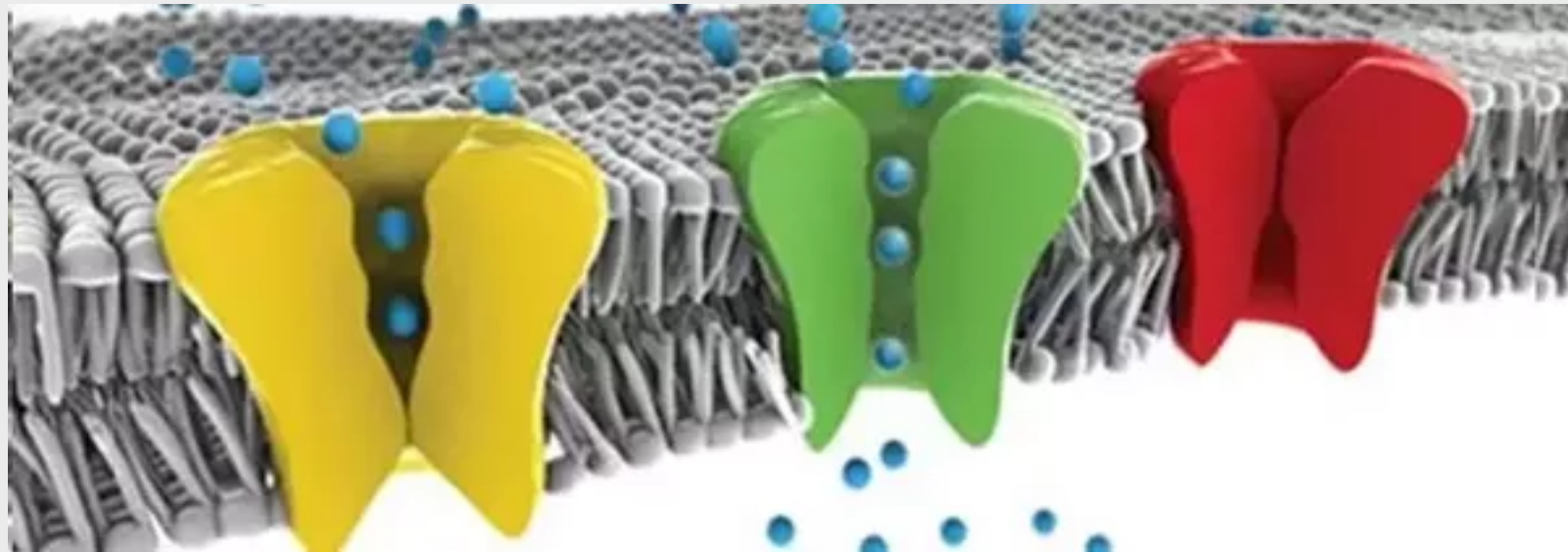
$$V_m = \frac{G_{Na} \cdot E_{eq_{Na}} + G_K \cdot E_{eq_K} + G_{Cl} \cdot E_{eq_{Cl}}}{G_{Na} + G_K + G_{Cl}}$$



at rest: $G_K \gg G_{Na} \longrightarrow V_m \approx -70mV$

since $E_{eq_{Cl}}$ is $-75mV$, Cl flux is almost zero at rest (neglected)

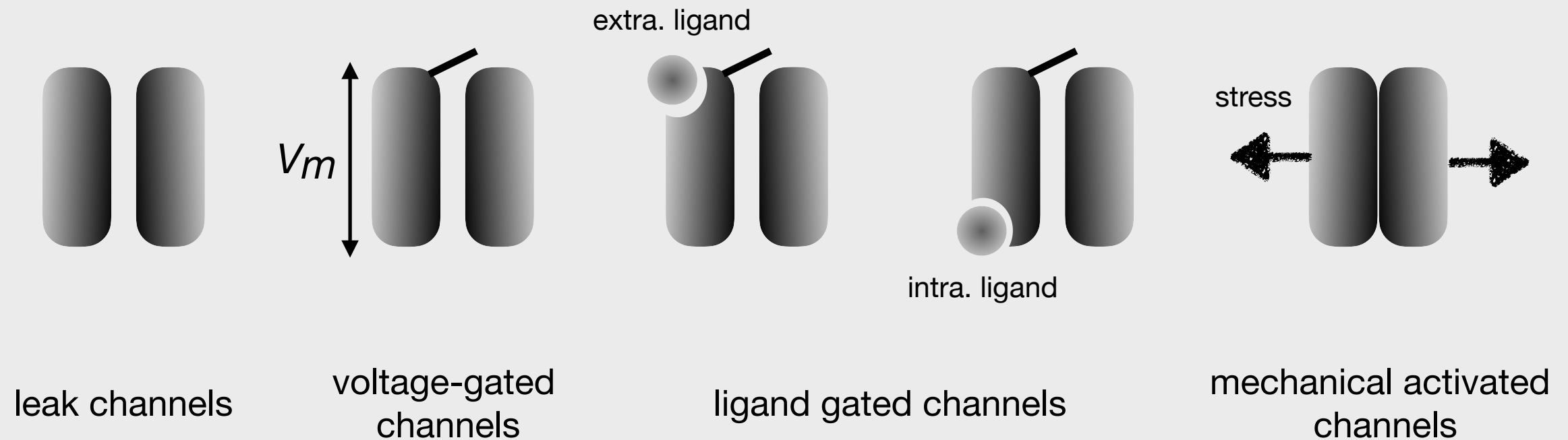
ion channels



ion channels are membrane proteins conducting ions... or not

- some of them are always open (leak channels)
- some of them open under specific conditions (intracellular or extracellular ligand, voltage , mechanical stress). They will remain open as long as the conditions are met
- some open but undergo an additional conformational change making them non-conductive (inactivation, desensitization). They will generate a transient ionic flux although the opening conditions are still present.

ion channels

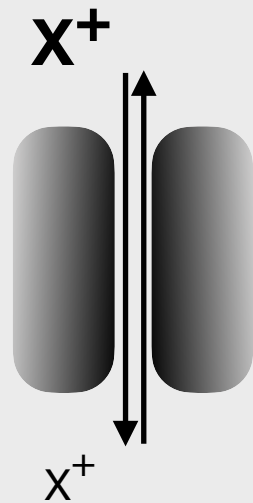


The main activation conditions define the family of the ionic channels, however :

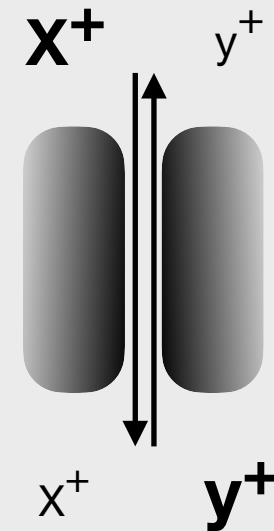
- every channels is modulated by intracellular mechanism (phosphorylation, nucleotide binding, Ca,...)
- properties of ligand gated or mechanically activated channels may be modulated by the membrane potential

equilibrium potential *versus* reversal potential

ion channels are selectively permeant to cations or anions but some are highly selective for a single ion while others are permeant for 2 or 3 different ions



$$I = G \cdot (V_m - E_{eq_{X^+}})$$



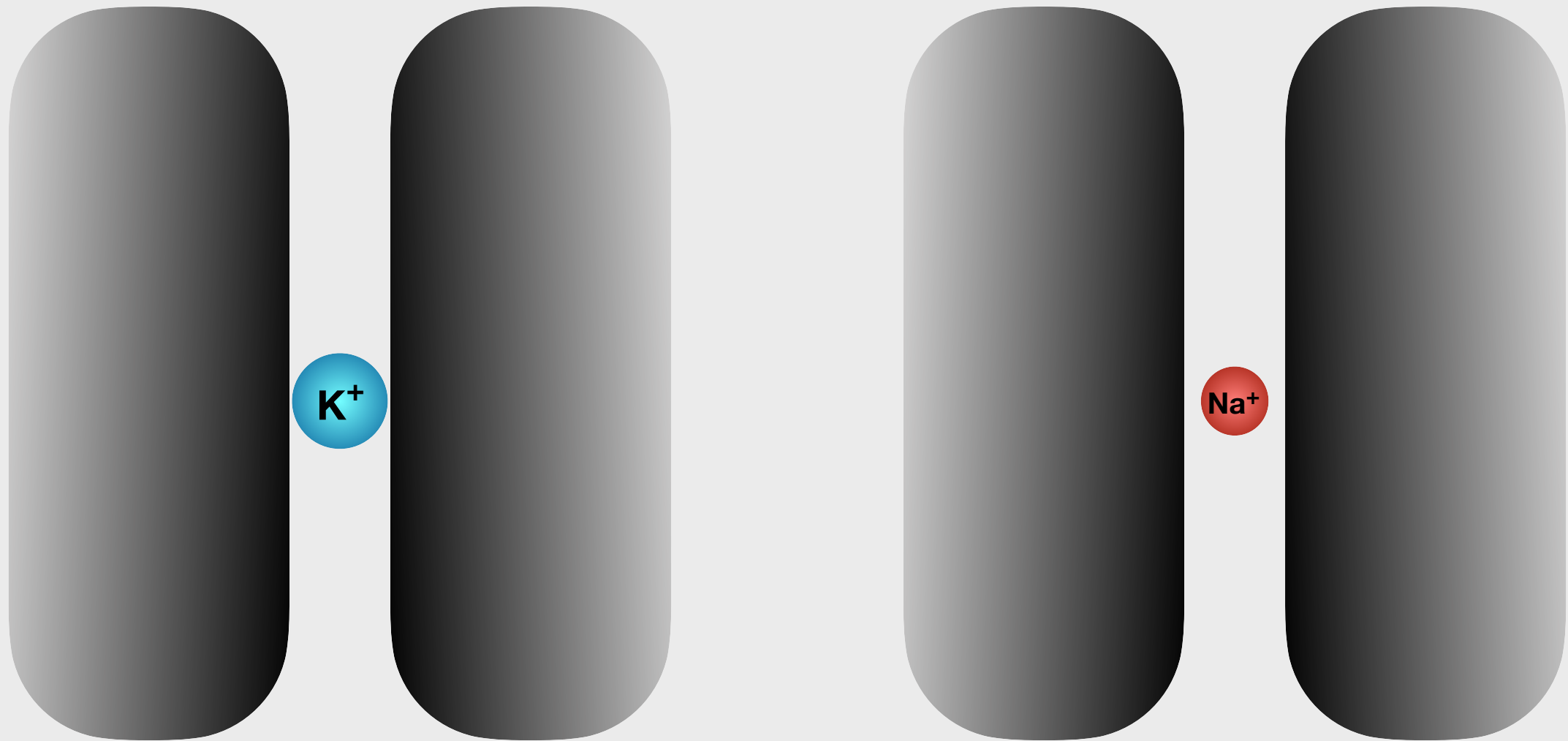
$$I = G \cdot (V_m - E_{rev})$$

with
$$E_{rev} = \frac{G_{X^+} E_{eq_{X^+}} + G_{Y^+} E_{eq_{Y^+}}}{G_{X^+} + G_{Y^+}}$$

$\frac{G_{X^+}}{G_{Y^+}}$ is the relative permeability to X and Y of the ion channel (permeability ratio)

golden rule of electrophysiology: when a channel opens, the ionic flux will only stop when the membrane potential is equal to the reversal potential (equilibrium potential for single ion) -> the membrane potential always evolves towards this value.

selectivity of ion channels



Molecular weight of Na^+ : ~23

Molecular weight of K^+ : ~39

One + charge

?



why smaller Na^+ ions do not pass through the selectivity filter of potassium channels?

selectivity of ion channels

H₂O is neutral but polar (the positive and negative electrical charges are unevenly distributed)



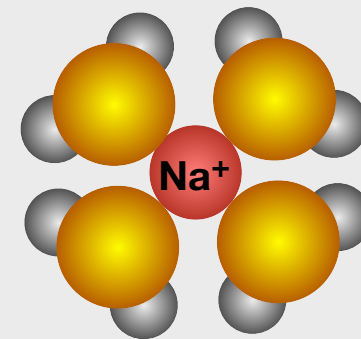
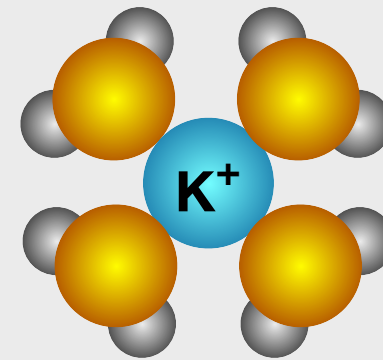
Ions in water are loosely bound to H₂O (hydration shell).



too large to fit into an ion channel.

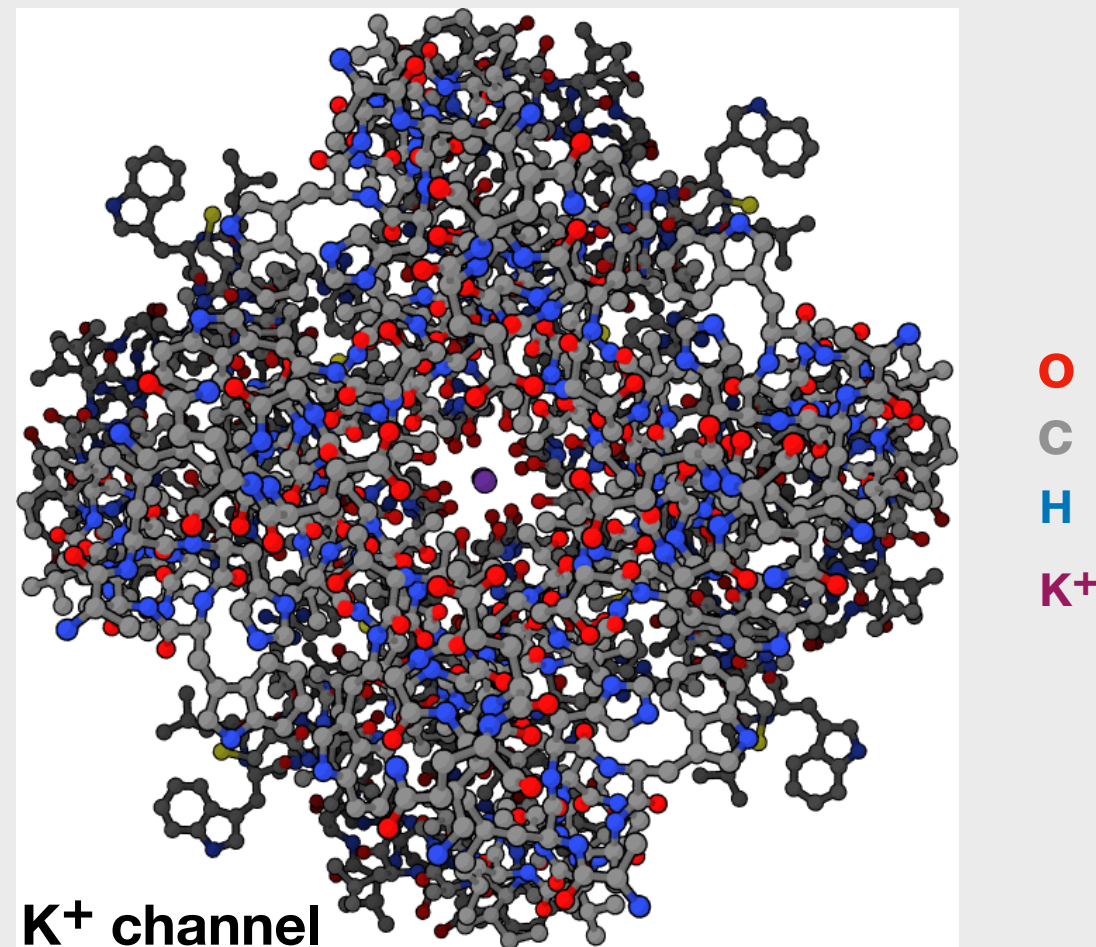


to remove H₂O, an ion must bind another molecule with a stronger bond than the hydrogen bond with water.



selectivity of ion channels

Carbonyl oxygen (C=O) are present in every protein including ion channels. The position in space of carbonyl oxygen varies depending on channel's folding



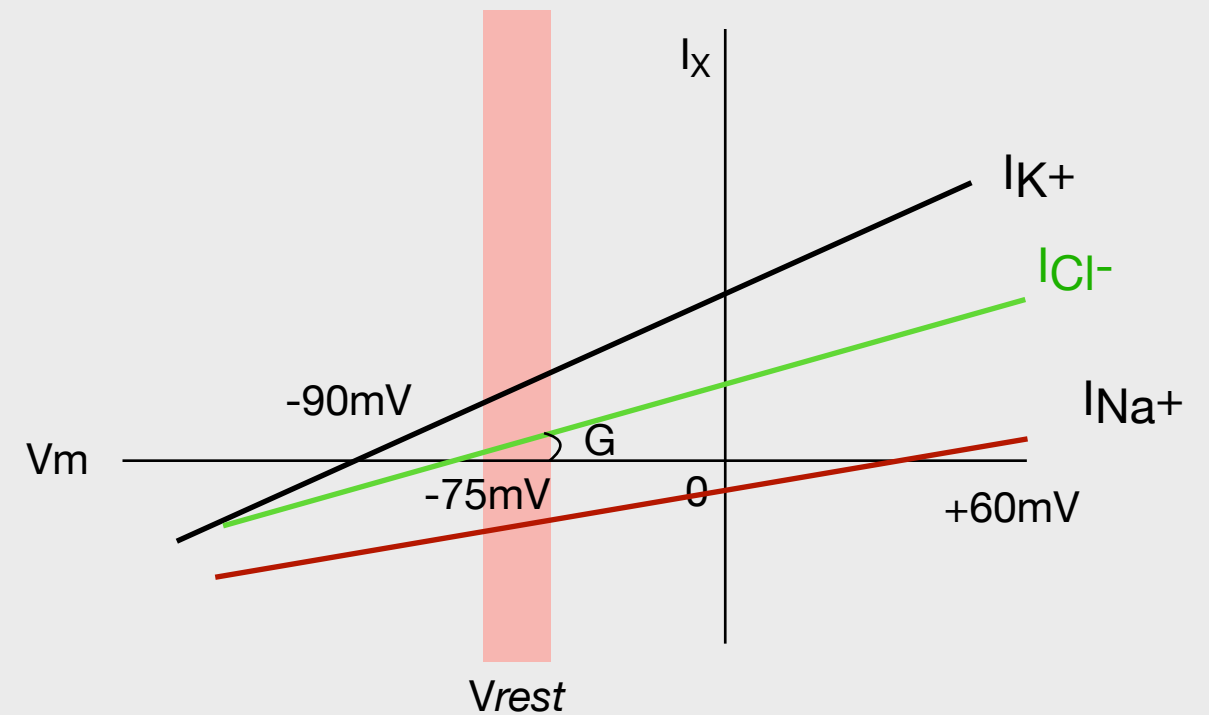
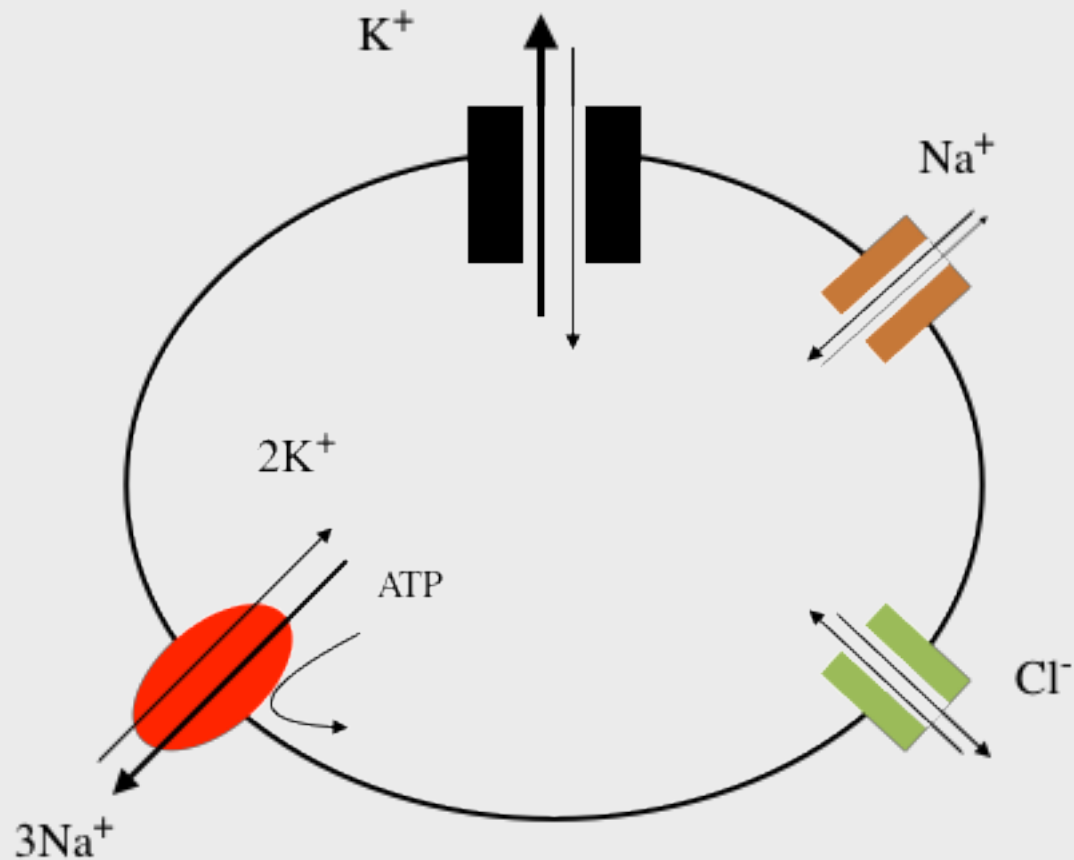
Space arrangement of carbonyl oxygen in the K⁺ channel pore can remove the hydration shell of the K⁺ ion but not the hydration shell of the Na⁺ ion (so it remains too large to fit into the pore)

rem : *selectivity mechanism can be more complex (Ca²⁺ channels for exemple) but space arrangement that allows to dehydrate the ion with its binding in the pore remains a general principle.*

leak channels (passive conductances)

$$I_X = G_X \cdot (V_m - E_{eq_X})$$

theoretically non gated by voltage $\longrightarrow G_X = \text{cste}$



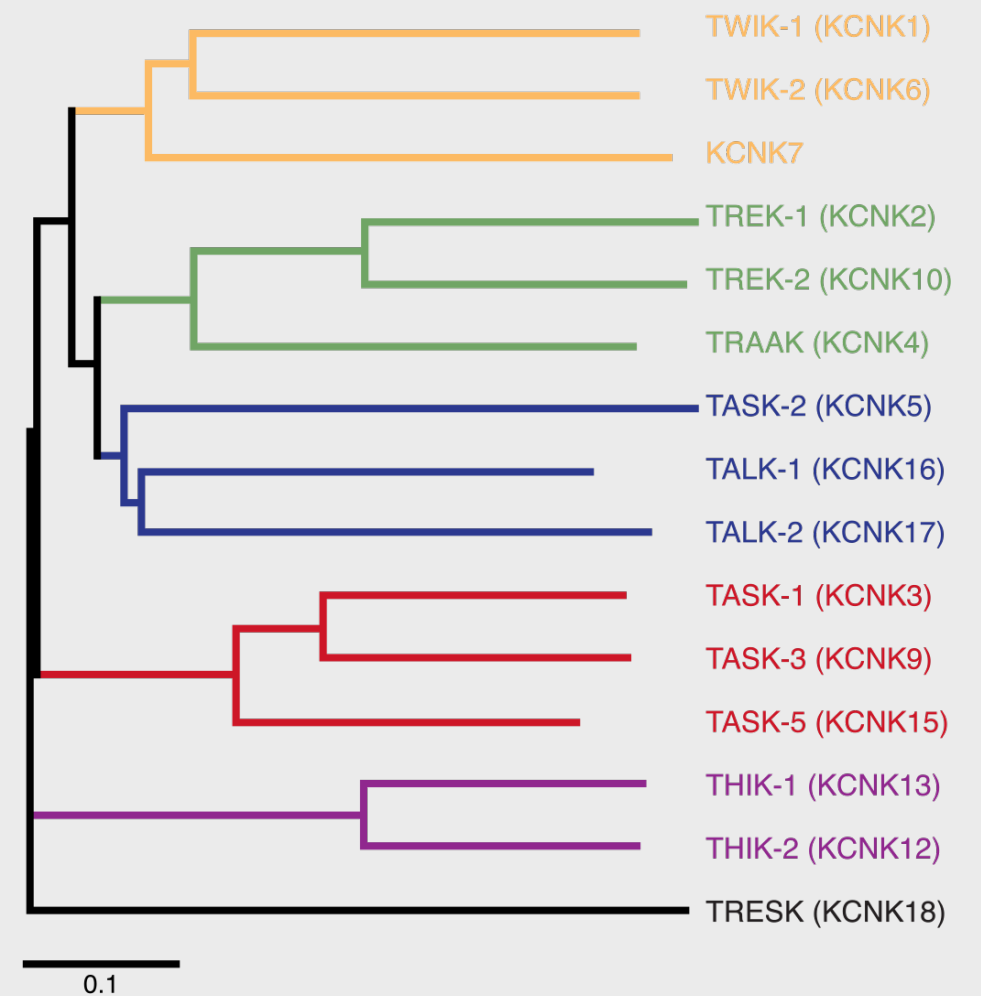
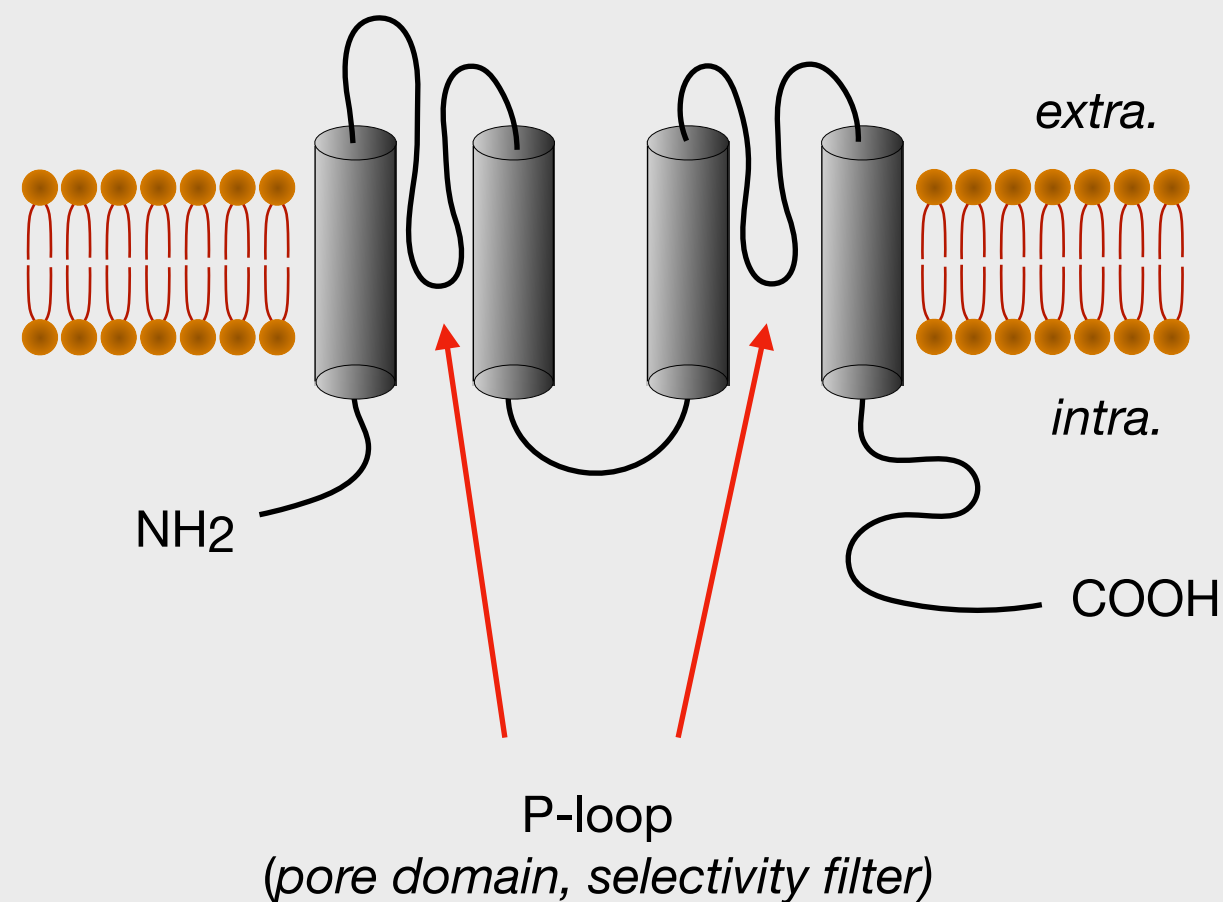
at V_{rest} :

- almost no Cl^- current (small entry of Cl^-)
- small Na^+ entry
- large K^+ outflow

leak channels (passive conductances)

- very little is known about Na^+ (*but see NALCN channels if you are interested*) and Cl^- leak conductances
- multiple families of K^+ leak channels have been studied in details (properties, structure, regulation)

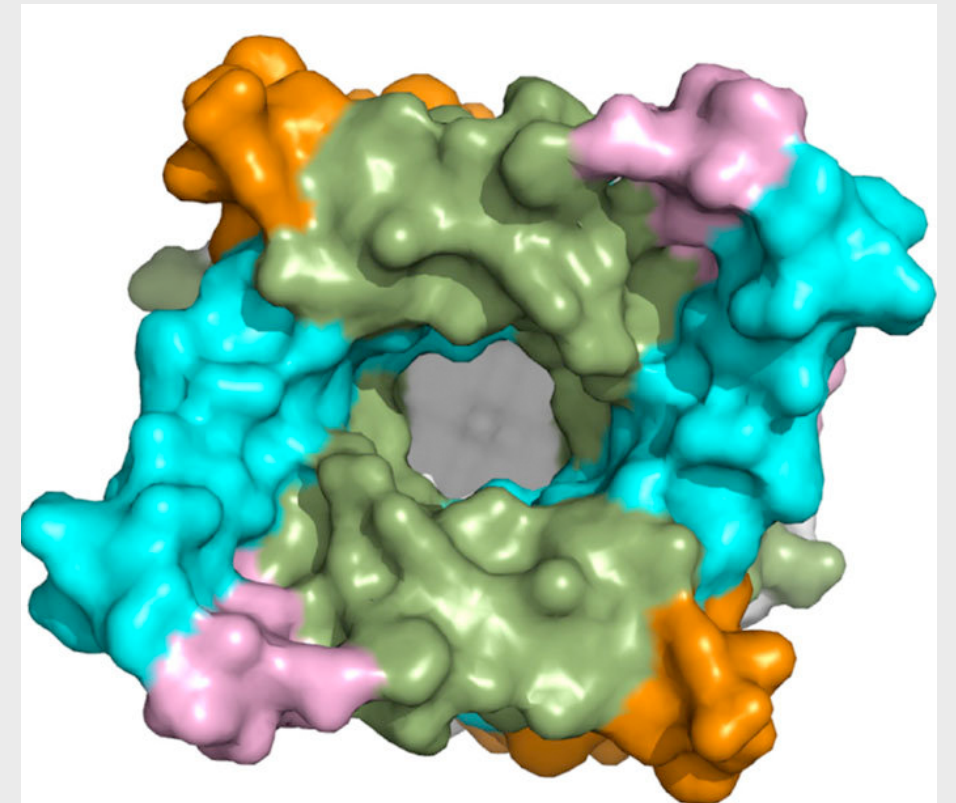
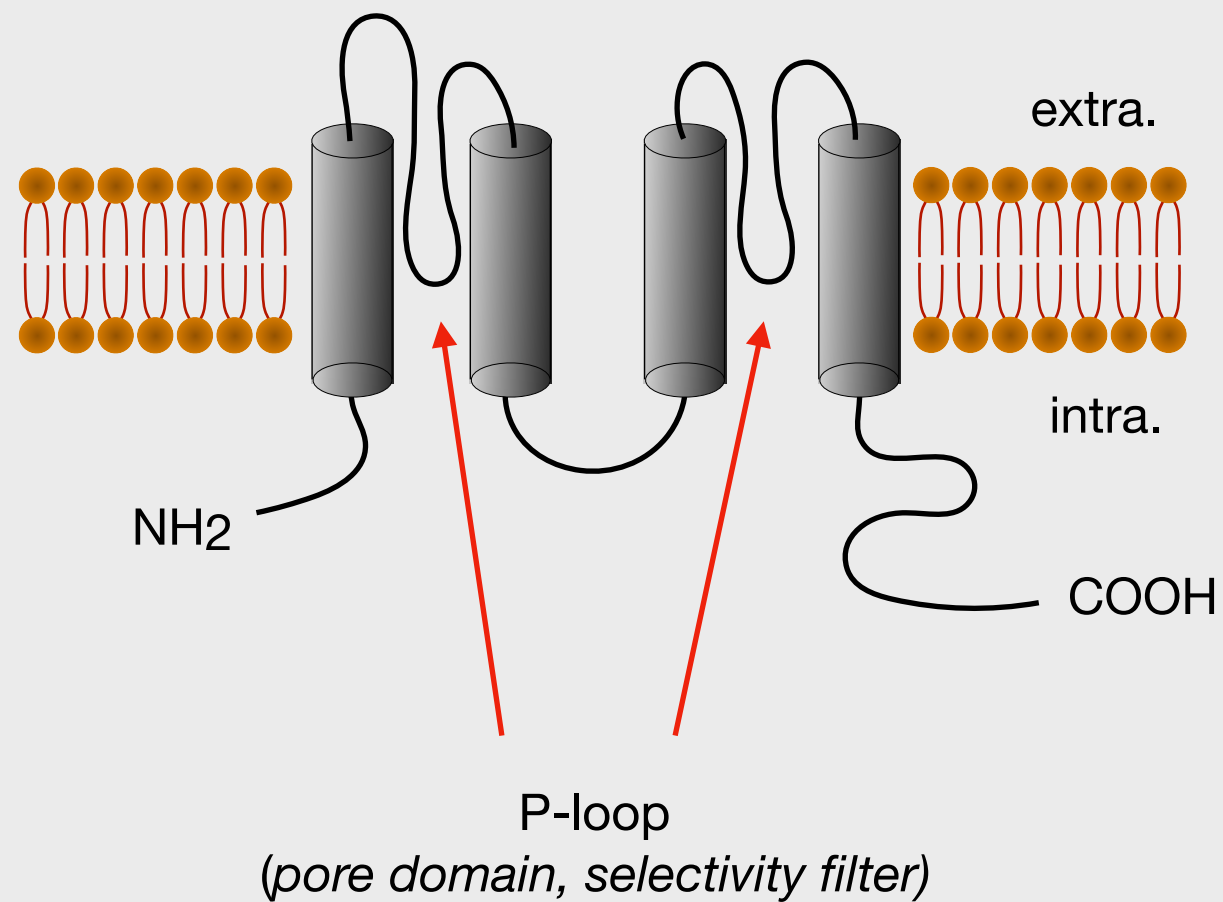
K₂P channels



leak channels (passive conductances)

- 2 K2P units assemble to form a channel

K2P channels

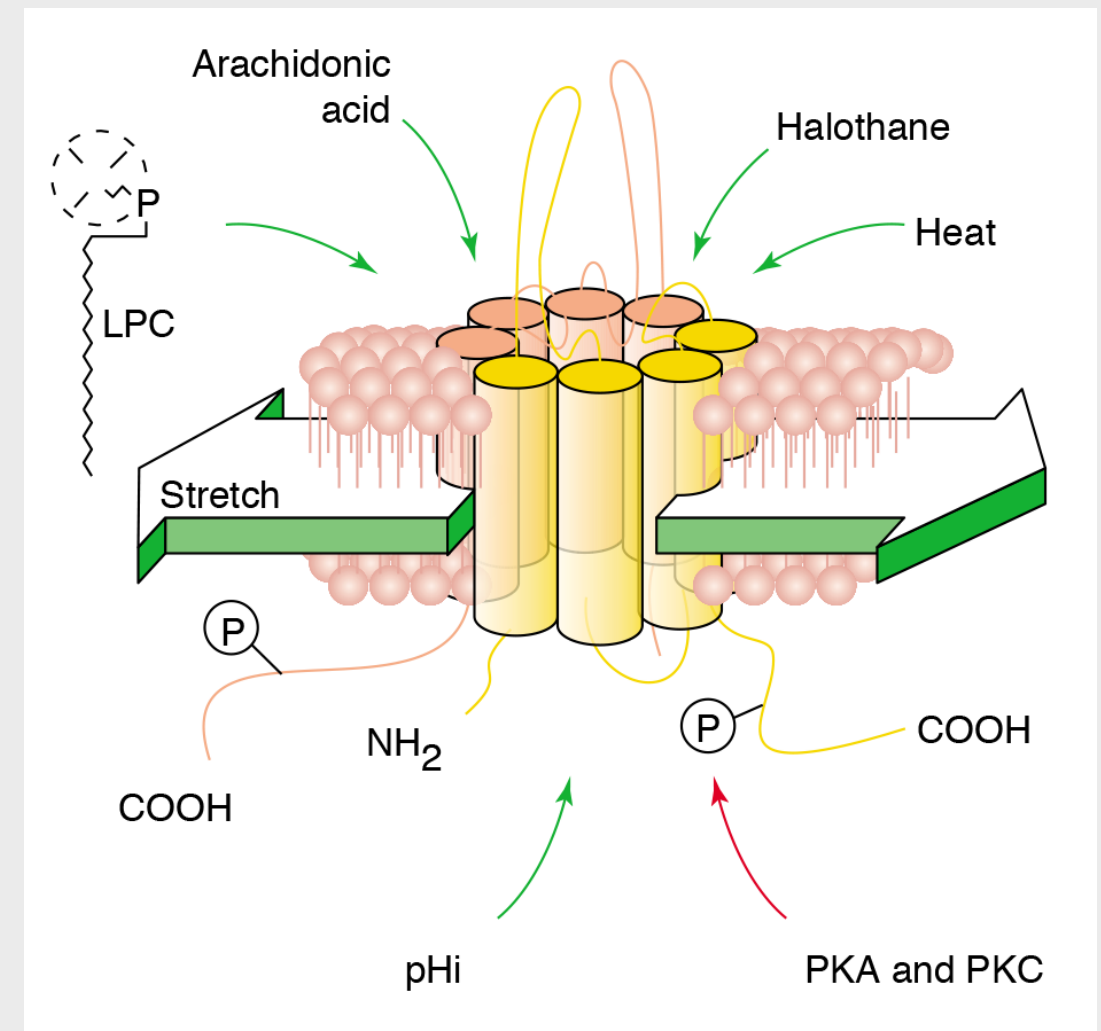
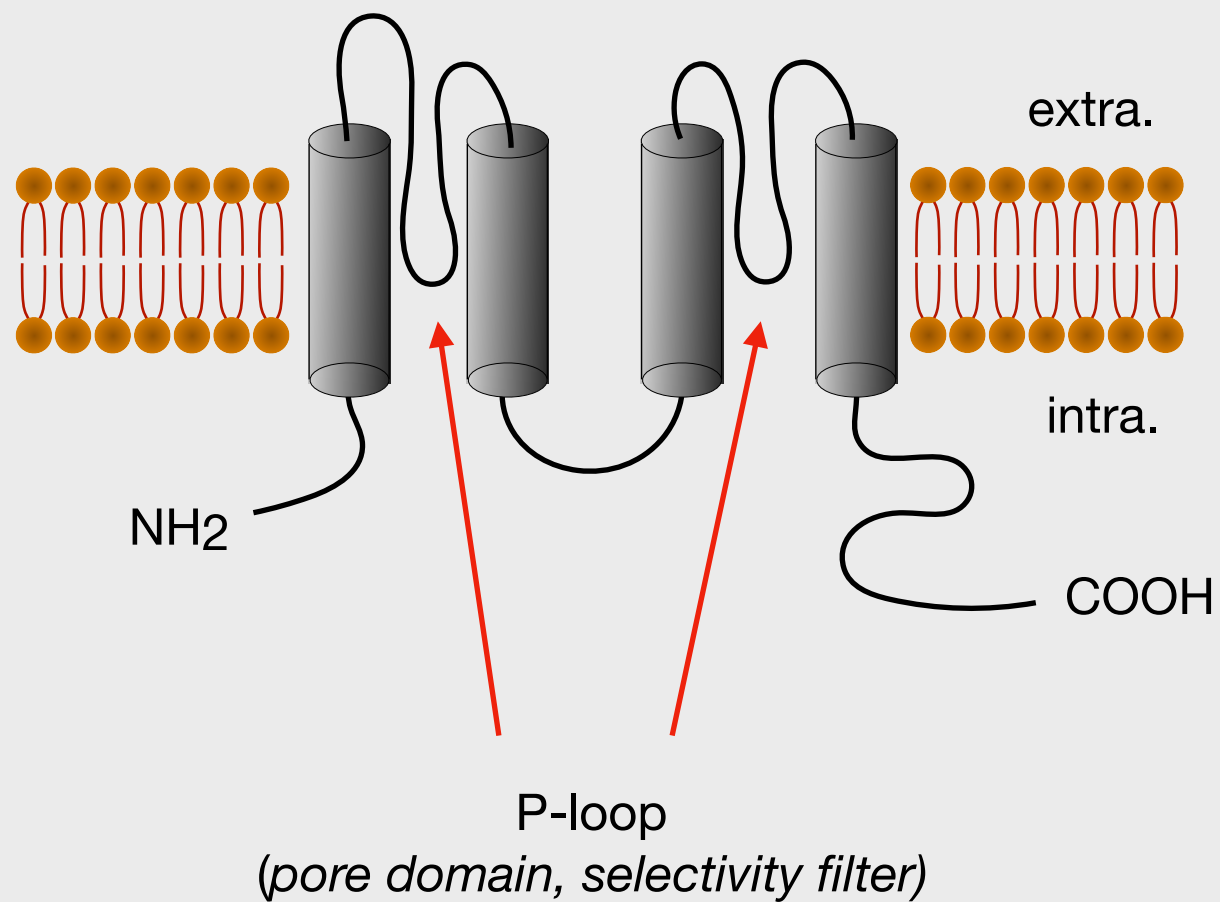


Reniguta & Daut, 2015

leak channels (passive conductances)

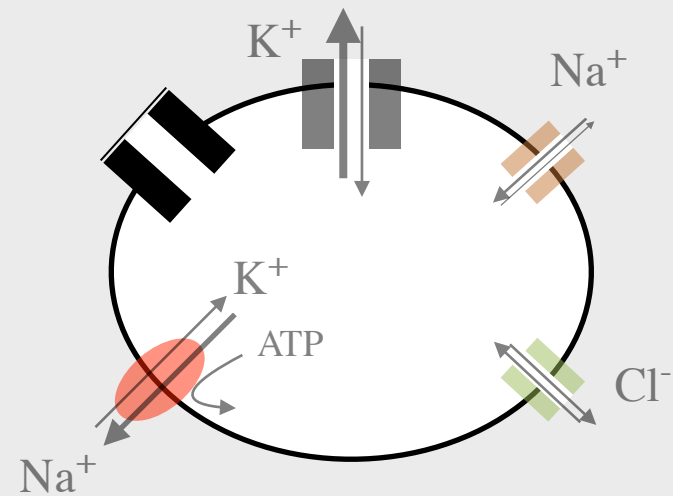
- 2 K2P units assemble to form a channel
- multiple regulations that change G

K₂P channels



Patel & Honoré, 2001

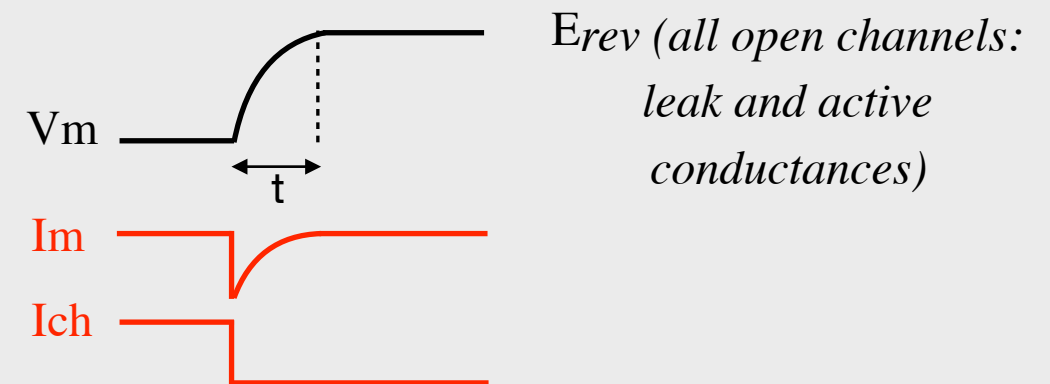
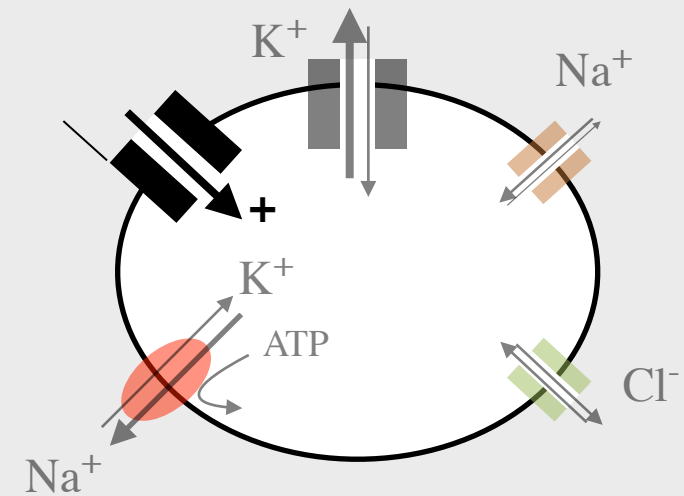
active conductances



$V_m = V_{rest}$ —

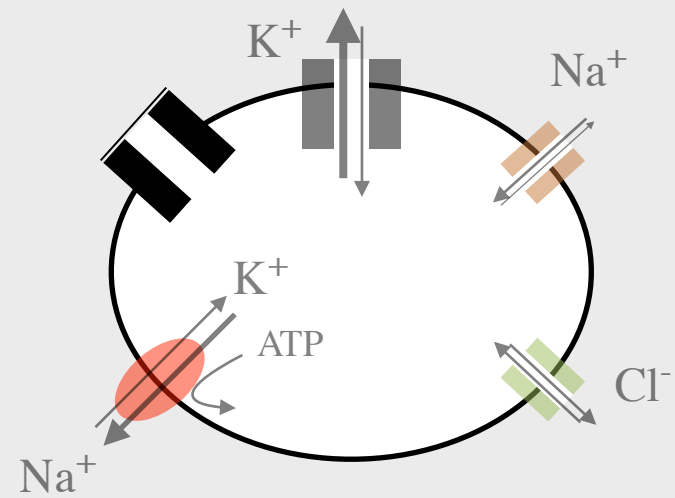
$I_m = 0$ —

$I_{ch} = 0$ —



rem : the time t to reach the steady state (and the speed of V_m variation) depends on the membrane capacitance (C_m) and the leak conductance (G_L ; or membrane resistance $R_m = 1/G_L$). The membrane potential exponentially increases with a time constant: $\tau = R_m \times C_m$.

active conductances

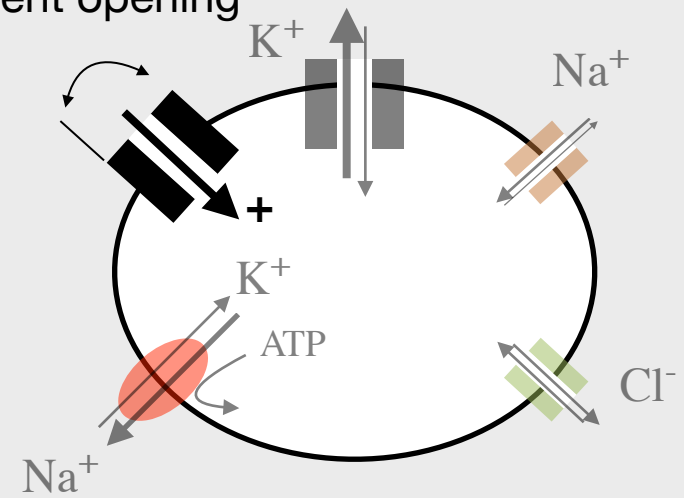


$V_m = V_{rest}$ —

$I_m = 0$ —

$I_{ch} = 0$ —

transient opening

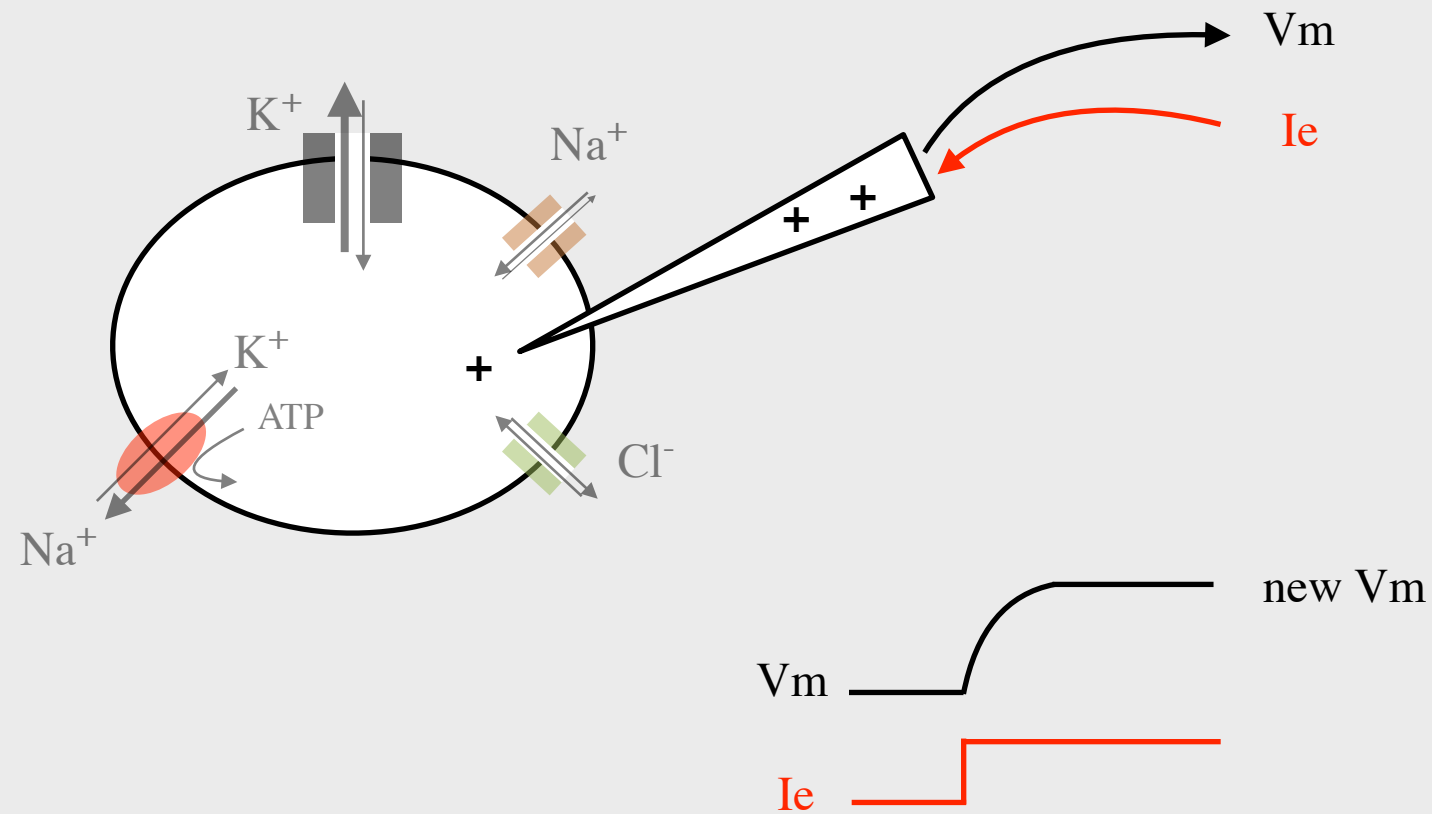


V_m — E_{rev}

I_m —

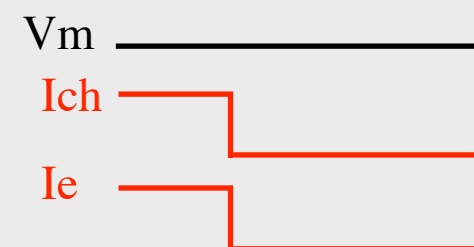
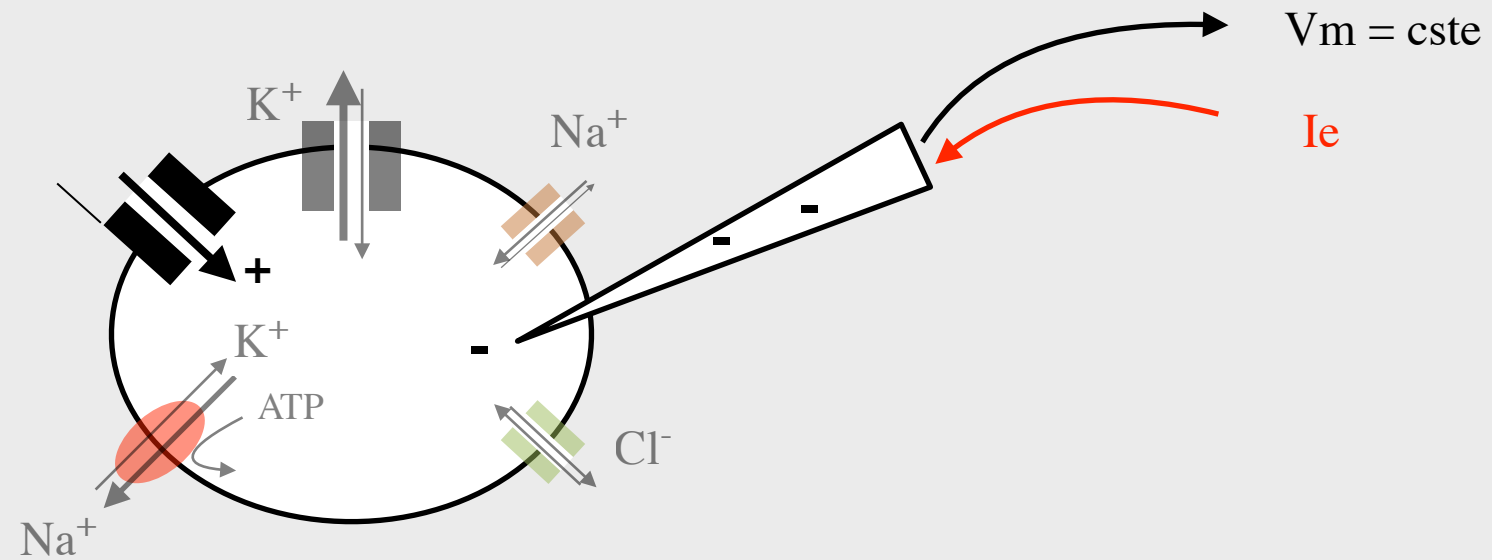
I_{ch} —

current clamp



by convention : injection of + charges through the electrode into the cell : $I_e > 0$
 not to confuse with membrane currents : $I_m > 0$ outward current of + charges

voltage clamp



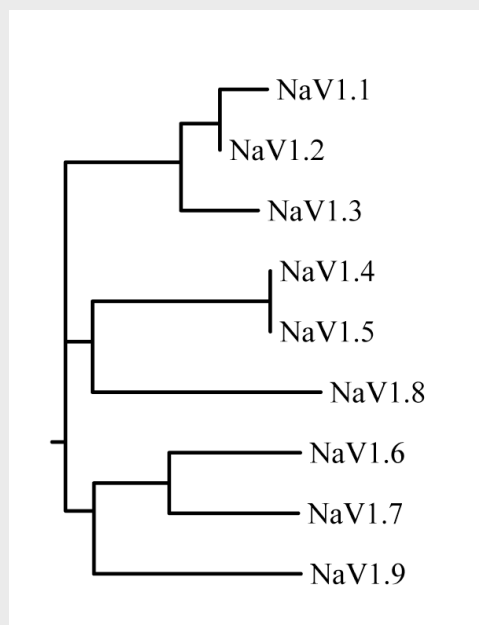
rem : it explains the convention for membrane current

active conductances : voltage-gated channels

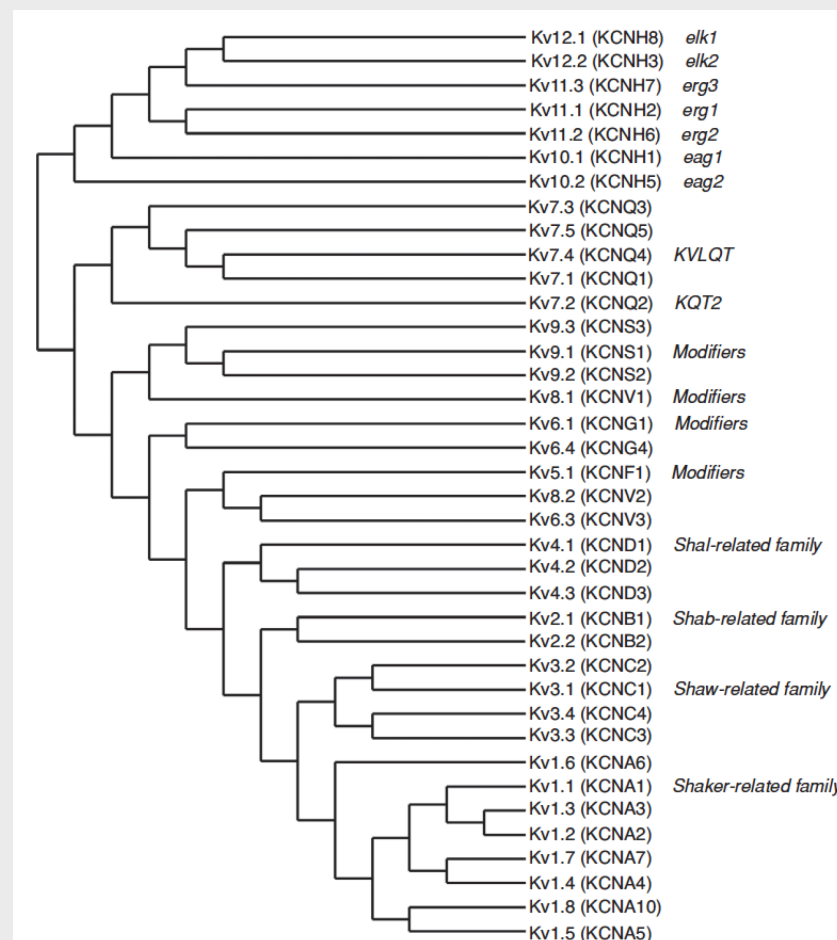
Ion	Ext. conc. (mM)	Int. conc. (mM)	Nernst potential (mV)
Na ⁺	150	5	60
K ⁺	3	150	-90
Ca ²⁺	2	50 nM	120
Cl ⁻	125	5	-75

selective for a single species of ions

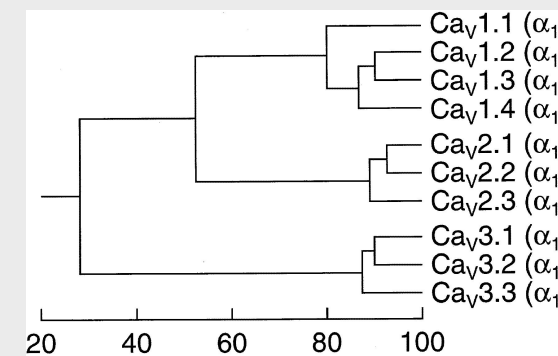
Na⁺



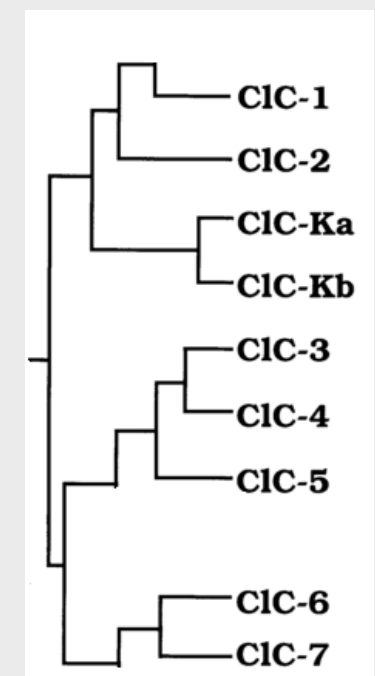
K⁺



Ca²⁺



Cl⁻



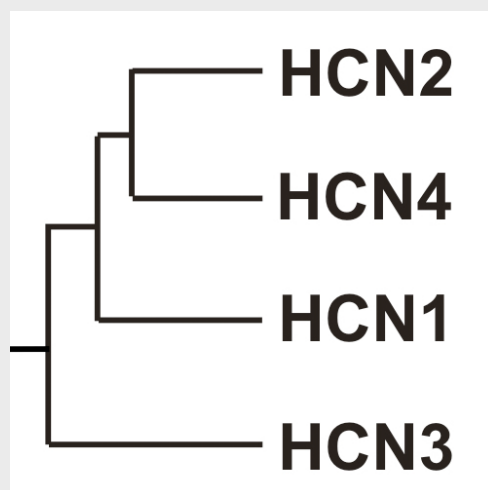
active conductances : voltage-gated channels

Ion	Ext. conc. (mM)	Int. conc. (mM)	Nernst potential (mV)
Na ⁺	150	5	60
K ⁺	3	150	-90
Ca ²⁺	2	50 nM	120
Cl ⁻	125	5	-75

selective for multiple species of ions, such as

Hyperpolarization activated Cyclic Nucleotide channel (Ih)

Na⁺ & K⁺

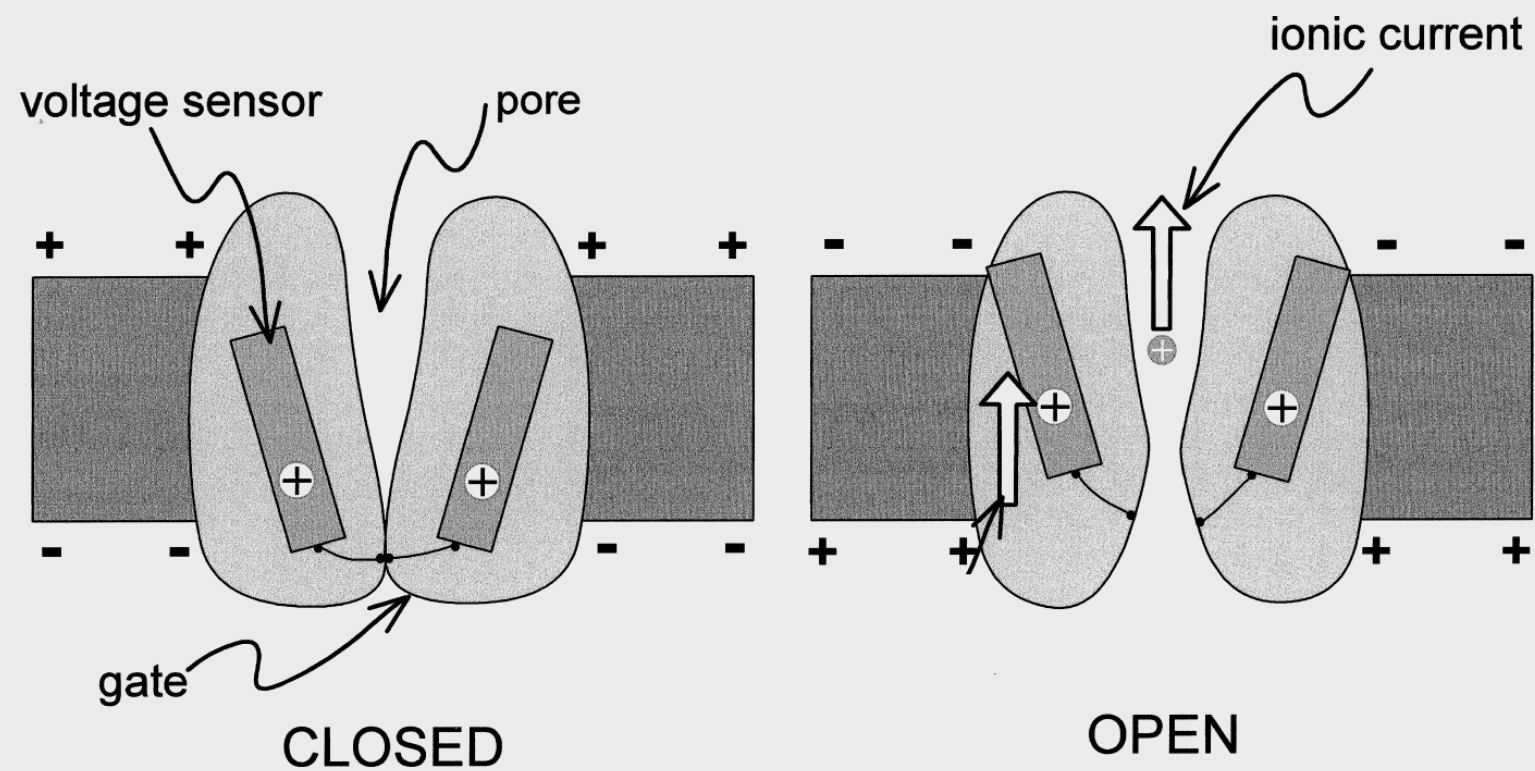


$$E_{rev} \approx -30mV$$

active conductances : voltage-gated channels

$$I_X = G_X(V_m) \cdot (V_m - E_{eq_X})$$

gated by voltage $\longrightarrow G_X$ varies according to V_m



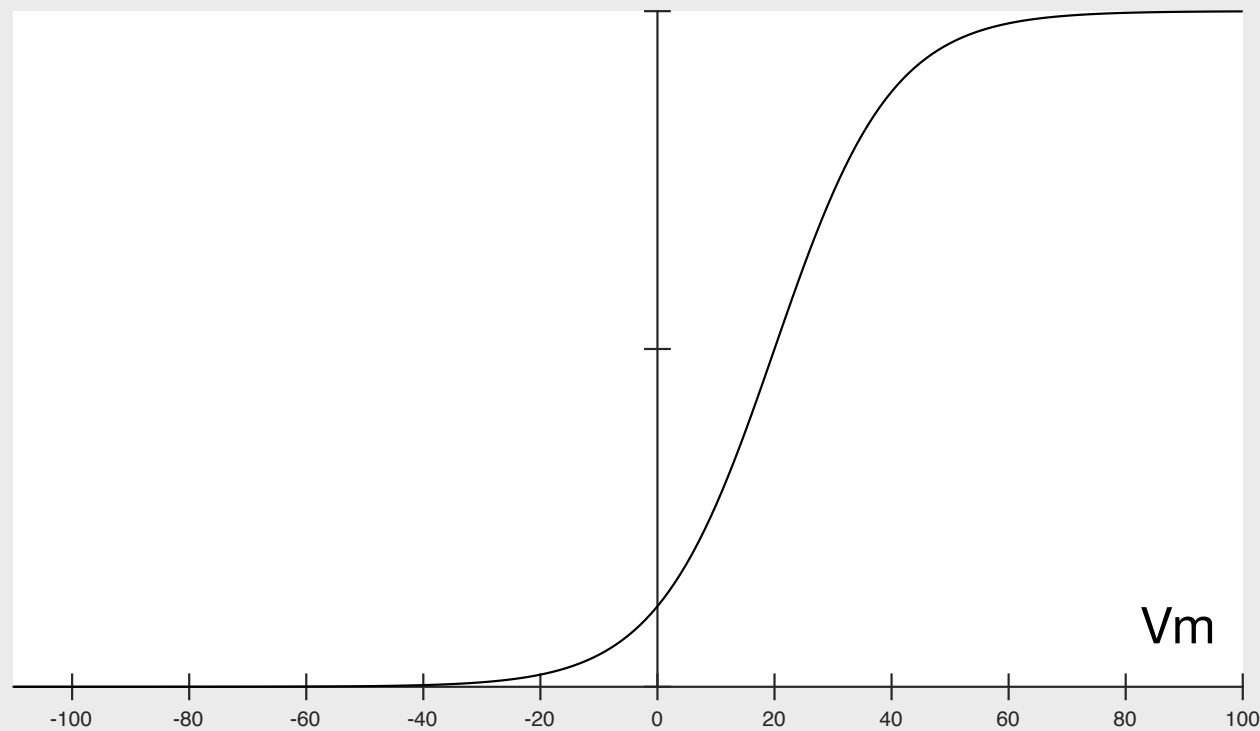
active conductances : voltage-gated channels

$$I_X = G_X(V_m) \cdot (V_m - E_{eq_X})$$

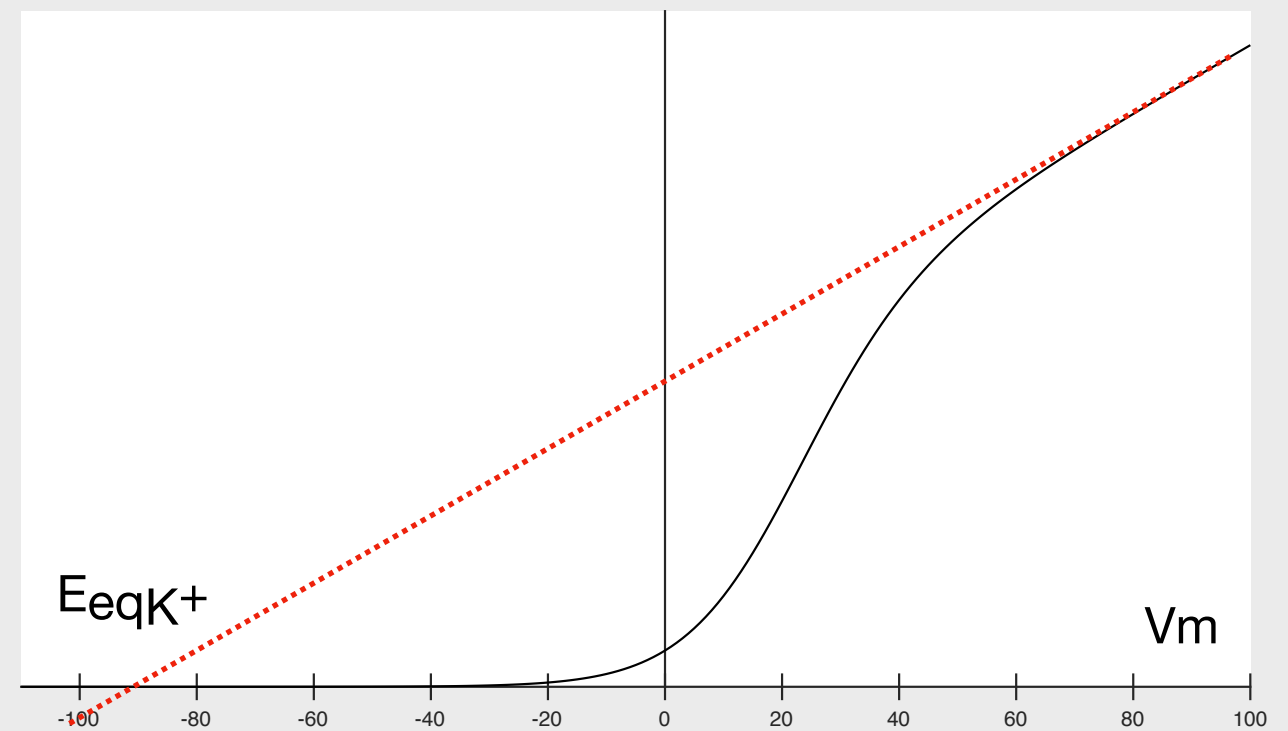
gated by voltage $\longrightarrow G_X$ varies according to V_m

K⁺ channels

$G_{K^+}(V_m)$



$I_{K^+}(V_m)$



outward current : entry of K⁺

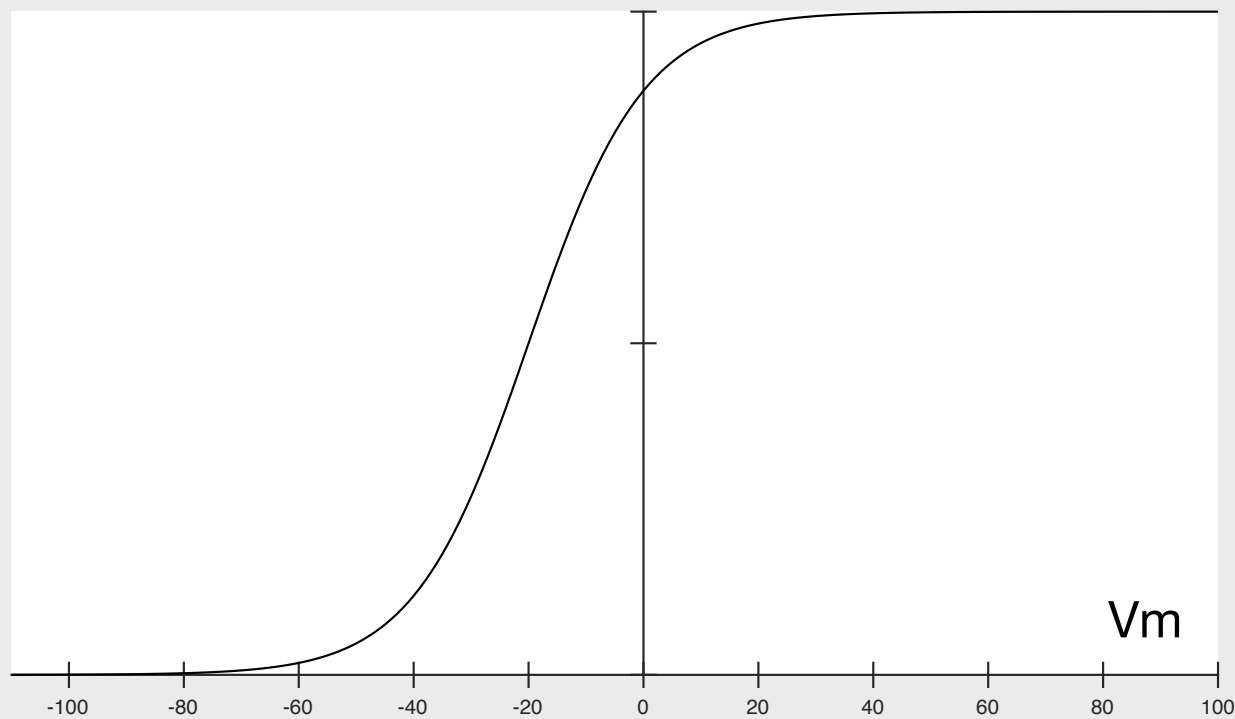
active conductances : voltage-gated channels

$$I_X = G_X(V_m) \cdot (V_m - E_{eq_X})$$

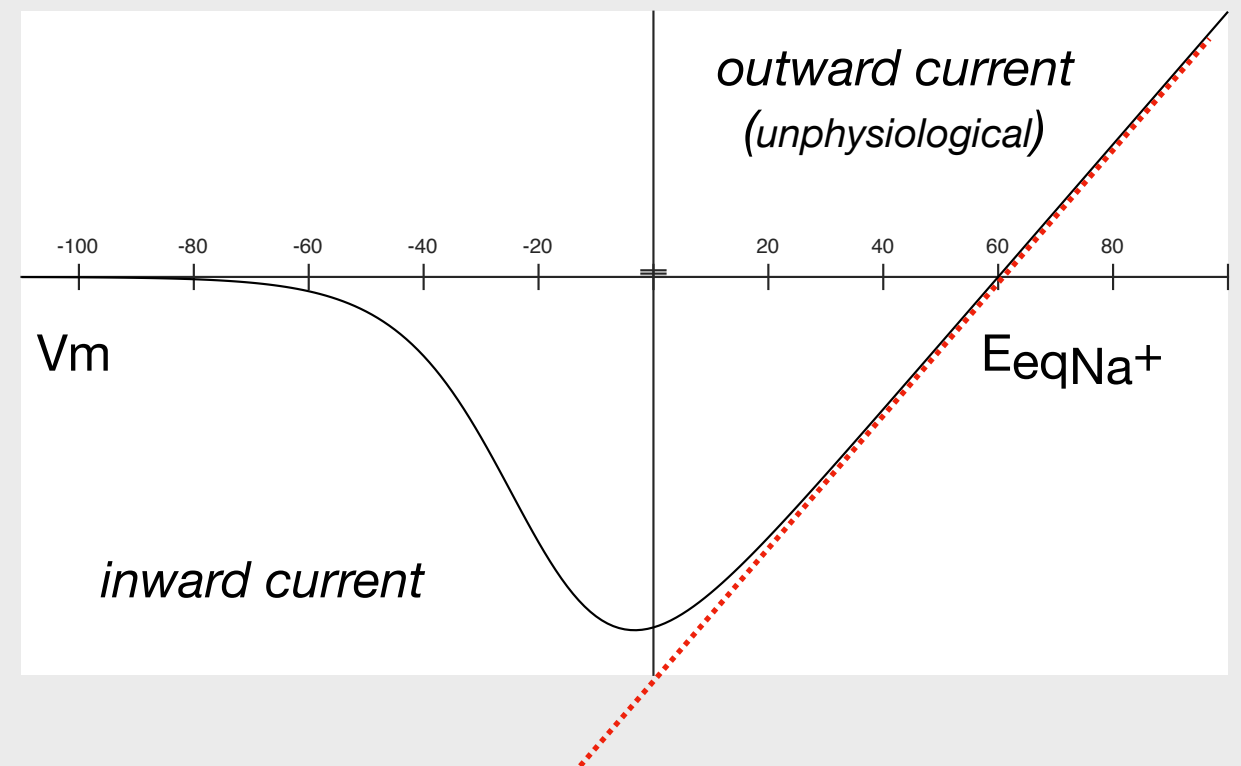
gated by voltage $\longrightarrow G_X$ varies according to V_m

Na⁺ channels

$G_{Na^+}(V_m)$



$I_{Na^+}(V_m)$

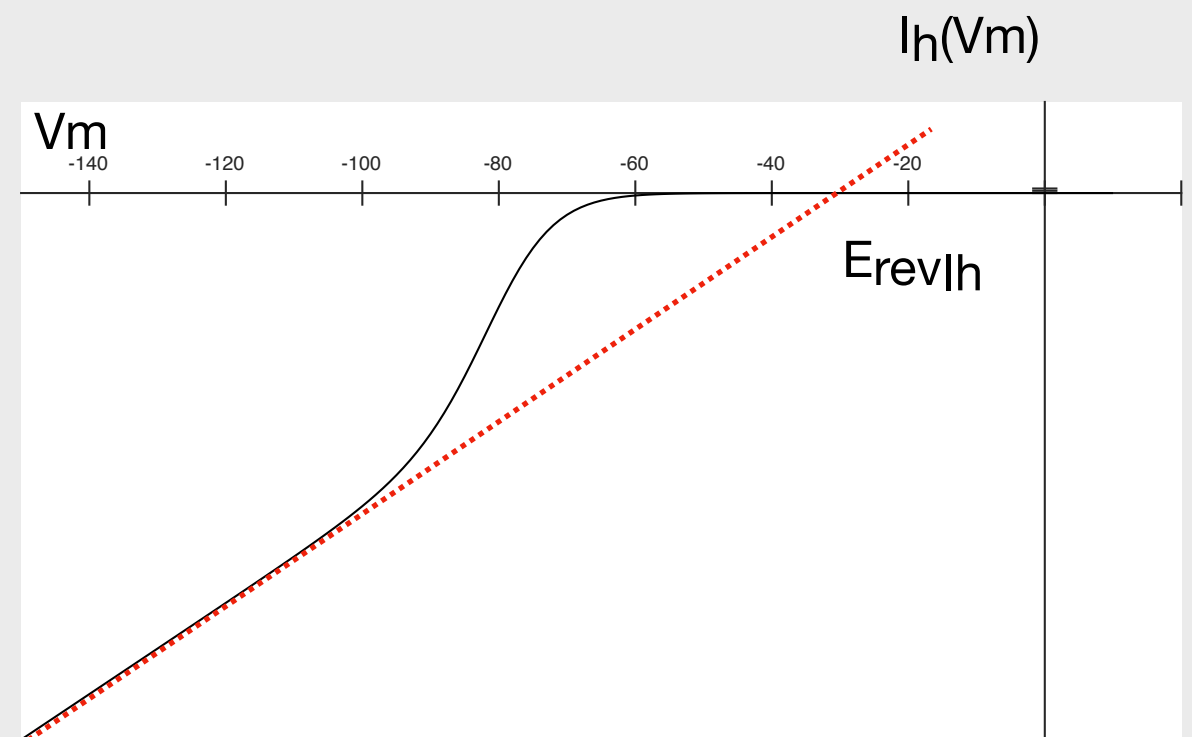
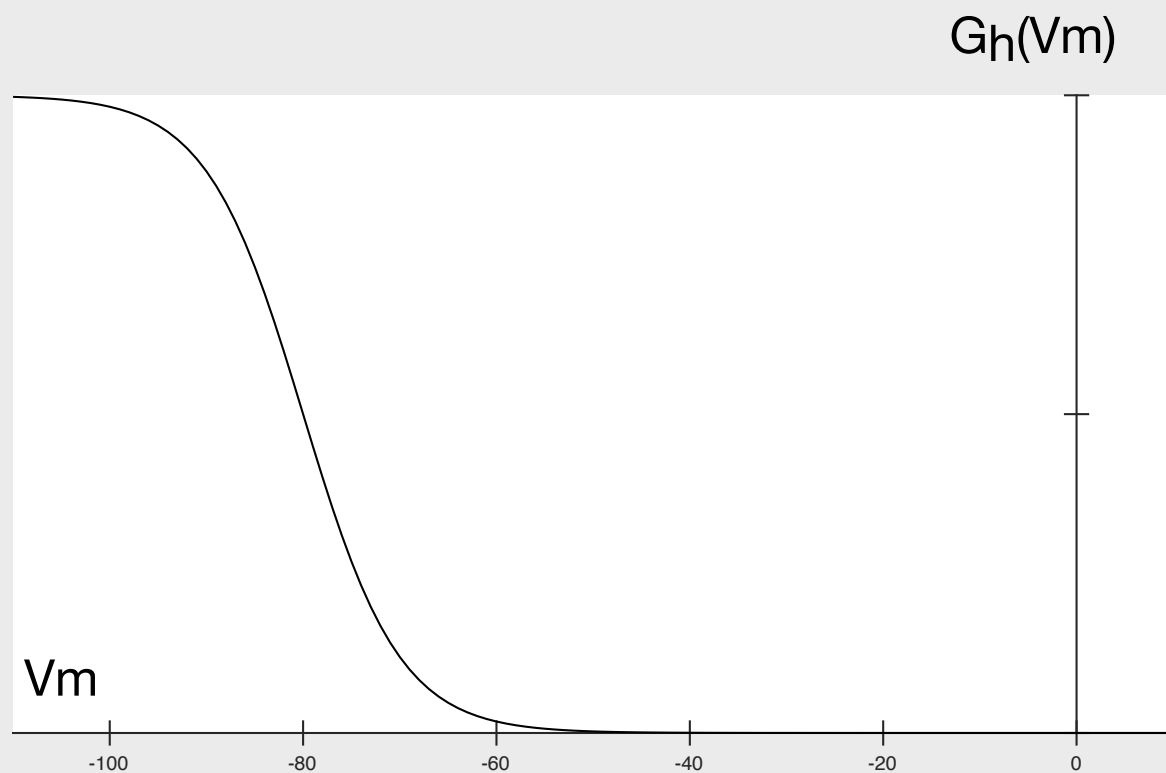


active conductances : voltage-gated channels

$$I_X = G_X(V_m) \cdot (V_m - E_{eq_X})$$

gated by voltage $\longrightarrow G_X$ varies according to V_m

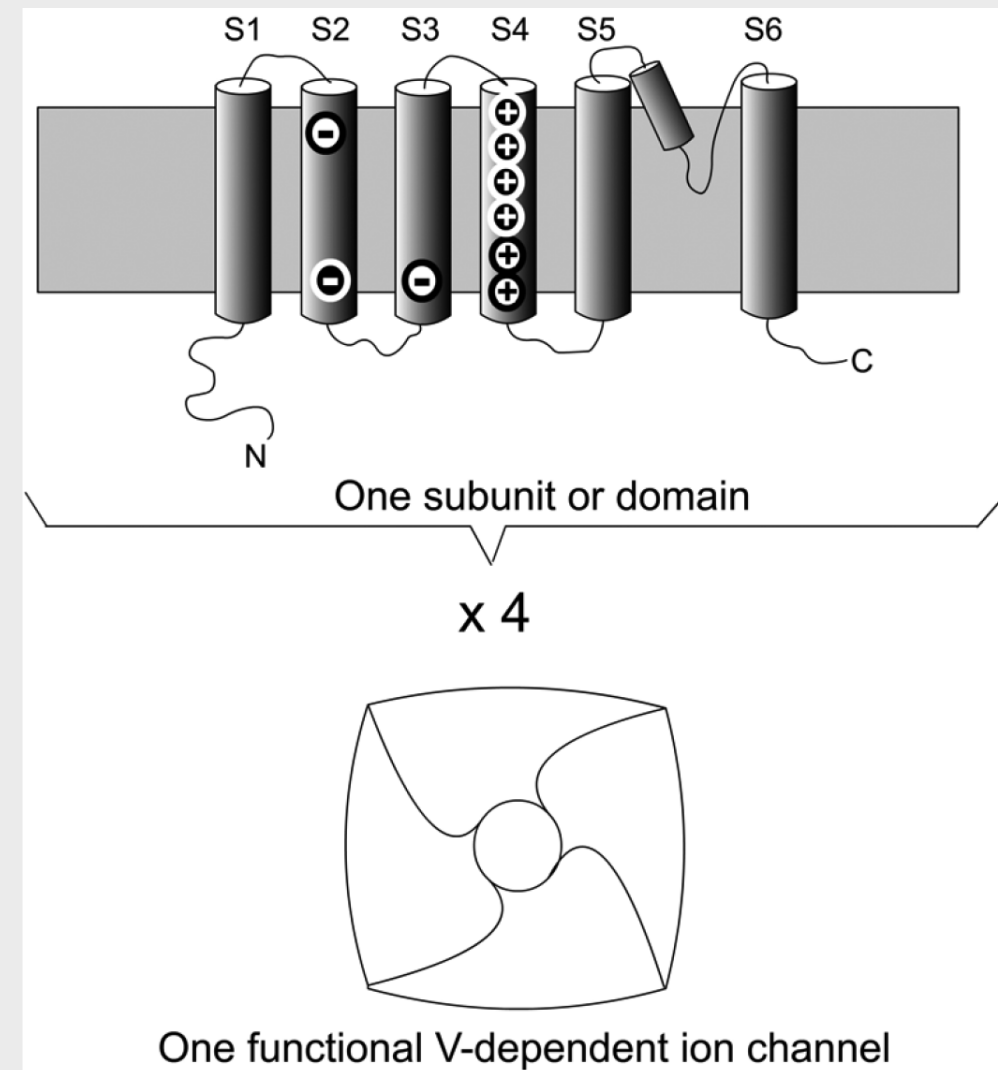
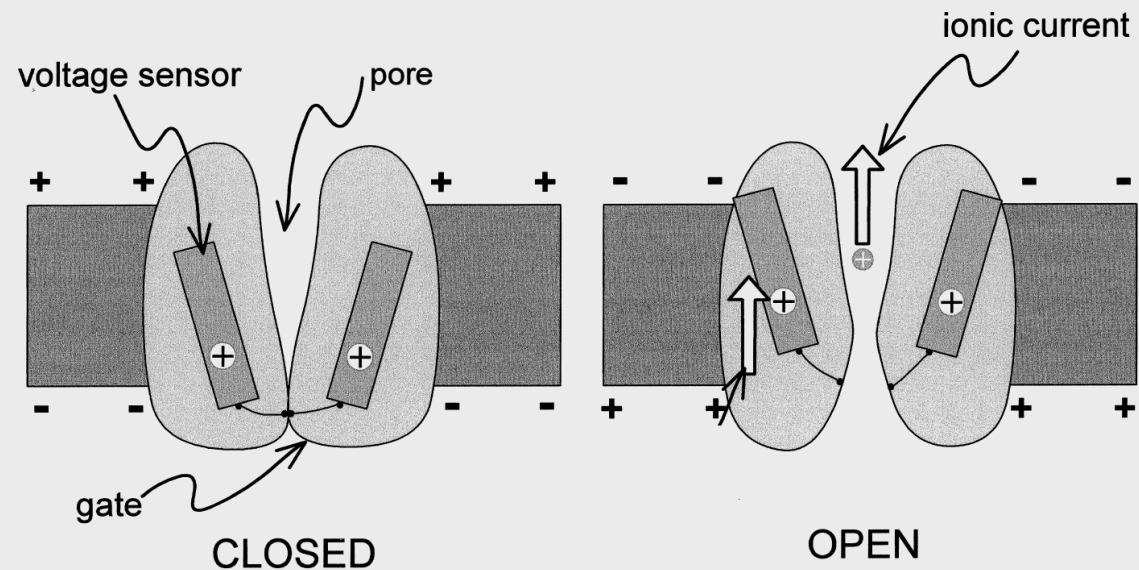
*H*CN channels (I_h)



*inward current
(opposing hyperpolarization)*

active conductances : voltage-gated channels

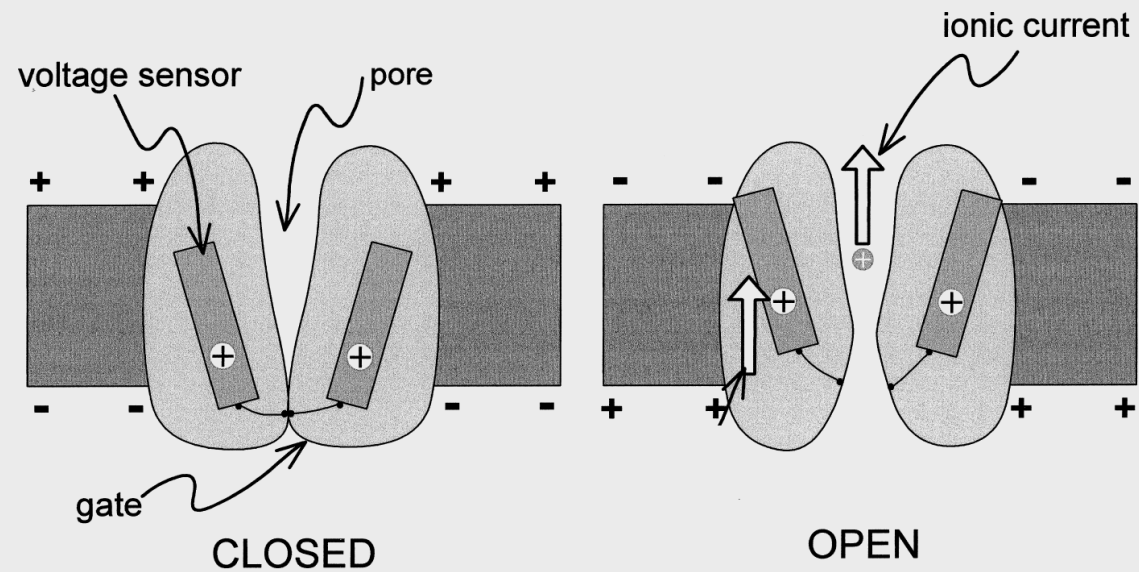
basic structure



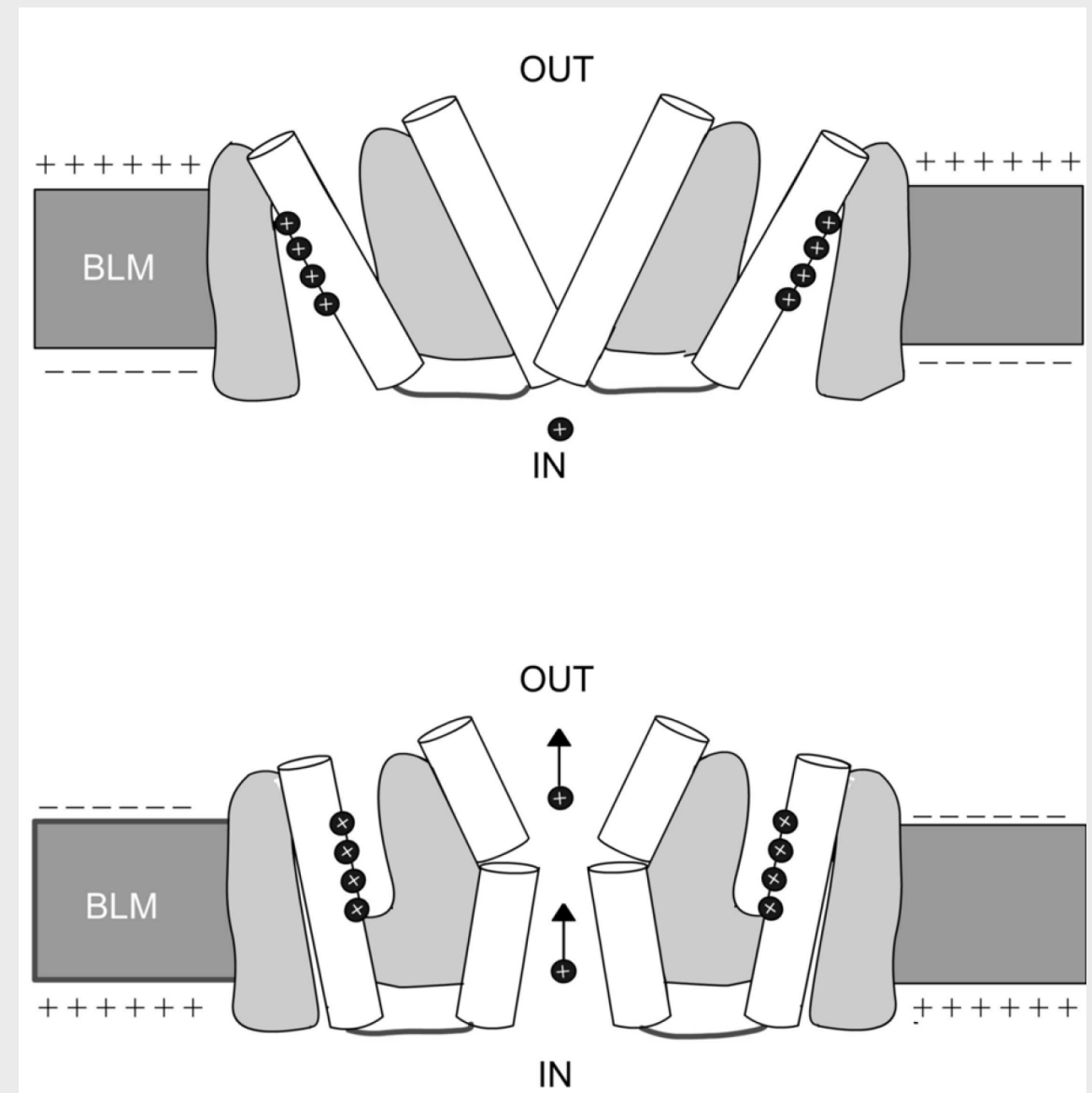
- 4 domains
- S4 contains many arginine (positively charged)
- 4 « gates »
- P-loop connecting S5 to S6

active conductances : voltage-gated channels

basic structure



- 4 domains
- S4 contains many arginine (positively charged)
- 4 « gates »
- P-loop connecting S5 to S6

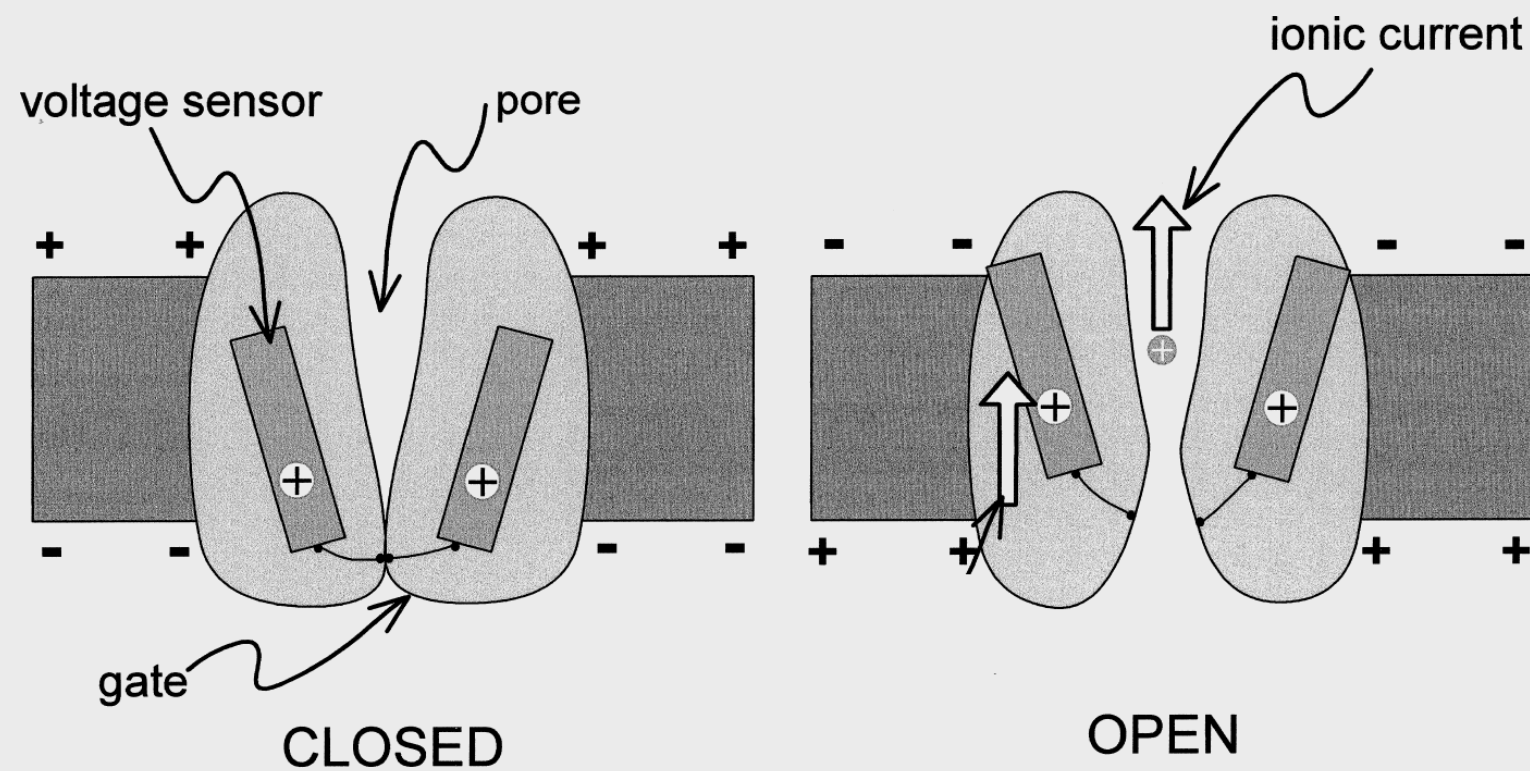


Bezanilla, 2005

active conductances : voltage-gated channels

$$I_X(t) = G_X(V_m, t) \cdot (V_m - E_{eq_X})$$

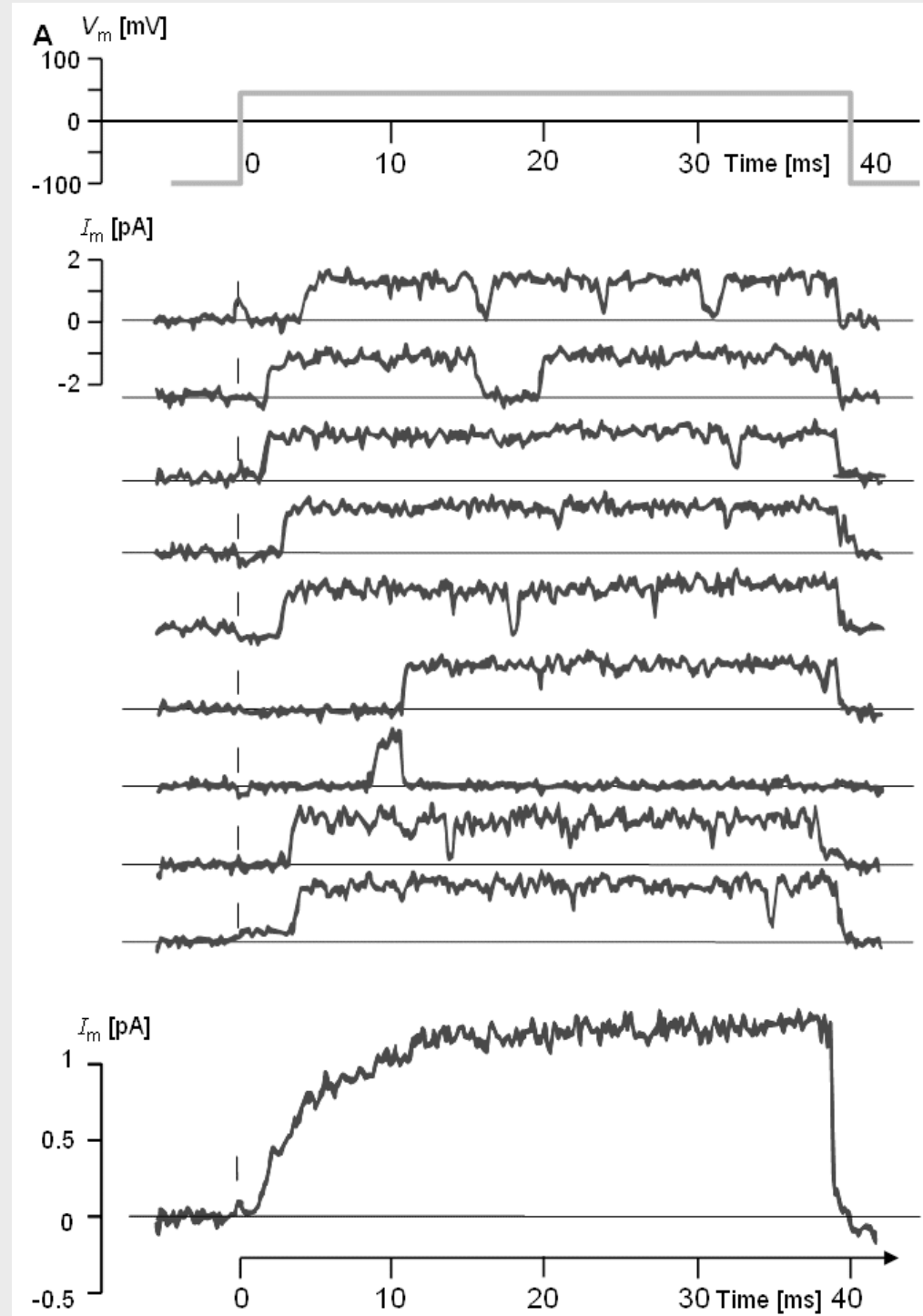
channel conformation changes take time
— —> G_X varies according to V_m and t



active conductances : voltage-gated channels

- not all channels open simultaneously
- all channels do not remain open all the time

voltage-clamp: single channel recordings



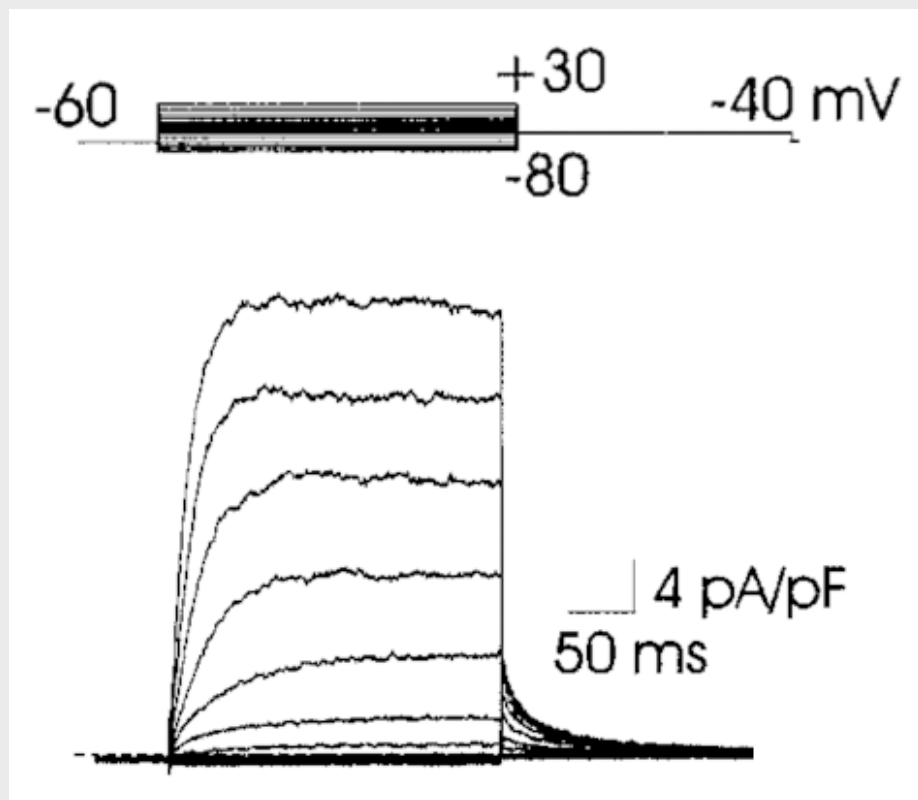
$$I_{K^+}(t) = G_{K^+}(V_m, t) \cdot (V_m - E_{eq_{K^+}})$$

active conductances : voltage-gated channels

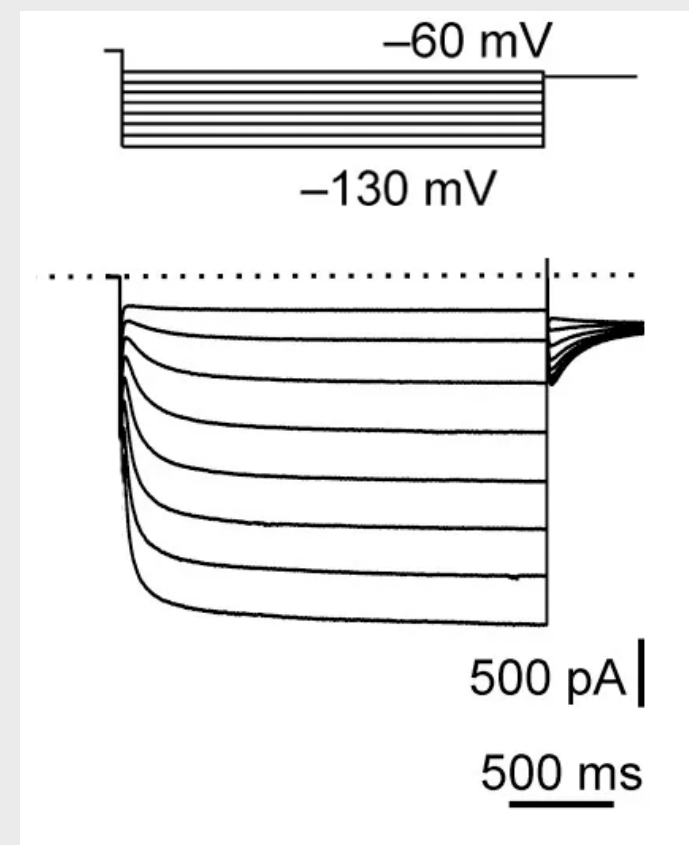
voltage-clamp: whole-cell recordings

- at a fixed V_m , a given maximal number of channels are open simultaneously
- the time required to reach this equilibrium state also varies with V_m (*activation kinetics*)

K^+ channels
(delayed rectifier)



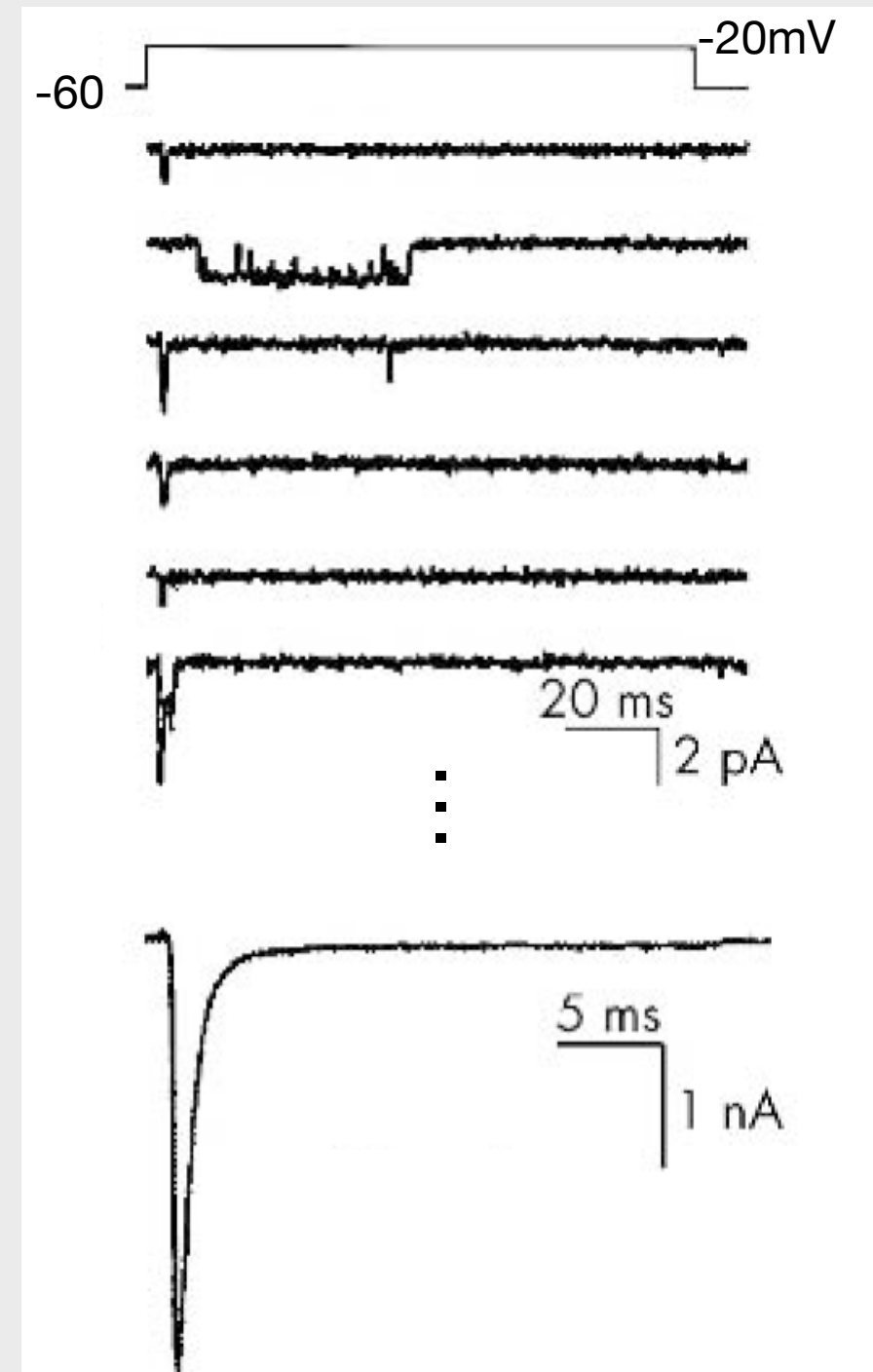
HCN channels (I_h)



active conductances : voltage-gated channels

voltage-clamp: single channel recordings

- not all channels open simultaneously
- all channels do not remain open all the time
- some channels open but undergo an additional conformational change making them non-conductive (inactivation,).
- they generate a transient ionic flux although the opening conditions are still present.



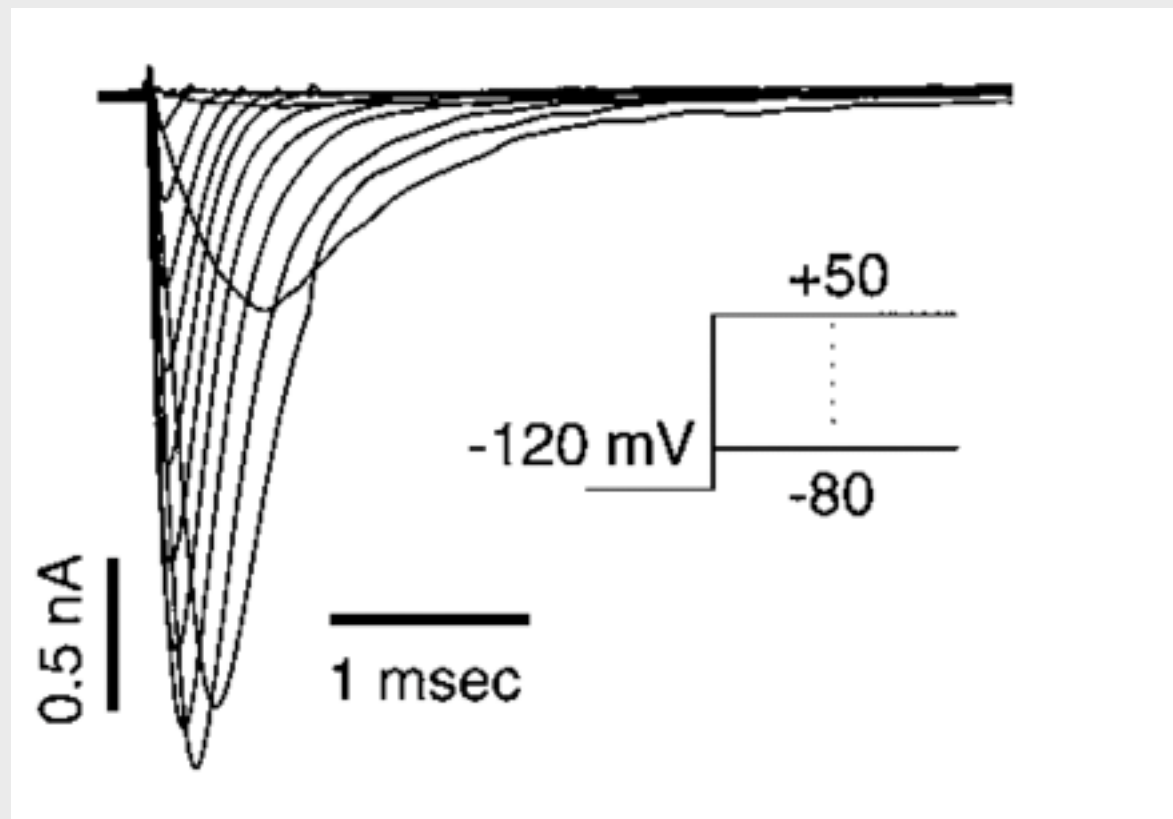
$$I_{Na^+}(t) = G_{Na^+}(V_m, t) \cdot (V_m - E_{eq_{Na^+}})$$

active conductances : voltage-gated channels

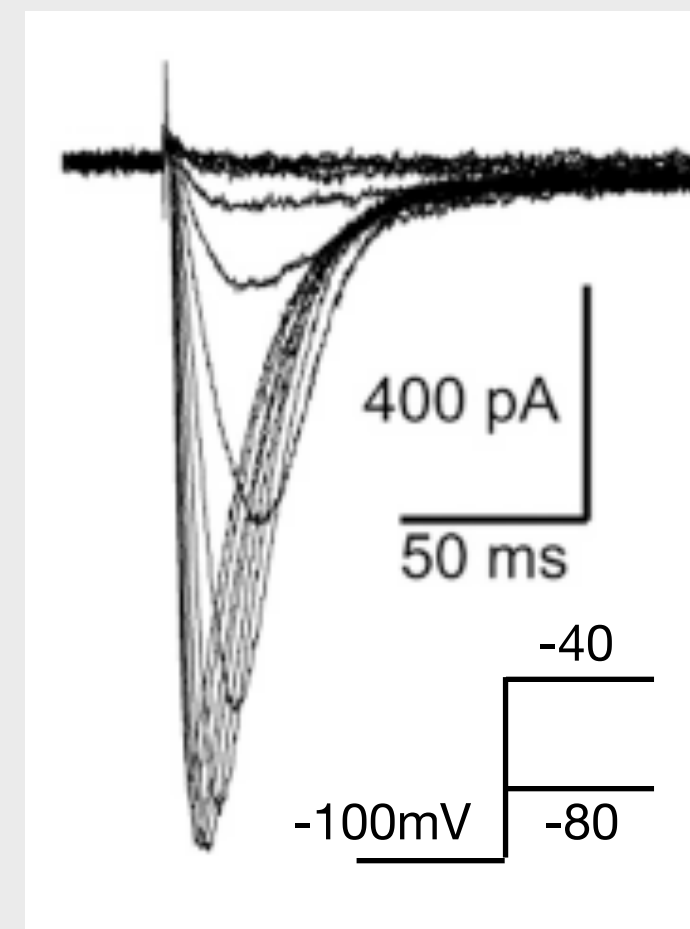
voltage-clamp: whole-cell recordings

- at a fixed V_m , a given number of channels are open simultaneously
- the time required to reach this equilibrium state also varies with V_m (*activation kinetics*)
- the time to reach complete inactivation also varies with V_m (*inactivation kinetics*)

Na^+ channels



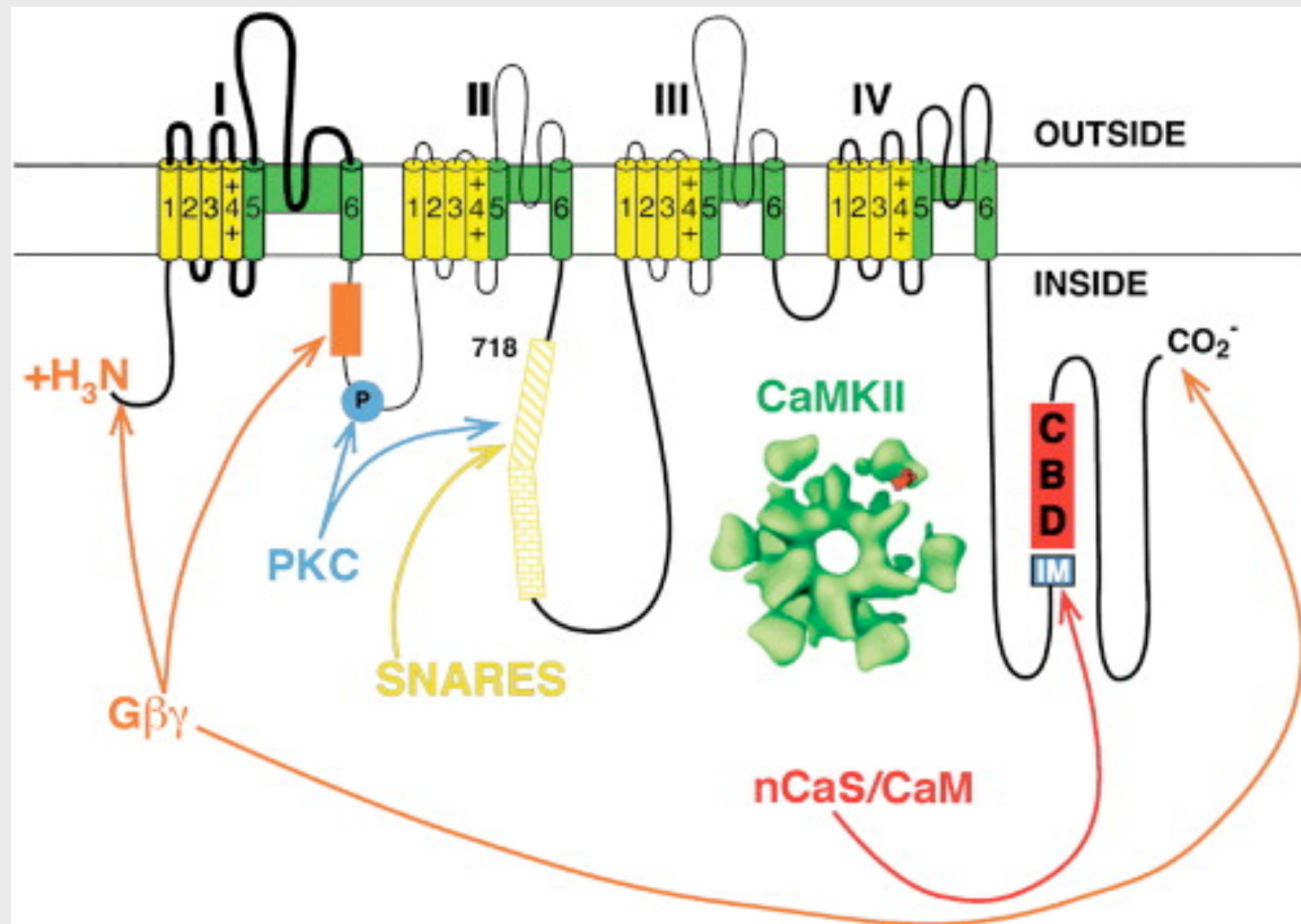
Ca^{2+} channels (T-type)



active conductances : voltage-gated channels

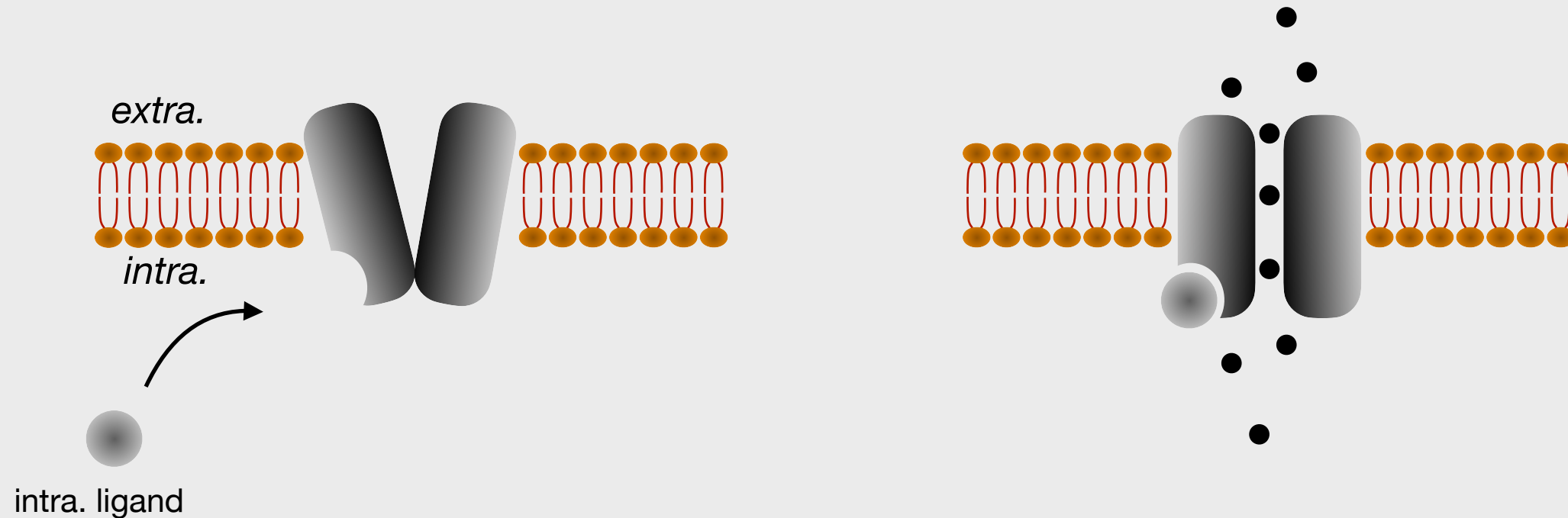
- opening of voltage-gated channels is defined by V_m
- however multiple regulations by intracellular mechanisms control their function

Ca^{2+} channels



active conductances : ligand-gated channels

intracellular ligand



intracellular *ligands* :

Ca²⁺ (K⁺ channels, Cl⁻ channels, ...)

ATP (K⁺ channels)

cyclic nucleotides (CNG)

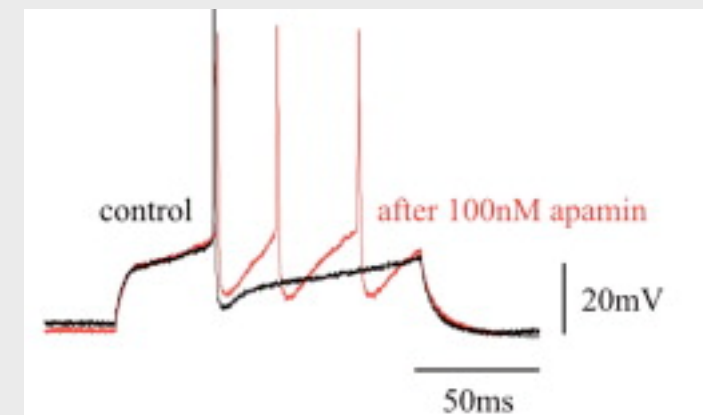
PIP2 (K⁺ channels)

G β (K⁺ channels)

...



change cell excitability according to action potential occurrence (Ca²⁺ entry), activation of metabotropic receptors, intracellular regulations, metabolism ...

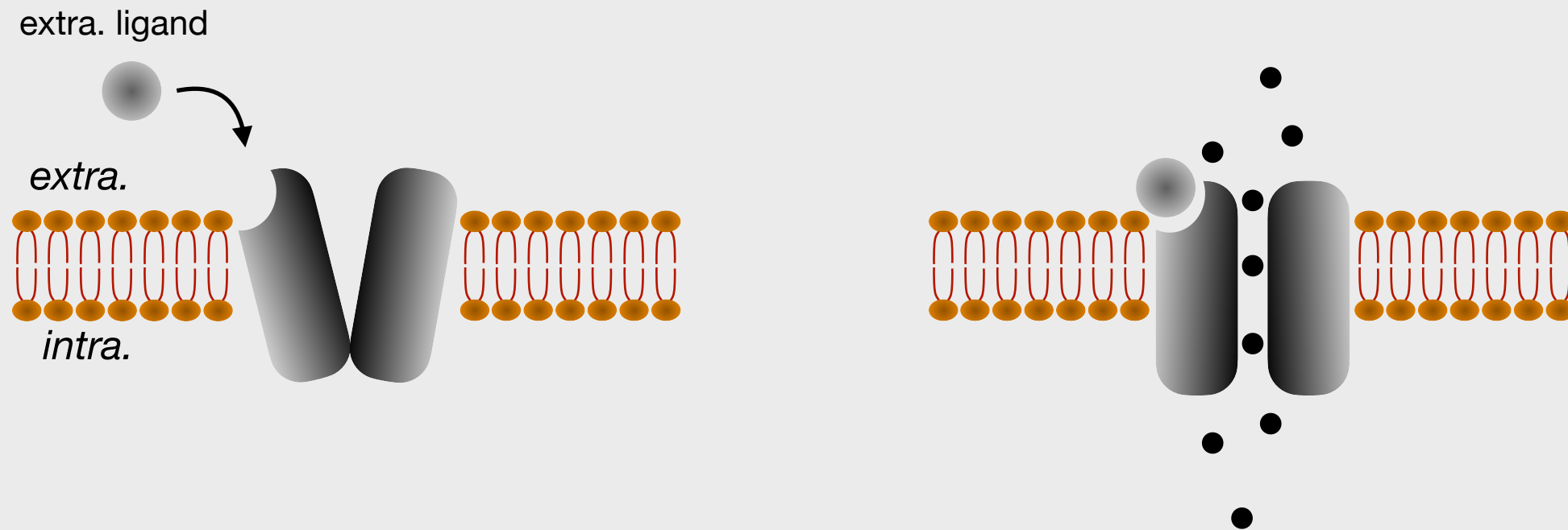


Aumann, 2008

rem: *although these channels are not gated by Vm, their properties are often modulated by Vm*

active conductances : ligand-gated channels

extracellular ligand



3 major families:

- 'Cys-loop' receptors: nicotinic receptor for acetylcholine (nAChR); γ -aminobutyric acid (GABA)_A receptor, 5-hydroxytryptamine (5-HT)₃ receptor (serotonin); glycine receptor
- glutamate receptors: α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA); kainate; N-methyl-D-aspartic acid (NMDA) receptors
- P2X receptors (ATP).

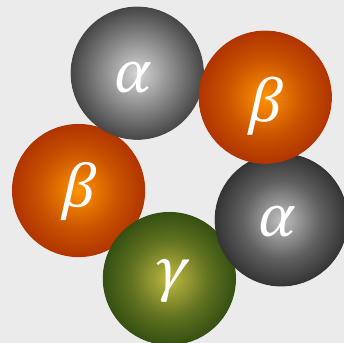


fast synaptic transmission : ACh, 5-HT, GABA, Glycine, Glutamate (, ATP ?)

active conductances : ligand-gated channels

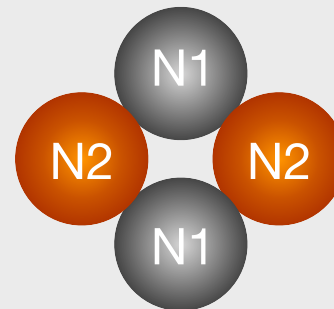
extracellular ligand

'Cys-loop' receptors
pentamer



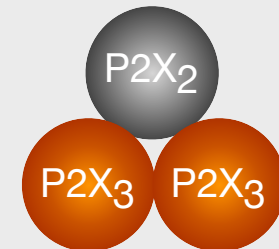
GABA_A-R

glutamate receptors
tetramer



NMDA-R

P2X receptors
trimer



3 major families:

- 'Cys-loop' receptors: nicotinic receptor for acetylcholine (nAChR); γ -aminobutyric acid (GABA)_A receptor, 5-hydroxytryptamine (5-HT)₃ receptor (serotonin); glycine receptor
- glutamate receptors: α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA); kainate; N-methyl-D-aspartic acid (NMDA) receptors
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fast synaptic transmission : ACh, 5-HT, GABA, Glycine, Glutamate (, ATP ?)

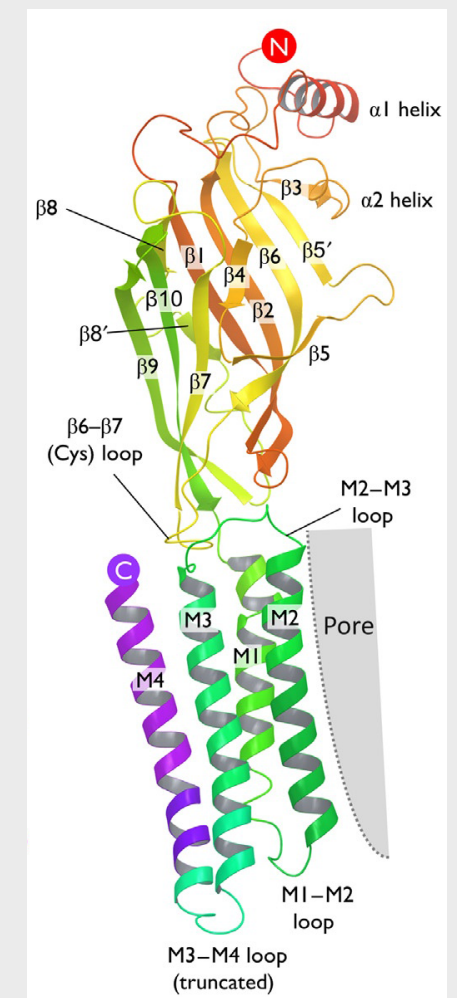
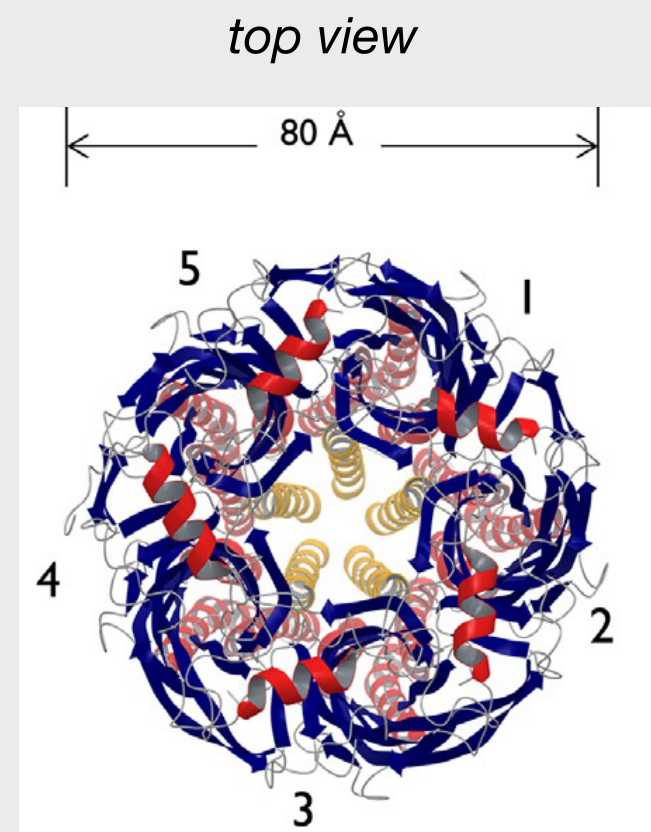
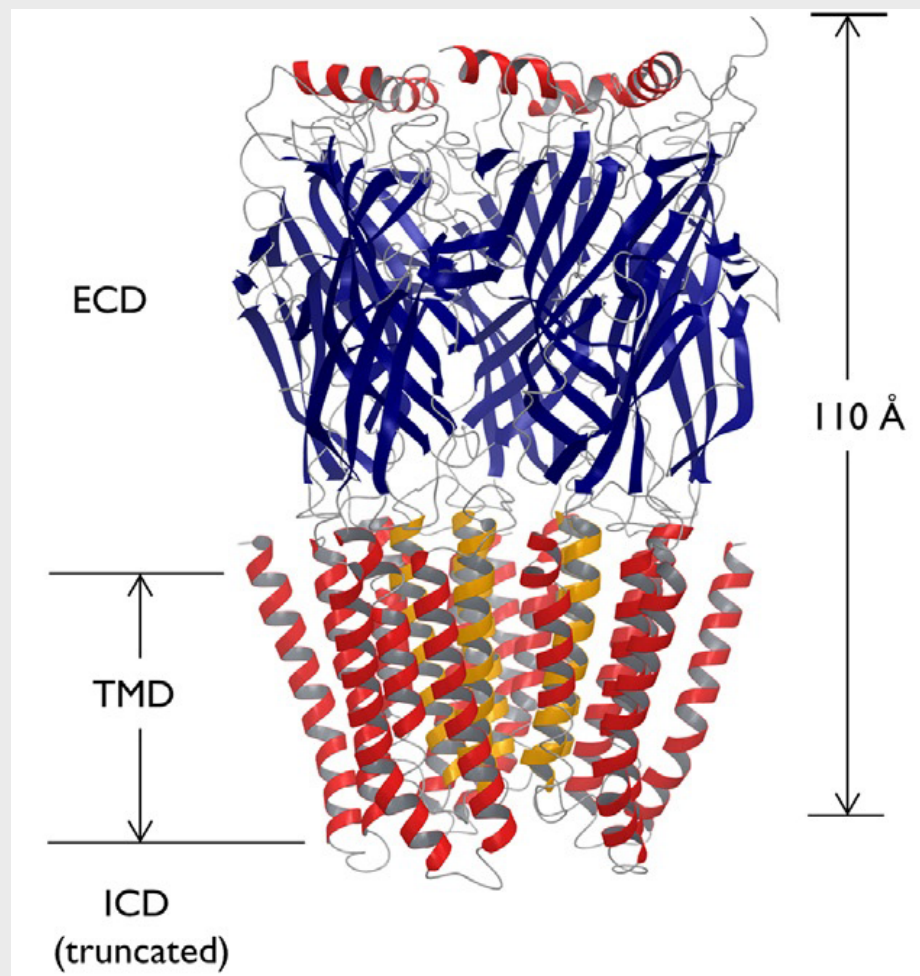
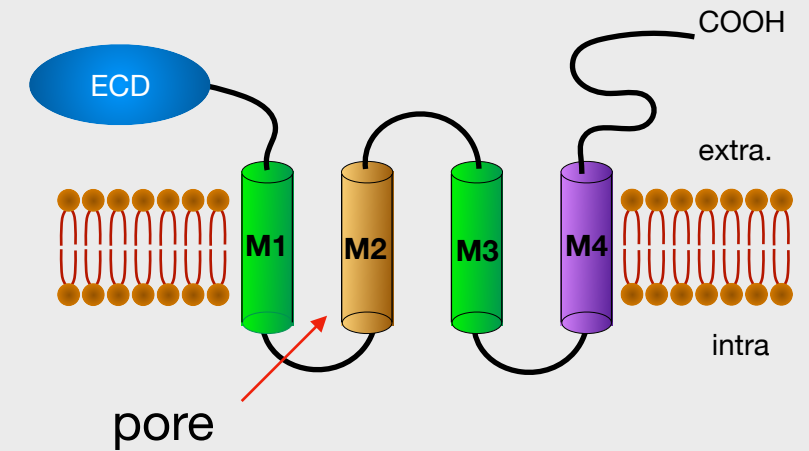
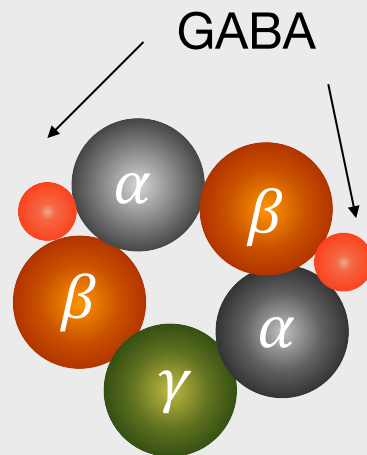
active conductances : ligand-gated channels

extracellular ligand

		subunits	permeability	Eeq/rev
nicotinic-R		$\alpha 1 \dots 10, \beta 1 \dots 4, \gamma, \varepsilon$	Na ⁺ , K ⁺ (, Ca ²⁺)	Erev≈0mV
5HT3-R		5HT3A...E	Na ⁺ , K ⁺ , Ca ²⁺	Erev≈0mV
GABA _A -R		$\alpha 1 \dots 6, \beta 1 \dots 3, \gamma 1 \dots 3, \delta, \pi, \theta, \rho 1 \dots 3$	Cl ⁻ , (HC03 ⁻)	Eeq≈-75mV
glycine-R		$\alpha 1 \dots 4, \beta$	Cl ⁻	Eeq≈-75mV
glutamate-R	AMPA	GluA1...4	Na ⁺ , K ⁺ (, Ca ²⁺)	Erev≈0mV
	NMDA	GluN1, GluN2A...D, GluN3A,B	Na ⁺ , K ⁺ , Ca ²⁺	Erev≈0mV
	kainate	GluK1...5	Na ⁺ , K ⁺ (, Ca ²⁺)	Erev≈0mV
P2X-R		P2X1...7	Na ⁺ , K ⁺ , Ca ²⁺	Erev≈0mV

active conductances : ligand-gated channels

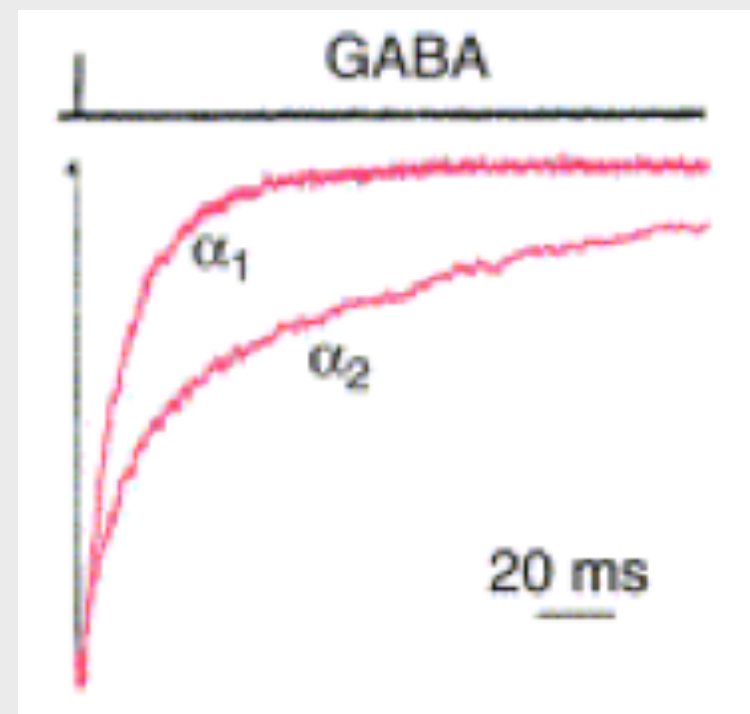
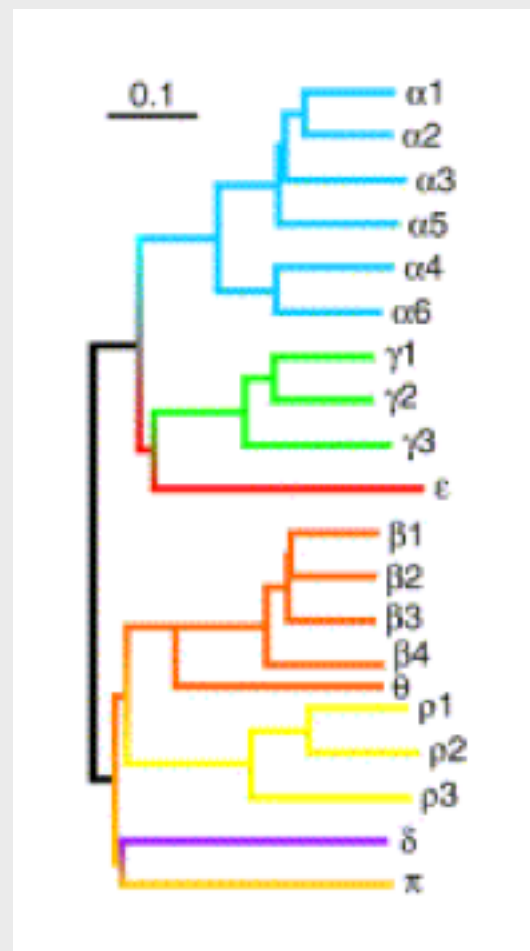
'Cys-loop' receptors: $GABA_A-R$



TMD: Transmembrane domain ; ECD: Extracellular domain ; ICD: Intracellular domain

active conductances : ligand-gated channels

'Cys-loop' receptors: GABA_A-R

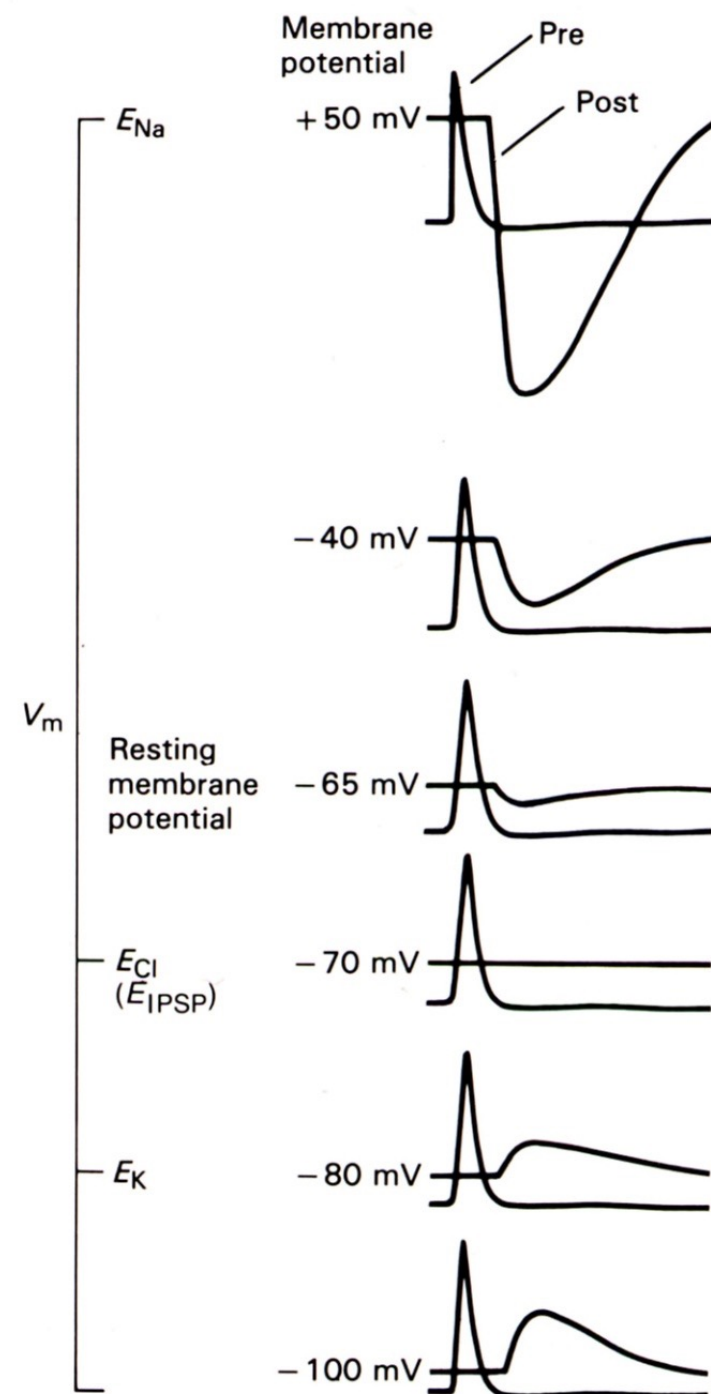


- fast opening
- deactivation kinetics vary according to the composition of the receptor

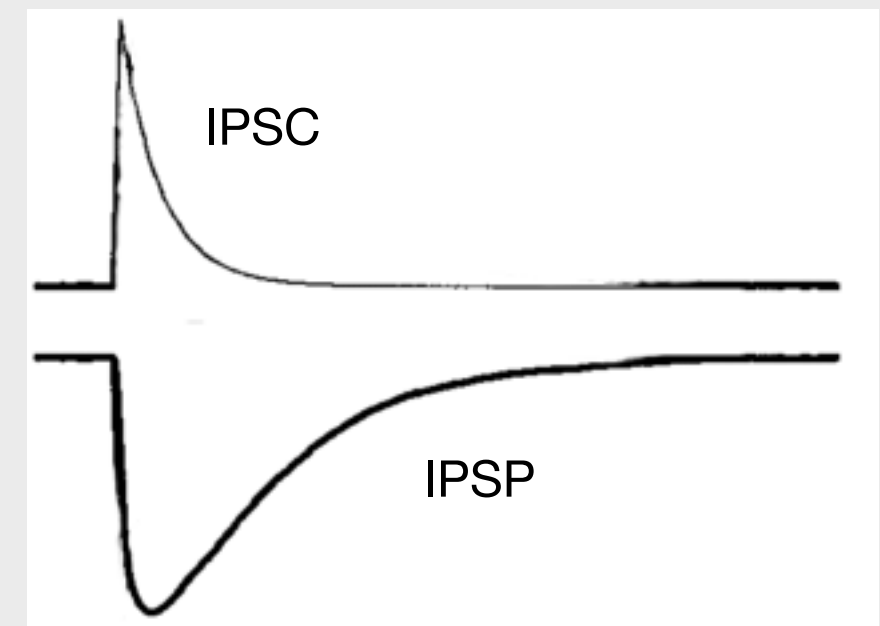
$E_{Cl} \approx -75\text{mV} < \text{threshold of AP} \longrightarrow \text{inhibiteur}$

active conductances : ligand-gated channels

'Cys-loop' receptors: GABA_A-R



$V_{rest} = -50 mV$

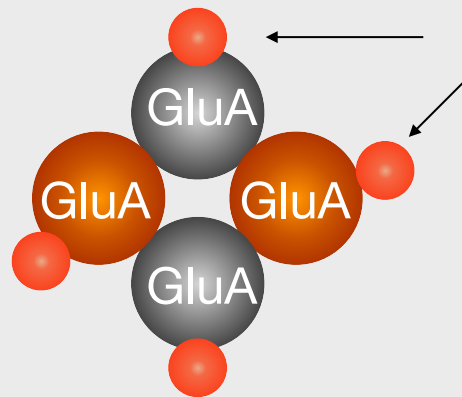


$E_{eqCl} \approx -75 mV < \text{threshold of AP} \longrightarrow \text{inhibiteur}$

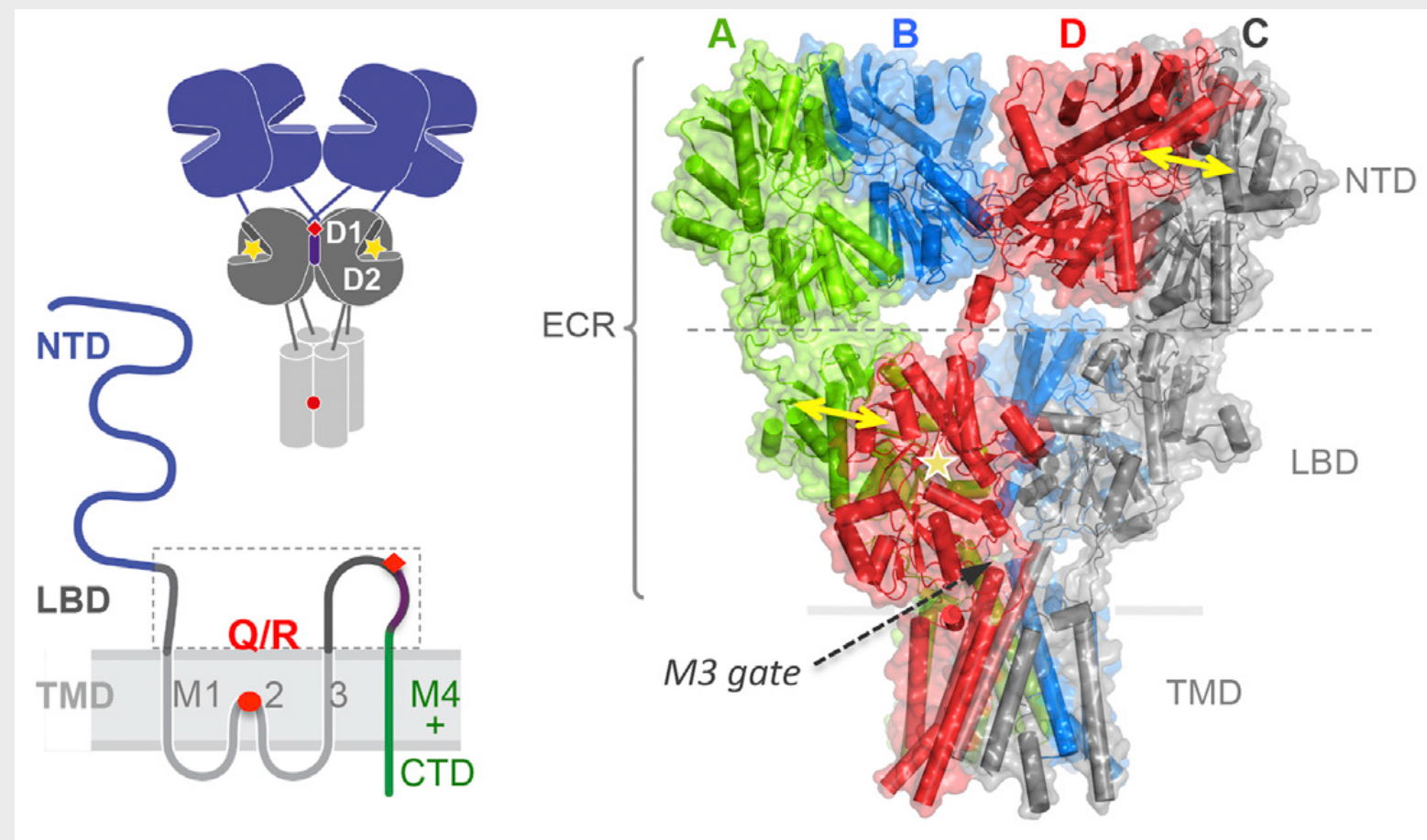
active conductances : ligand-gated channels

glutamate receptors: AMPA-R

homo or heterotetramer



glutamate (2 or 4 glutamate binding -> different open states, different conductances)



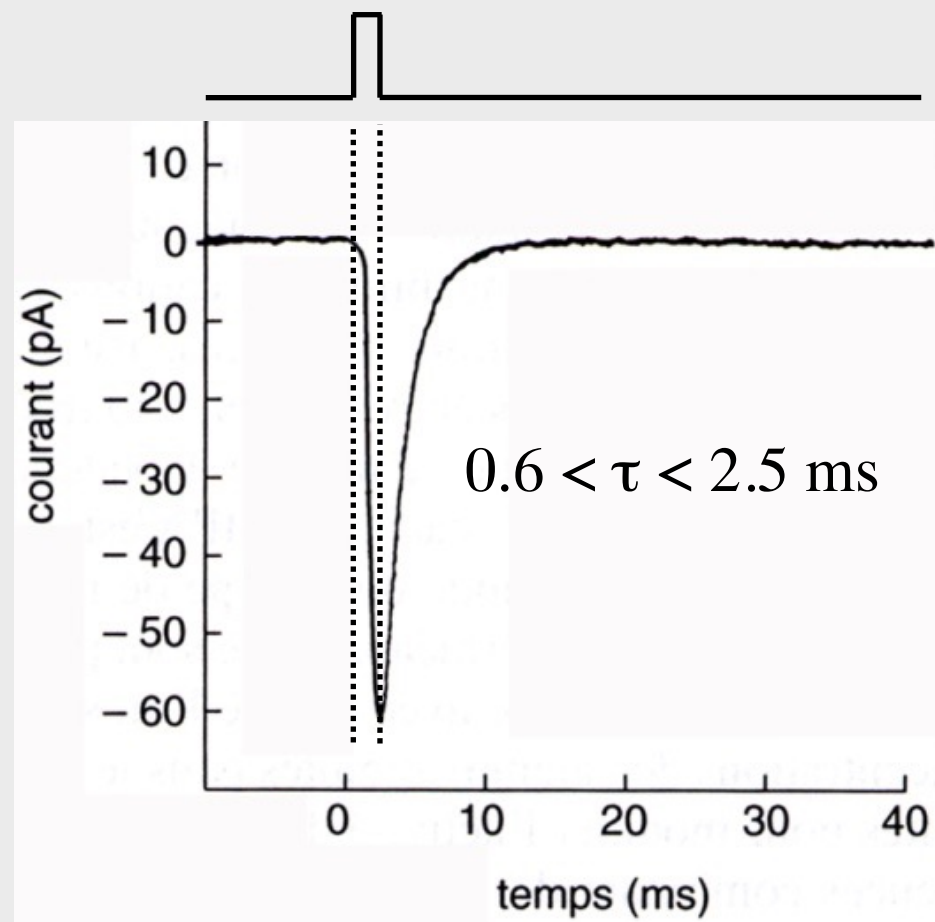
LBD: ligand binding domain

active conductances : ligand-gated channels

glutamate receptors: AMPA-R

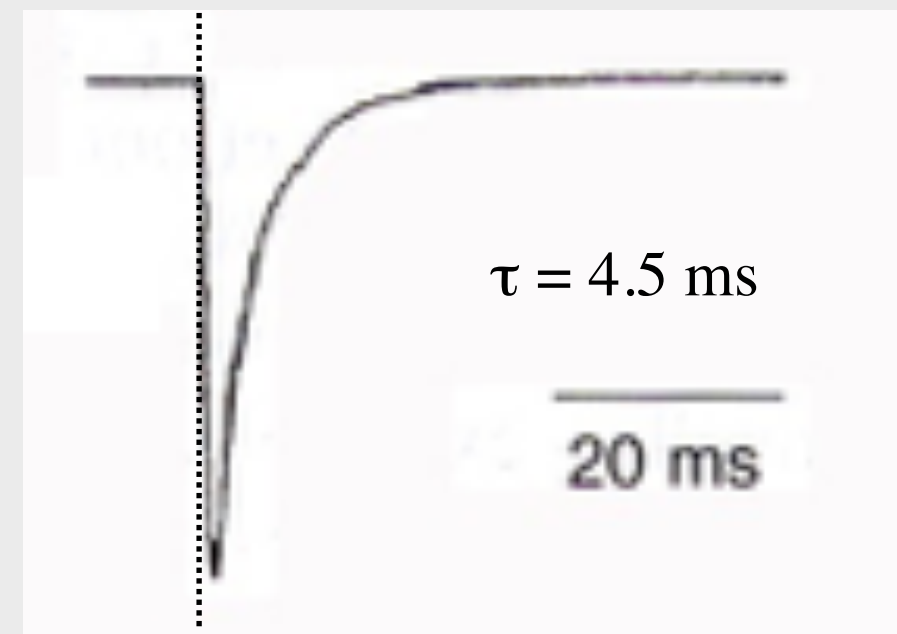
$E_{rev} \approx 0 \text{ mV}$

short application of glutamate



deactivation

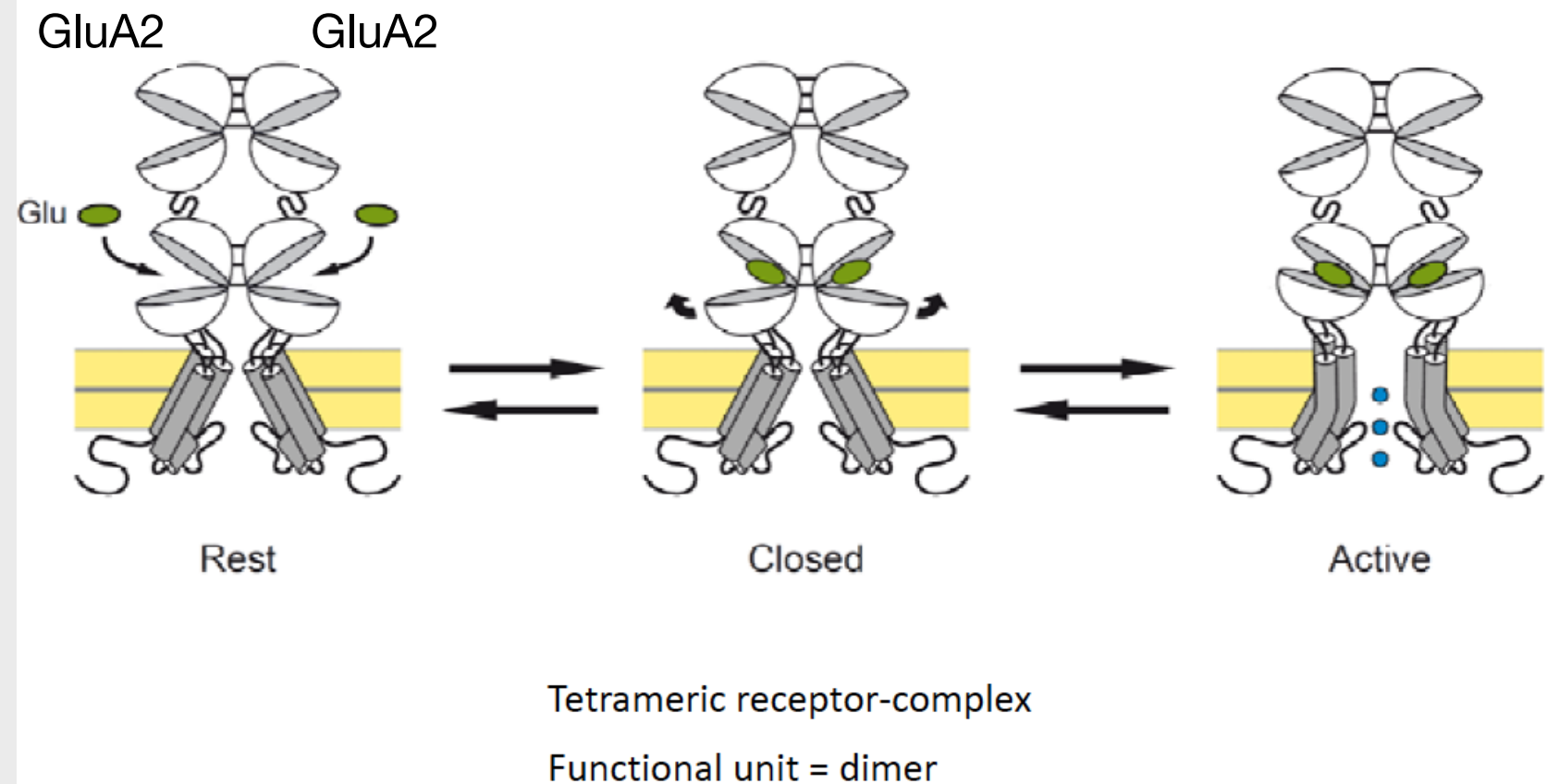
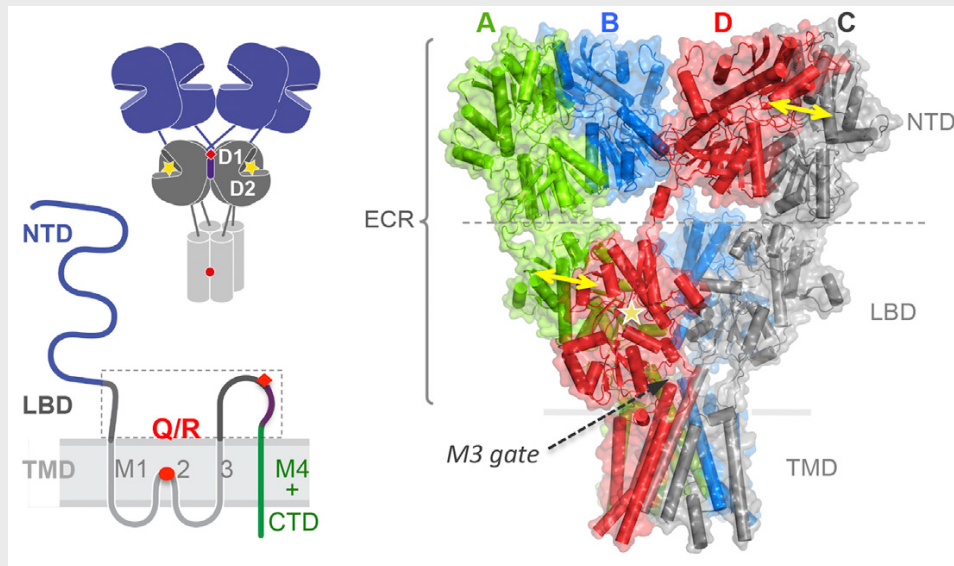
sustained application of glutamate



desensitization

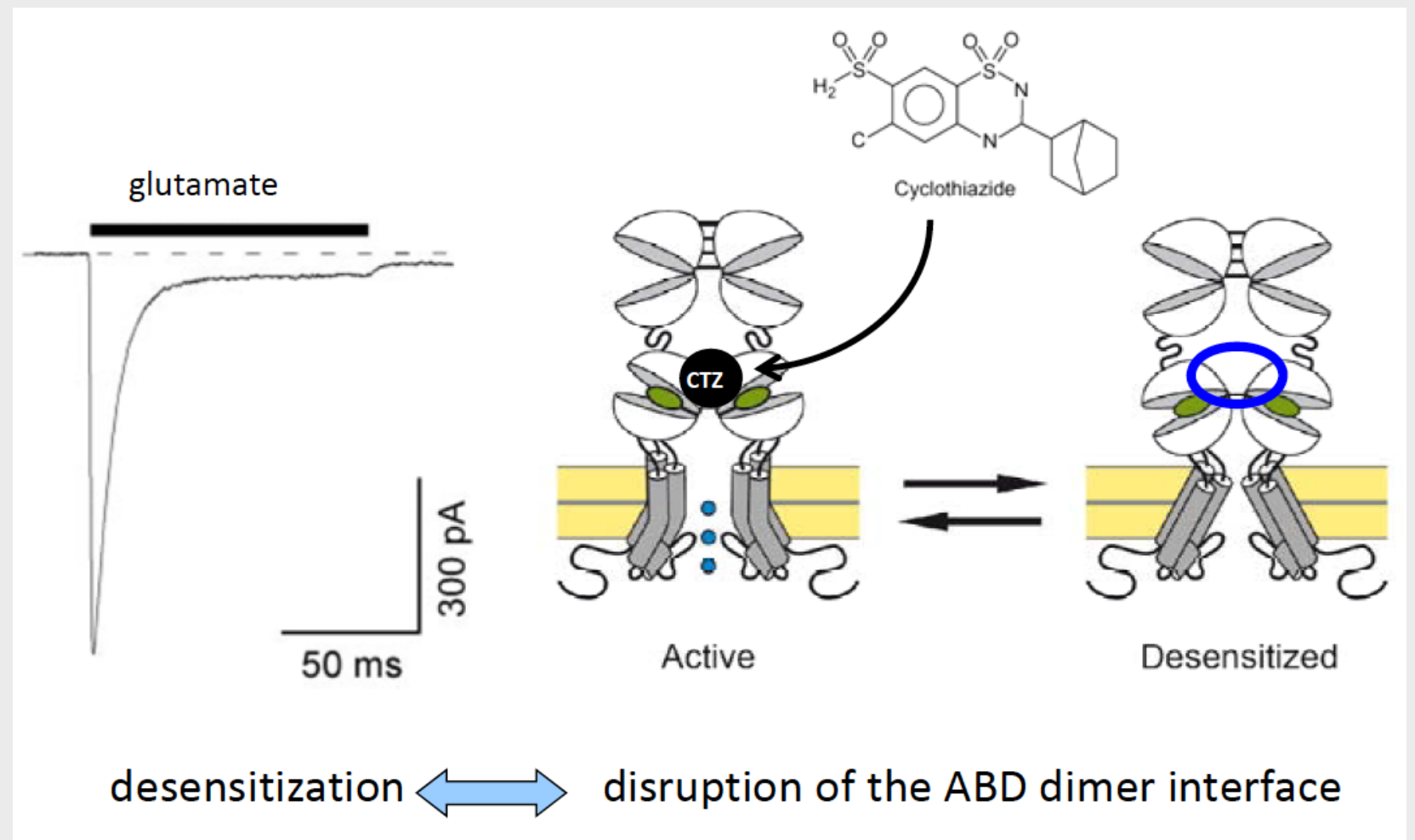
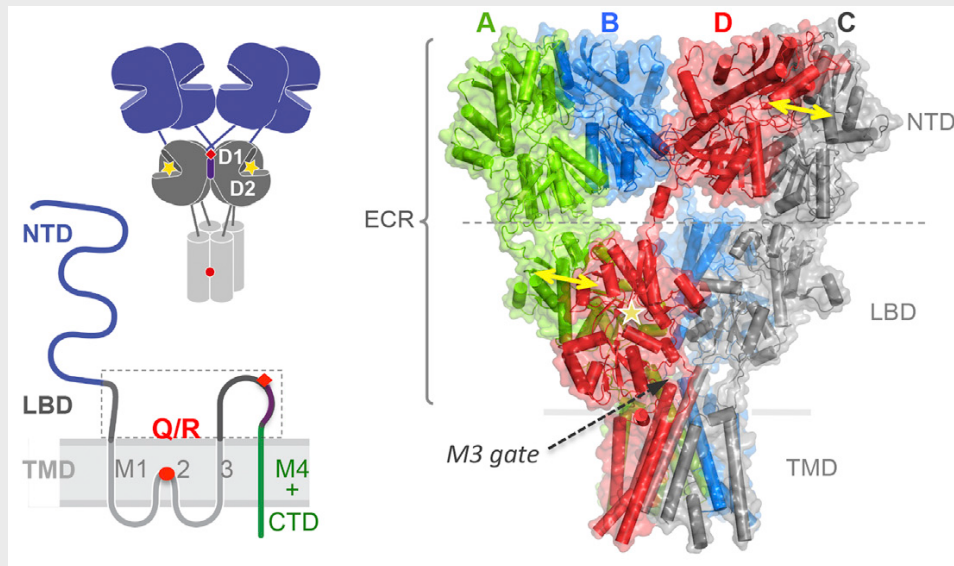
active conductances : ligand-gated channels

glutamate receptors: AMPA-R



active conductances : ligand-gated channels

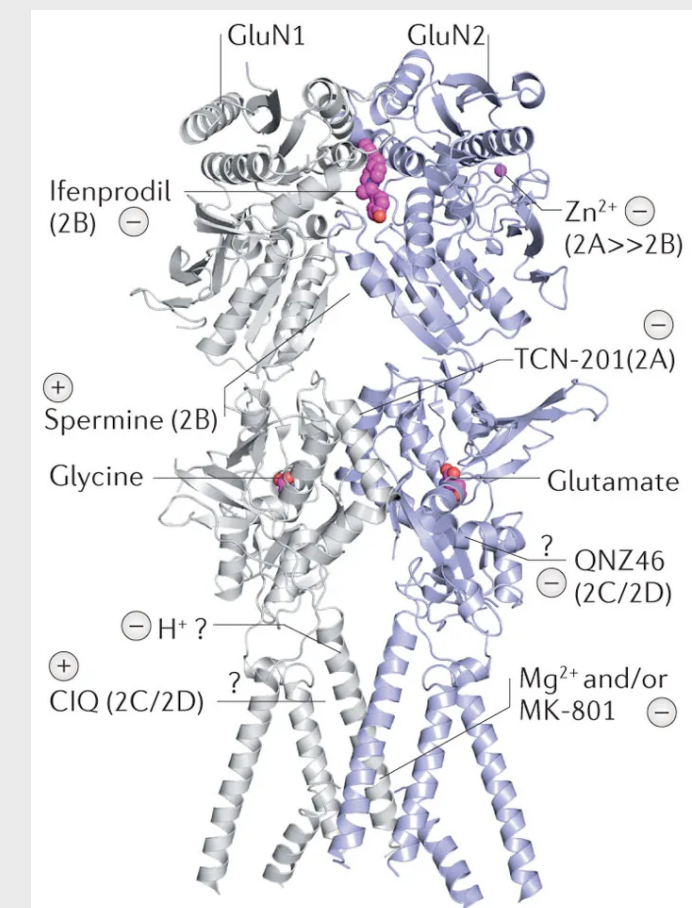
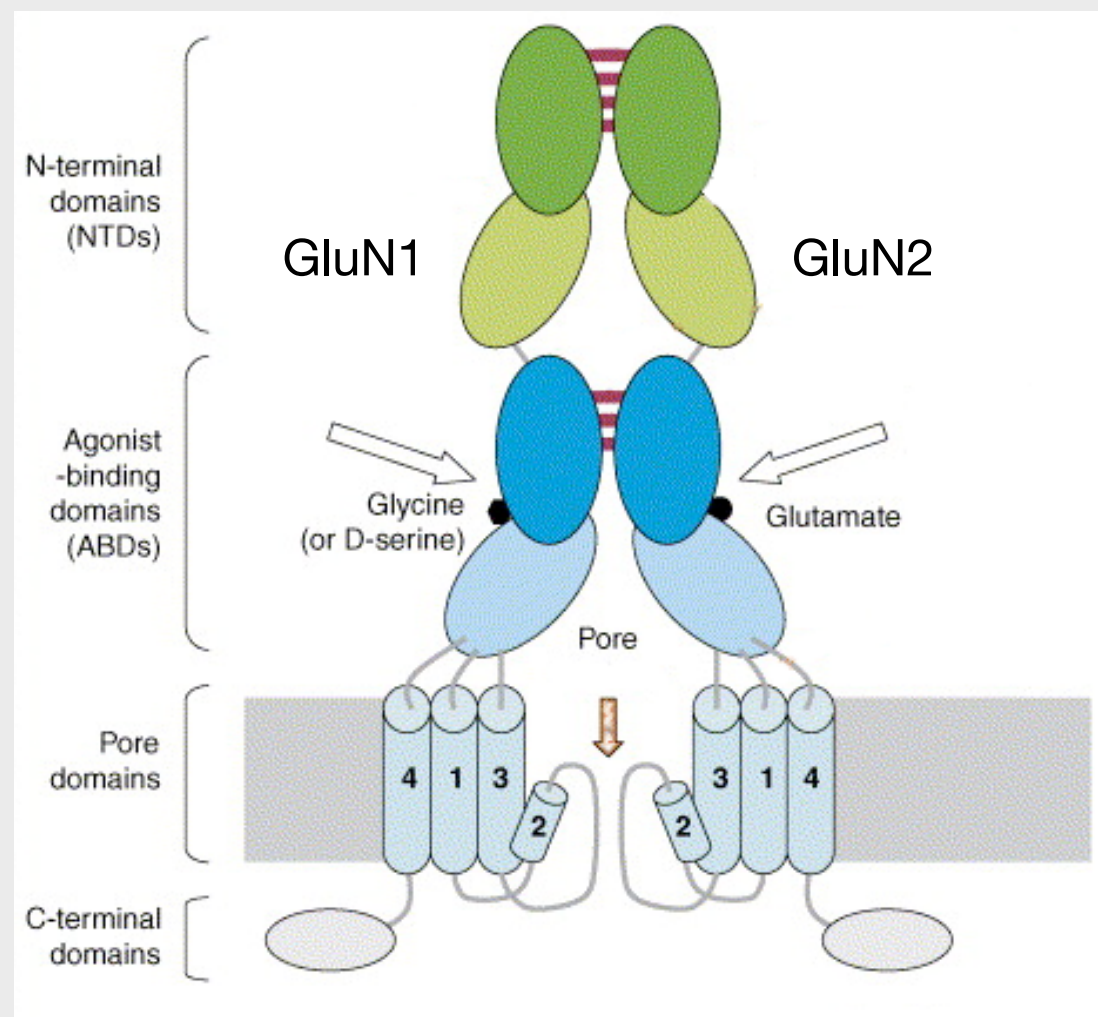
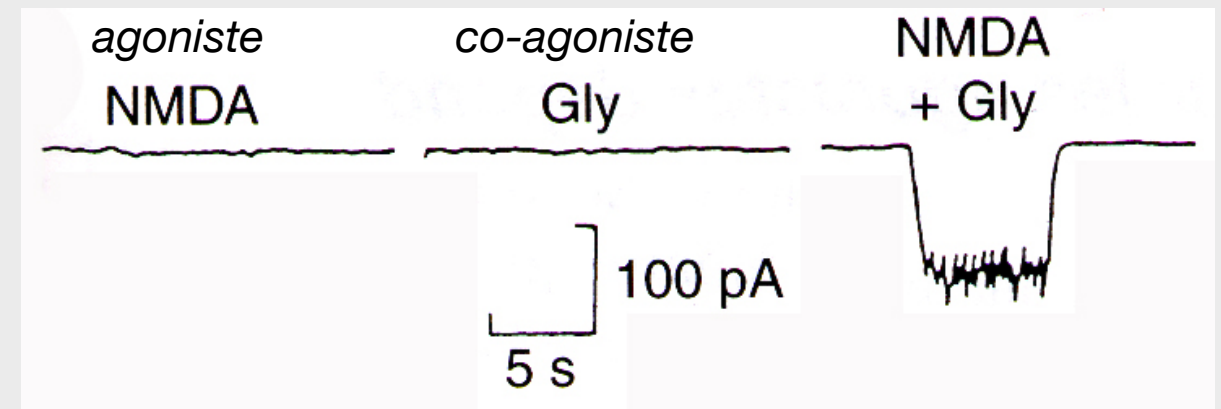
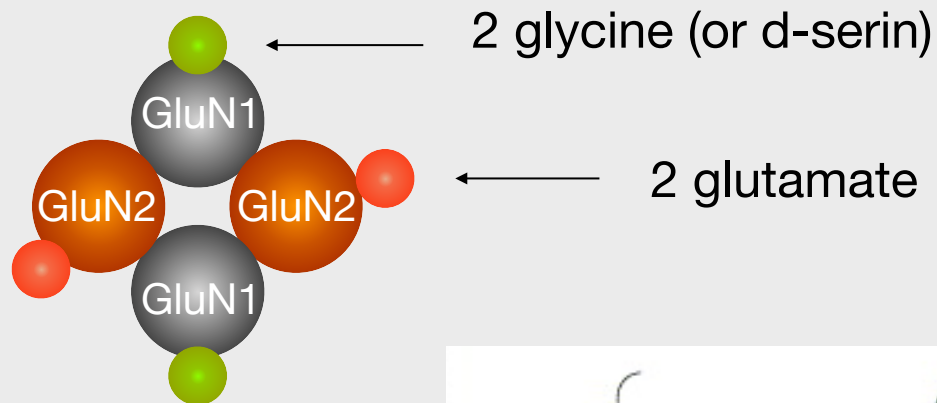
glutamate receptors: AMPA-R



active conductances : ligand-gated channels

glutamate receptors: NMDA-R

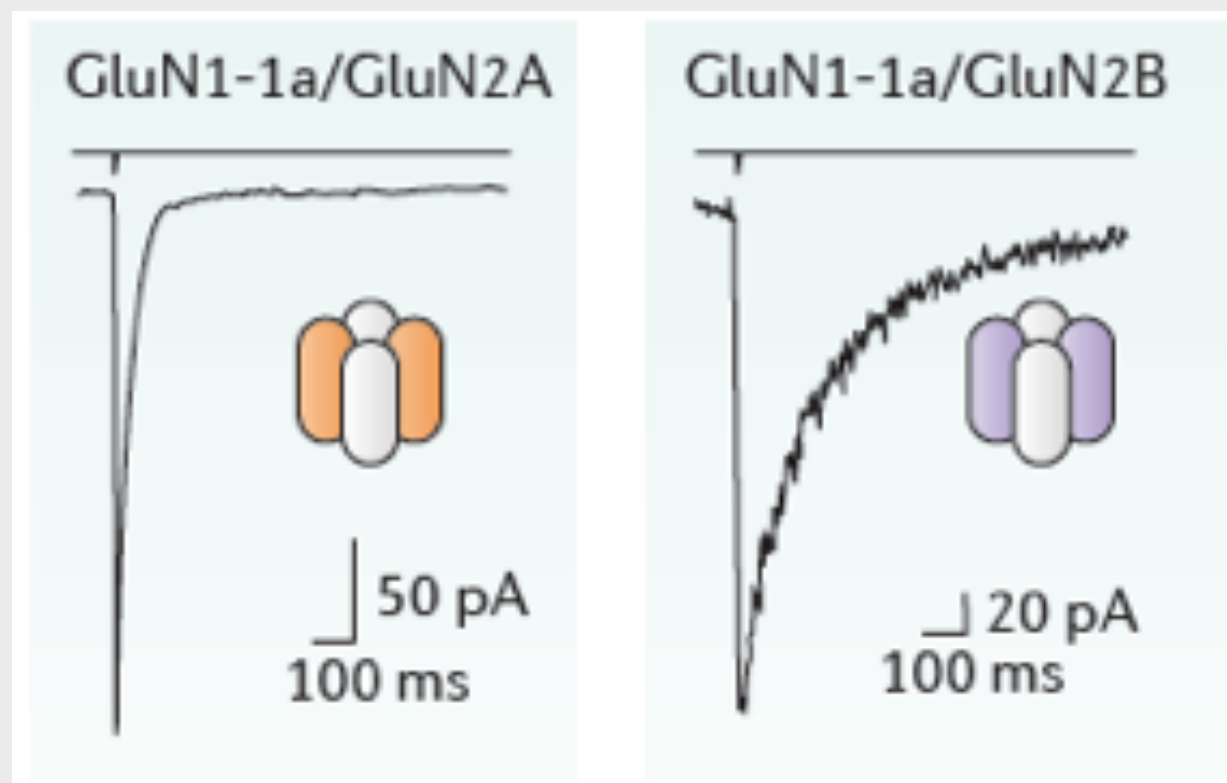
heterotetramer: GluN1 + GluN2 (or 3)



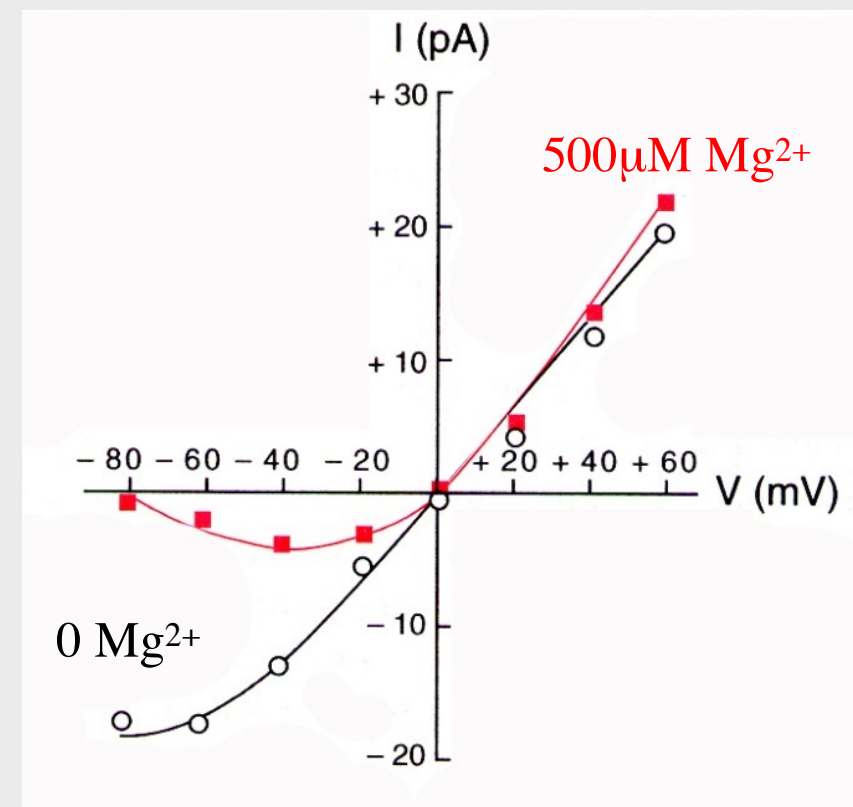
active conductances : ligand-gated channels

glutamate receptors: NMDA-R

$E_{rev} \approx 0\text{mV}$



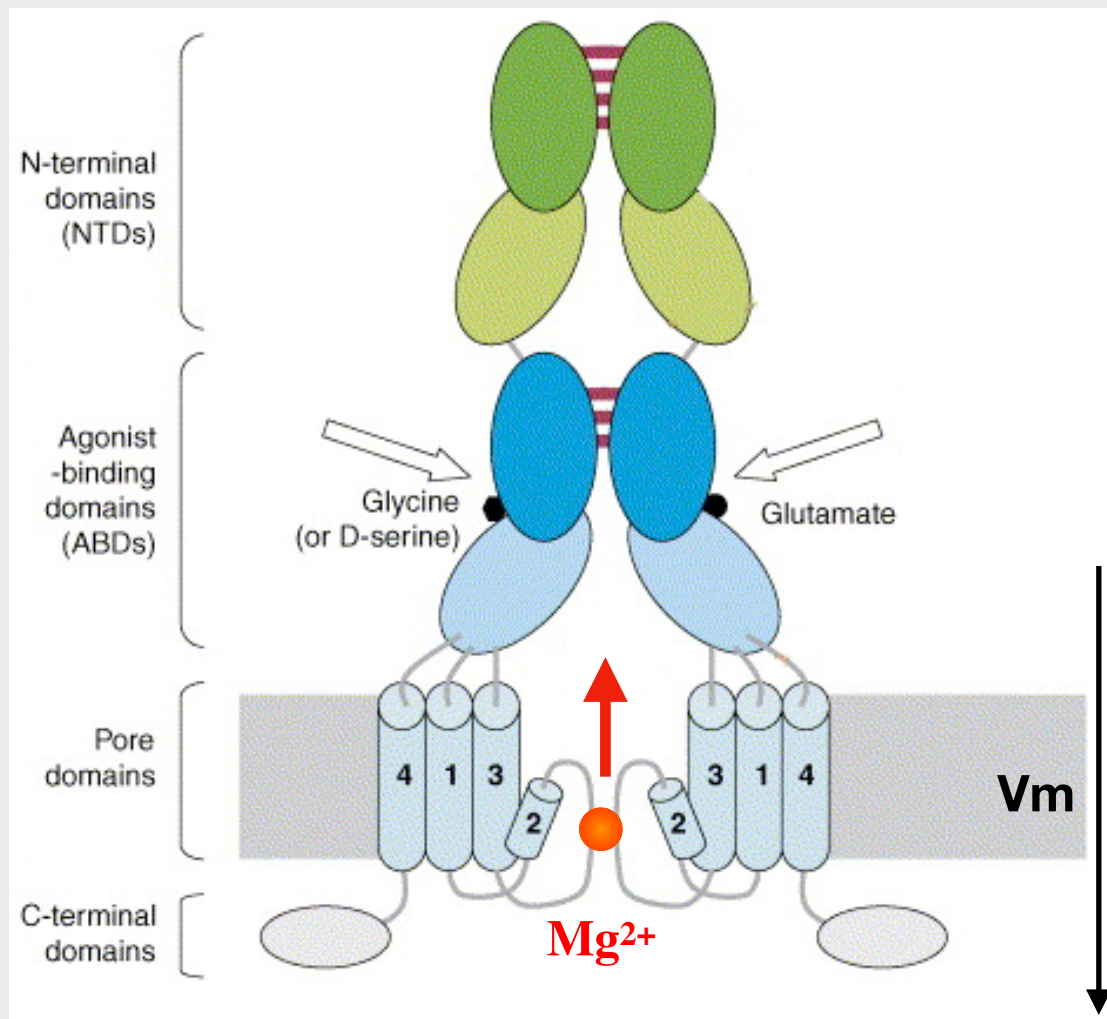
slow deactivation (*kinetic vary according to subunit*)



clear voltage-dependency induced by the Mg^{2+} block

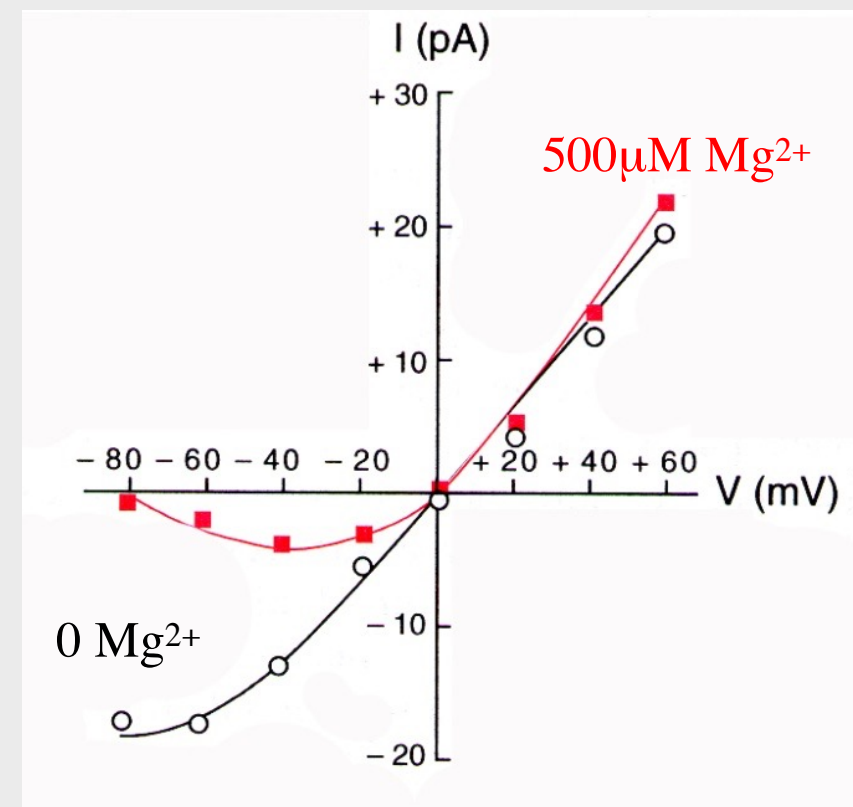
active conductances : ligand-gated channels

glutamate receptors: NMDA-R



V_m needs to be positive enough to expel the Mg^{2+} ion from the pore

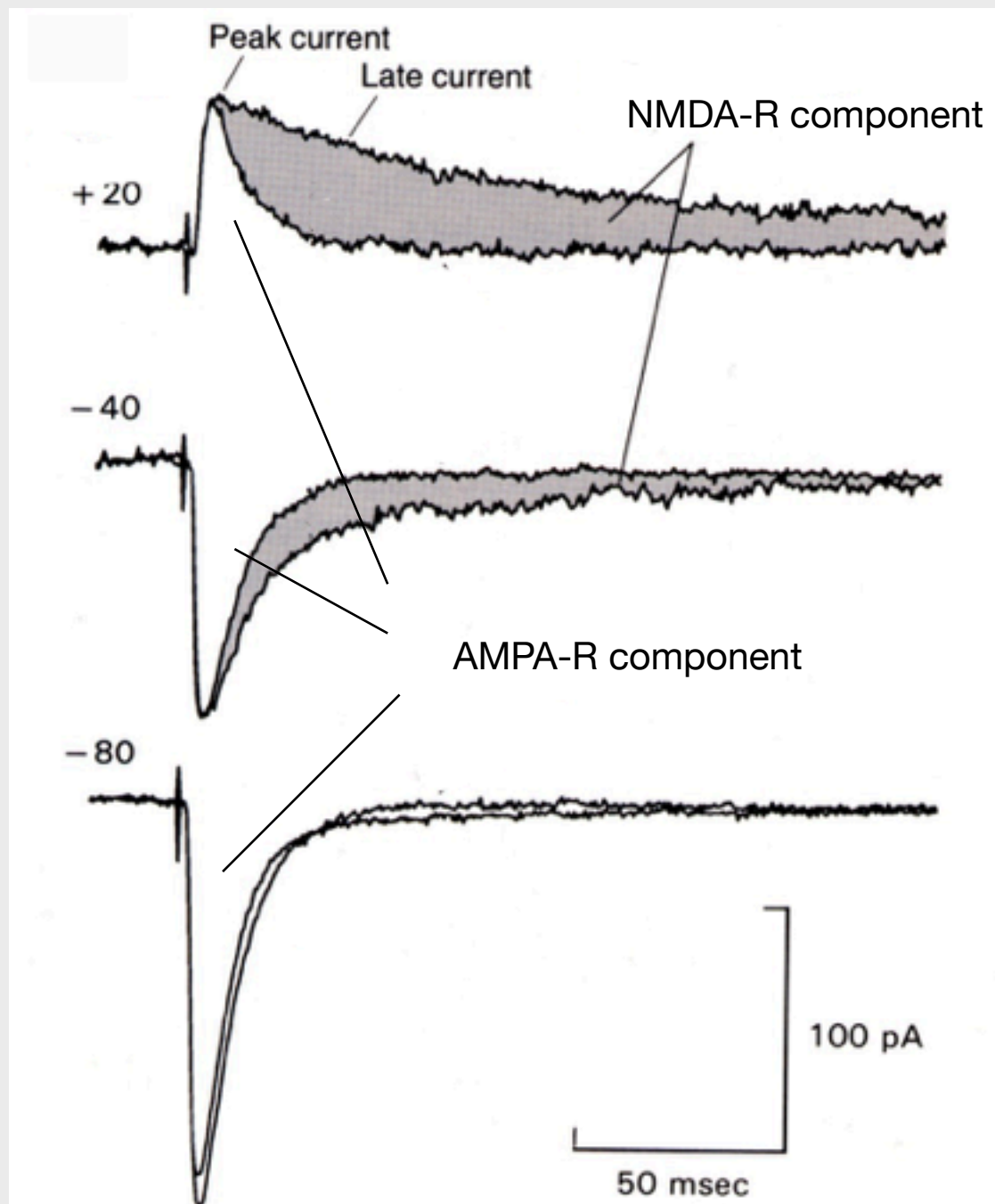
$E_{rev} \approx 0mV$



clear voltage-dependency induced by the Mg^{2+} block

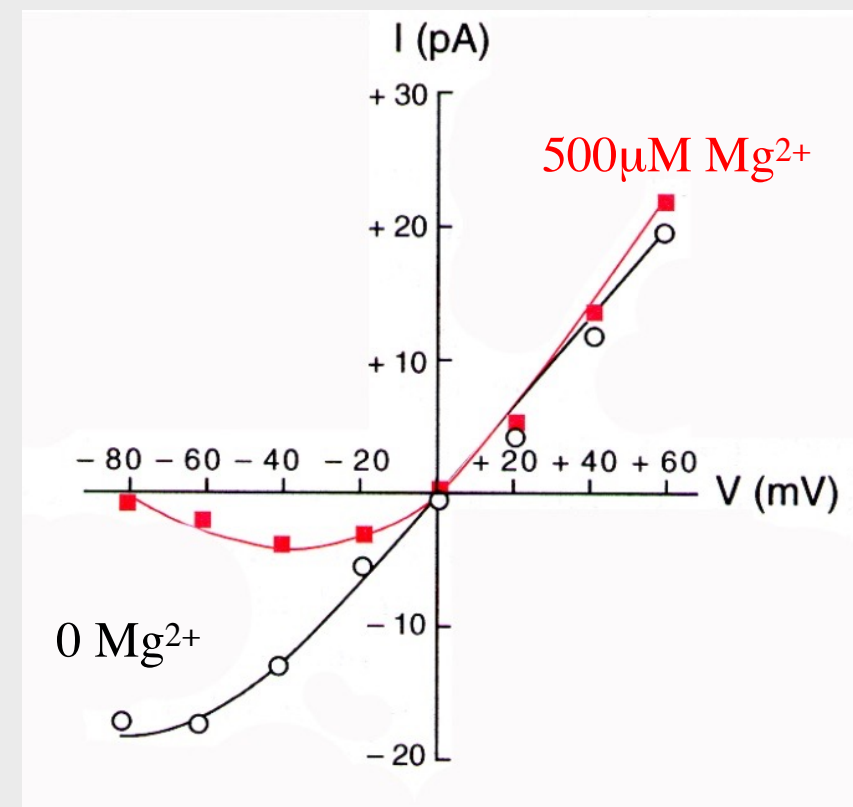
active conductances : ligand-gated channels

glutamate receptors: NMDA-R

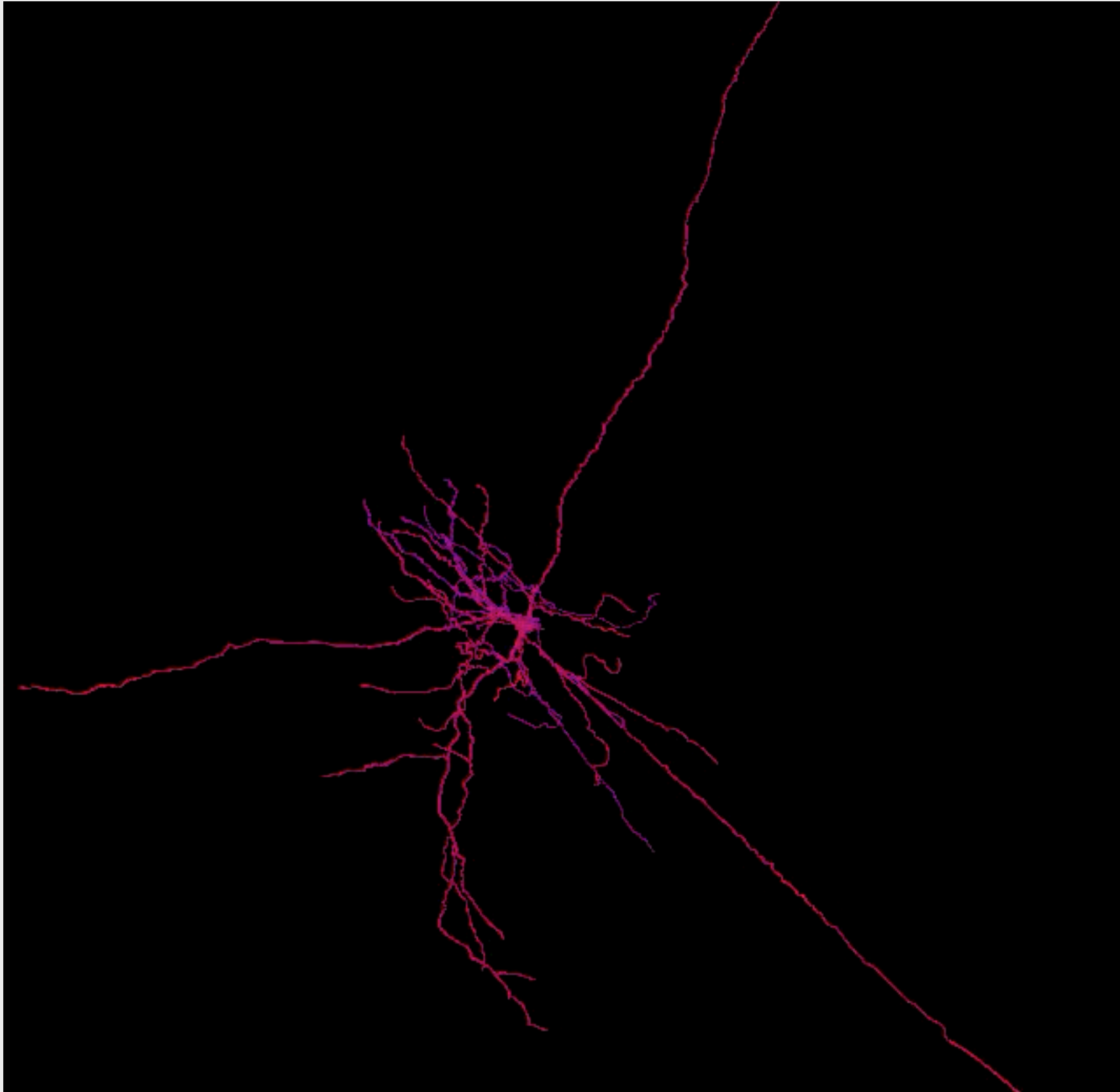


V_m needs to be positive enough to expel the Mg^{2+} ion from the pore

$E_{rev} \approx 0 mV$



clear voltage-dependency induced by the Mg^{2+} block



neocortical layer VI pyramidal neuron (*simulation : A. Destexhe*)